# The Projection Spectral Layer Theory (PSLT): A Rank-2 Computable Closure for the Three-Generation Structure and Higgs Signal Strength

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We present a comprehensive, computable EFT-level "closed chain" for the Projection Spectral Layer Theory (PSLT) in which a self-consistent three-generation structure emerges on the baseline closure from competing spectral entropy, kinetics, and visibility. The framework is built on a minimal Einstein–Yang–Mills–Higgs (EYMH) effective action with non-minimal curvature coupling, where the background geometry induces a layer-indexed spectral scale $\omega_N$. The theoretical closure is achieved by unifying three key modules: (i) a **Micro-degeneracy module** ($g_N$) implemented in baseline `fp_2d_full` mode (bounded microcanonical low-$N$ window + energy-gap tail prescription, phase-space normalized at $N = 3$); (ii) a **Rank-2 Computable Kinetic module** ($\Gamma_N$) defined as the largest positive eigenvalue of a $2 \times 2$ growth matrix with explicit WKB tunneling suppression; and (iii) a **Visibility module** ($B_N$) implemented in baseline fully normalized EFT-operator profile mode (`b_mode=eft_operator_norm`), where overlap-extracted flavor-layer couplings and mediator scales feed finite one-loop + LL-RG-normalized operator kernels, with high-$N$ saturation ($B_{N>3} = 1$) to avoid double-counting with the $g_N$ regulator; a risk-weighted profile-anchored runtime-direct visibility parity branch is also available via `cell_direct_runtime_release_tuned`, using a canonical D-only anchor profile $\alpha(D)$ rather than a fixed global blend, but it is still not interpreted as a strict all-direct closure. The resulting normalized layer probability $P_N(t_{\mathrm{coh}})$ demonstrates, within this baseline closure, a stable "winner" phase diagram with clear dominance of layers $N = 1, 2, 3$ over a significant region of the geometric parameter space $(D, \eta)$. We verify the closure against two benchmarks: (1) internal stability of the three-generation hierarchy (Generation Ratio $> 90\%$ over 92.7% of the sampled $(D, \eta)$ grid), and (2) a scan-level EFT/Wilson-matched compatibility test with the ATLAS Run-3 $H \to \mu\mu$ signal strength ($\mu_{\mu\mu}^{\mathrm{obs}} = 1.4 \pm 0.4$). In the current UV+LL-RG baseline mode (`hll_observable_mode=eft_wilson_uv_rge`), the illustrative acceptance band is about 15.0% of the scanned grid for $\chi^2 < 4$, providing a sharper falsifiable constraint on $(D, \eta)$.

**CONTENTS**

## I. INTRODUCTION

The existence of exactly three generations of fermions is one of the most persistent puzzles in the Standard Model (SM) of particle physics. While the SM accommodates three families through the CKM and PMNS mixing matrices, it offers no dynamical reason for why the number is three, nor why the mass hierarchy spans such a vast range. Traditional attempts to explain this structure often rely on complex discrete symmetries or string logic compactifications that, while elegant, can be difficult to falsify directly.

The Projection Spectral Layer Theory (PSLT) proposes a different paradigm: *Generation structure is a spectral consequence of the underlying projection geometry of spacetime.* In this framework, "generations" are not ad-hoc copies but rather distinct *spectral layers* ($N = 1, 2, 3, \dots$) of vacuum excitations, selected by a competition between microstate degeneracy (entropy) and geometric formation rates (kinetics). This structural viewpoint is complementary to recent Clifford-algebra work where three families emerge from an intrinsic $S_3$ organization rather than manual generation copying [1]; the emphasis here is different: a map-level computable closure with explicit scan-level falsifiability.

In this work, we consolidate the complete theoretical framework of PSLT into a single, falsifiable "closed chain". We upgrade earlier effective descriptions to a **computable** rank-2 dynamical system with an action-derived extraction subchain and an action-informed global scan chain, where $\chi_{LR}(D)$ and $\tilde{A}_\ell(D)$ are now injected on the full scan grid from direct action-derived profiles. The core deliverable is a map from geometric control parameters $(D, \eta)$—representing dual-center separation and overlap—to an experimentally observable layer probability $P_N$.

Our primary contributions are:

1. **Theoretical Motivation**: We present three convergent arguments for the micro-degeneracy $g_N$ (Route A: Mock-modular, Route B: ER=EPR, Route C: QG Template), and implement a scan-ready baseline through the bounded `fp_2d_full` profile with controlled high-$N$ tail prescription.

2. **Rank-2 Computable Kinetics**: We replace heuristic growth rates with a rigorous eigenvalue problem for the formation matrix $\mathbf{M}_N$, incorporating numerical WKB tunneling suppression.

3. **Fully Normalized EFT Visibility**: We promote $B_N$ to an EFT-operator-normalized baseline built from overlap-extracted flavor-layer couplings and mediator scales with finite one-loop + LL-RG normalization; overlap-only and Yukawa laws are retained only as legacy comparators/fallbacks, while runtime-direct visibility extraction is release-qualified

only in a risk-weighted profile-anchored parity branch (`cell_direct_runtime_release_tuned`, canonical D-only anchor profile $\alpha(D)$), not as a strict all-direct closure.

4. **Verification**: We demonstrate stable three-generation dominance at $\mathcal{R}_3 > 90\%$ and a constrained EFT/Wilson-matched-compatible band for LHC $H \to \mu\mu$.

## II. THEORETICAL FRAMEWORK

### A. The Closed Chain Equation

The central output of the PSLT framework is the normalized layer occupancy probability $P_N(t_{\rm coh})$, defined by the competition between entropic weight ($W_{\rm entropy}$) and kinetic formation ($W_{\rm kinetic}$). The master equation is:

$$\boxed{P_N(t_{\rm coh}; D, \eta) = \frac{W_N(t_{\rm coh})}{\sum_K W_K(t_{\rm coh})}, \qquad W_N = B_N \cdot g_N \cdot \left(1 - \mathrm{e}^{-\Gamma_N(D,\eta) t_{\rm coh}}\right).} \tag{1}$$

This equation unifies the three critical modules:

- $g_N$: **Micro-degeneracy** (Entropy). The number of available microstates in layer $N$.
- $\Gamma_N$: **Dynamical Rate** (Kinetics). The rate at which these states can form/tunnel from the vacuum.
- $B_N$: **Visibility Factor** (Observation). The coupling strength of layer $N$ to the observable sector.

Figure 1 illustrates the logical flow of the PSLT framework.

### B. Action-Derived Chain at a Glance

For implementation-level clarity, Figure 2 summarizes the action-derived operator chain and its interfaces to scan-level diagnostics in one page. The executable, clone-level Standard Operating Procedure (SOP) is provided in Appendix A.

### C. Geometric Foundation: Minimal EYMH + Curvature

We start from a minimal Einstein–Yang–Mills–Higgs (EYMH) effective action supplemented by a non-minimal curvature coupling $\xi R|\Phi|^2$:

$$S = \int \mathrm{d}^4x \sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R - \frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu} - (D_\mu\Phi)^\dagger(D^\mu\Phi) - \lambda(|\Phi|^2 - v^2)^2 - \xi R|\Phi|^2\right]. \tag{2}$$

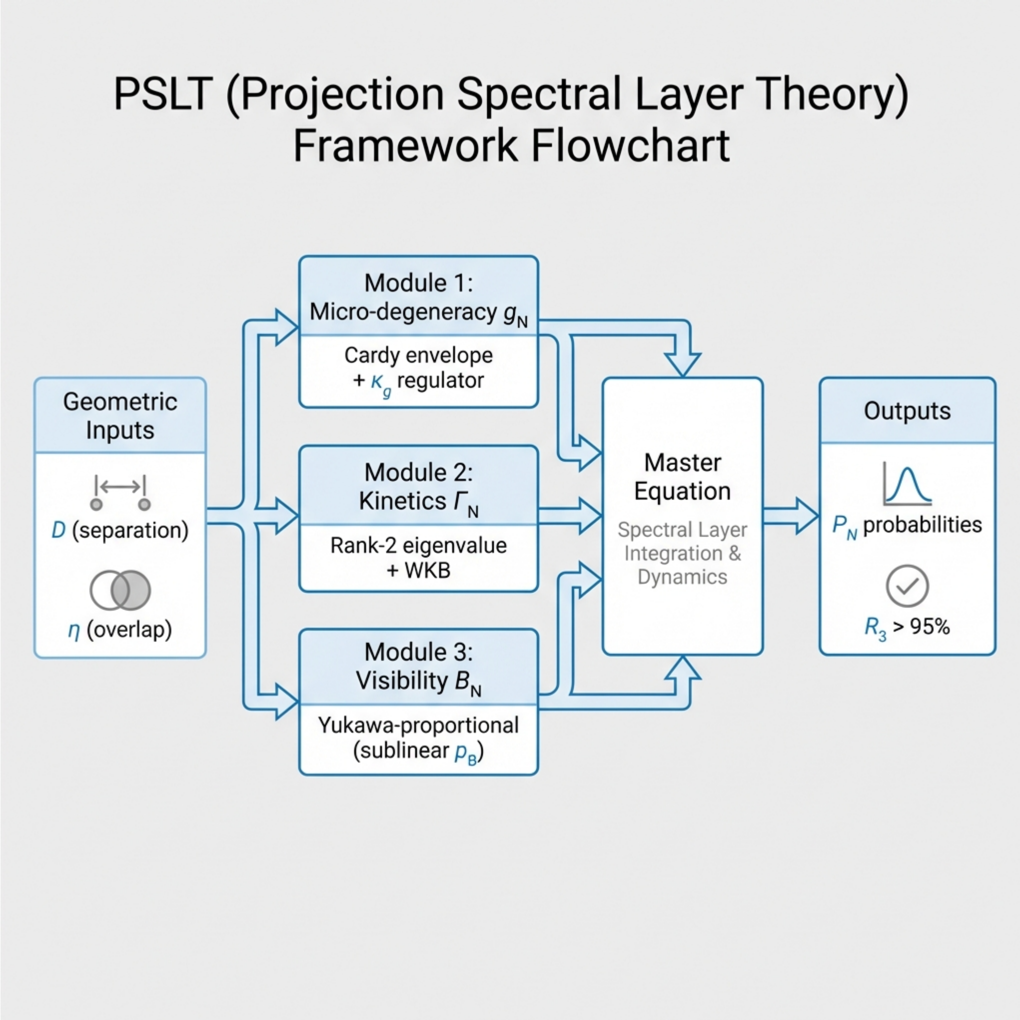


FIG. 1. Schematic overview of the PSLT framework. Geometric inputs $(D, \eta)$ feed into three modules (micro-degeneracy $g_N$, kinetic rate $\Gamma_N$, visibility $B_N$), which combine in the master equation to yield the observable layer probabilities $P_N$ and the three-generation ratio $\mathcal{R}_3$. In the baseline chain, high-$N$ control is implemented by the bounded phase-space-tail prescription in `fp_2d_full`, while $B_{N>3}$ saturates to unity.

PSLT posits that the background geometry is a *projection* from a higher-dimensional manifold, inducing a locally conformally flat metric:

$$g_{\mu\nu}(x) = \Omega^2(x)\eta_{\mu\nu}. \tag{3}$$

Under conformal rescaling $\Phi \to \Omega^{-1}\tilde{\Phi}$, the scalar equation of motion reduces to a Schrödinger-like eigenproblem:

$$[-\nabla^2 + V_{\text{eff}}(x)]\psi_N = \omega_N^2 \psi_N, \tag{4}$$

where the effective potential $V_{\text{eff}}$ is determined by the conformal factor $\Omega(x)$ and the specific projection geometry (e.g., stereographic projection of dual centers).

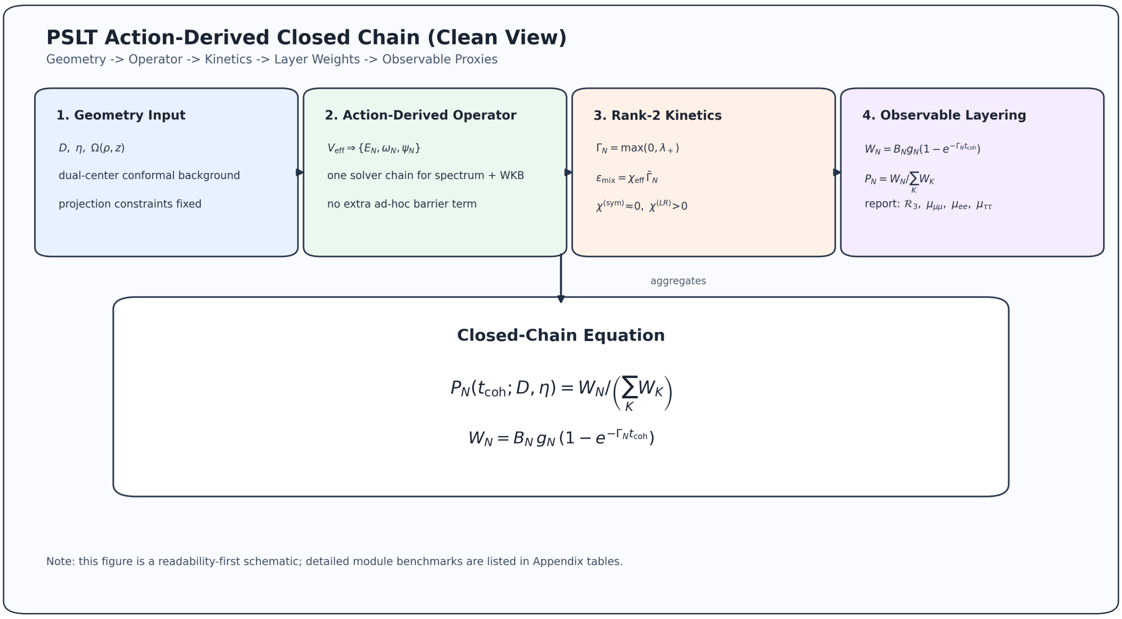


FIG. 2. One-page action-derived chain overview used in this manuscript: geometry input → operator extraction → scan diagnostics → observable EFT/Wilson checks. This diagram is aligned with the repository reproducibility entrypoint and Appendix A.

The layer index $N$ emerges naturally as the principal quantum number of this spectrum. In Section III, we derive the spectral scale $\mu(D)$ explicitly from a two-center geometry; the result supports a power-law scaling $\mu(D) \propto D^{-\gamma}$ with $\gamma \approx 0.1$ in the demonstrator regime. For analytic tractability, we adopt the **hydrogenic proxy**:

$$\omega_N(D) \simeq \mu(D) \left(1 - \frac{1}{2N^2}\right), \qquad \mu(D) = \mu_0\, D^{-\gamma}, \tag{5}$$

where $\mu_0$ and $\gamma$ are determined numerically from the geometry (Section III). This form captures the essential $N$-dependence from the hydrogenic spectrum while allowing the $D$-dependence to be set by explicit calculation. For transparency, we additionally provide a bound-state exact benchmark for $(\omega_1, \omega_2)$ at $D = \{6, 12, 18\}$ with coarse/mid/fine convergence from the same action-derived operator chain (Appendix B 9, Table X). These benchmark values use $E = \omega^2 - m_0^2 < 0$ and should not be mixed with the generalized localized-extraction eigenvalues $\lambda$ in Appendix D.

### D. Model-Chain Assumption Status

For transparency, Table I separates action-derived components from EFT-level surrogates currently used in the global scan.

TABLE I. Status map of the PSLT closed chain in the present manuscript.

| Component | Current Status | Used in Global Scan | Next Upgrade Path |
|---|---|---|---|
| $\Omega(\rho, z; D)$ two-center geometry | Projected-Poisson two-source Green solution | Yes | Derive the same source from an explicit higher-dimensional parent |
| $V_{\text{eff}}$, low modes, WKB action | Action-derived and numerically converged | Yes | Extend the same solver chain to the full $(D, \eta, N)$ map |
| $g_N$ phase-space mode (`fp_2d_full`) | Baseline `fp_2d_full`; direct $g_N$ parity released in `cell_direct_runtime`; risk-weighted visibility parity in `cell_direct_runtime_release_tuned` | Yes: `full_direct`, `cell_direct_runtime`, `release_tuned` | Lower the anchor profile $\alpha(D)$ toward strict all-direct closure without reopening gate drift |
| $\chi_N^{(LR)}(D)$ localized channel | Localized-direct `Dgrid60` profile from 2D fine extraction; release-production parity path available in `cell_direct_runtime` | Yes: `full_direct`, `cell_direct_runtime` | Push the same $\alpha(D)$ toward strict all-direct closure while preserving parity |
| $\Gamma_{N,\ell}$ channel normalization ($A_\ell$) | Action-derived $\tilde{A}_\ell(D)$ baseline with stabilized localized-direct support; legacy constant-$A_\ell$ kept only as comparator | Yes: `full_direct` | Move to fully localized channel-resolved direct extraction over $(D, \eta, N)$ |
| Open-system $\chi_{\text{eff}}$ (Lindblad) | Scan-ready diagnostic with profiled $\gamma_\phi(D), \gamma_{\text{mix}}(D)$ and calibrated `open_system_micro` | Diagnostic only | Derive a microscopic EYMH bath $(L_k, \gamma_k)$ before promotion |
| $B_N$ visibility profile from EFT normalization | Baseline `eft_operator_norm`; runtime-direct operator extraction promoted only in the risk-weighted release branch | Yes | Reduce mean anchoring and approach strict all-direct visibility closure |
| $H \to \mu\mu$ mapping [Eq. (56)] | EFT/Wilson-matched baseline with UV-tree, finite-one-loop, and LL-RG audits | Yes | Extend the current closure to fuller UV-to-EFT matching from the EYMH action |

## III. GEOMETRY-TO-SPECTRUM WORKED EXAMPLE

This section demonstrates the complete derivation chain $\Omega(x; D) \to V_{\text{eff}}(x) \to \omega_N(D)$, establishing the geometry-to-spectrum correspondence as a computational reality rather than an ansatz.

### A. Two-Center Harmonic Conformal Factor

The geometric origin of the dual-center conformal factor is formulated as a projected Poisson constraint on the static spatial slice:

$$\nabla^2\Omega(\mathbf{x}) = -4\pi\,\sigma(\mathbf{x}), \qquad \sigma(\mathbf{x}) \equiv a[\rho_\varepsilon(\mathbf{x} - \mathbf{x}_+) + \rho_\varepsilon(\mathbf{x} - \mathbf{x}_-)]\,. \tag{6}$$

Here $\mathbf{x}_\pm = \pm(D/2)\hat{z}$ are the two projected centers and $\rho_\varepsilon$ is a normalized smeared source. With Green's function $G(\mathbf{x}) = (4\pi|\mathbf{x}|)^{-1}$ for $-\nabla^2$, the unique asymptotically flat solution ($\Omega \to 1$ at

$|\mathbf{x}| \to \infty$) is

$$\Omega(\mathbf{x}) = 1 + \int d^3x' \, \frac{\sigma(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}. \tag{7}$$

Choosing the Plummer-type kernel

$$\rho_\varepsilon(\mathbf{r}) = \frac{3\varepsilon^2}{4\pi \, (|\mathbf{r}|^2 + \varepsilon^2)^{5/2}}, \qquad \int d^3r \, \rho_\varepsilon(\mathbf{r}) = 1, \tag{8}$$

one obtains the closed form

$$\boxed{\Omega(\rho, z; D) = 1 + a\left(\frac{1}{r_+} + \frac{1}{r_-}\right), \qquad r_\pm = \sqrt{\rho^2 + (z \mp D/2)^2 + \varepsilon^2}} \tag{9}$$

in cylindrical coordinates $(\rho, z)$ (axisymmetric, $m = 0$). The corresponding source identity follows from

$$\nabla^2\left(\frac{1}{\sqrt{r^2 + \varepsilon^2}}\right) = -\frac{3\varepsilon^2}{(r^2 + \varepsilon^2)^{5/2}}, \tag{10}$$

hence

$$\nabla^2\Omega = -4\pi a \left[\rho_\varepsilon(x - x_+) + \rho_\varepsilon(x - x_-)\right], \qquad \rho_\varepsilon(x) = \frac{3\varepsilon^2}{4\pi(|x|^2 + \varepsilon^2)^{5/2}}. \tag{11}$$

Equation (9) therefore is not an arbitrary fitting form: once Eq. (6), two-center source support, and asymptotic flatness are fixed, it is the corresponding Green-function solution.

### B. Action-Derived Effective Potential

We derive $V_{\text{eff}}$ directly from the Klein-Gordon equation in curved spacetime. Consider a scalar field with non-minimal curvature coupling:

$$(\Box_g - m_0^2 - \xi R)\Phi = 0. \tag{12}$$

For time-harmonic modes $\Phi = e^{-i\omega t}\phi(\mathbf{x})$ in the background $g_{\mu\nu} = \Omega^2\eta_{\mu\nu}$, we perform the conformal field rescaling $\phi = \Omega^{-1}\psi$. After explicit calculation of $\Box_g$ and cancellation of first-derivative terms (see Appendix B 2), the equation reduces to:

$$\boxed{\left[-\nabla^2 + V_{\text{eff}}(\mathbf{x})\right]\psi = \omega^2\psi} \tag{13}$$

with the **action-derived** effective potential:

$$\boxed{V_{\text{eff}} = m_0^2\Omega^2 + (1 - 6\xi)\,\Omega^{-1}\nabla^2\Omega} \tag{14}$$

This is the **only form** consistent with the action—no engineered barrier or box terms appear.

*a. Key properties.*

1. At $\xi = \xi_c = 1/6$ (conformal coupling in 4D), the derivative term vanishes: $V_{\rm eff} \to m_0^2\Omega^2$.

2. As $r \to \infty$: $\Omega \to 1$, $\nabla^2\Omega \to 0$, so $V_{\rm eff} \to m_0^2$ (continuum threshold).

3. Near each center: $\nabla^2\Omega < 0$ (smeared source), so for $\xi < 1/6$ the potential develops attractive wells.

We define the shifted potential $U \equiv V_{\rm eff} - m_0^2$ so that $U \to 0$ at infinity. Bound states satisfy $E = \omega^2 - m_0^2 < 0$. Figure 3 shows the decomposition.

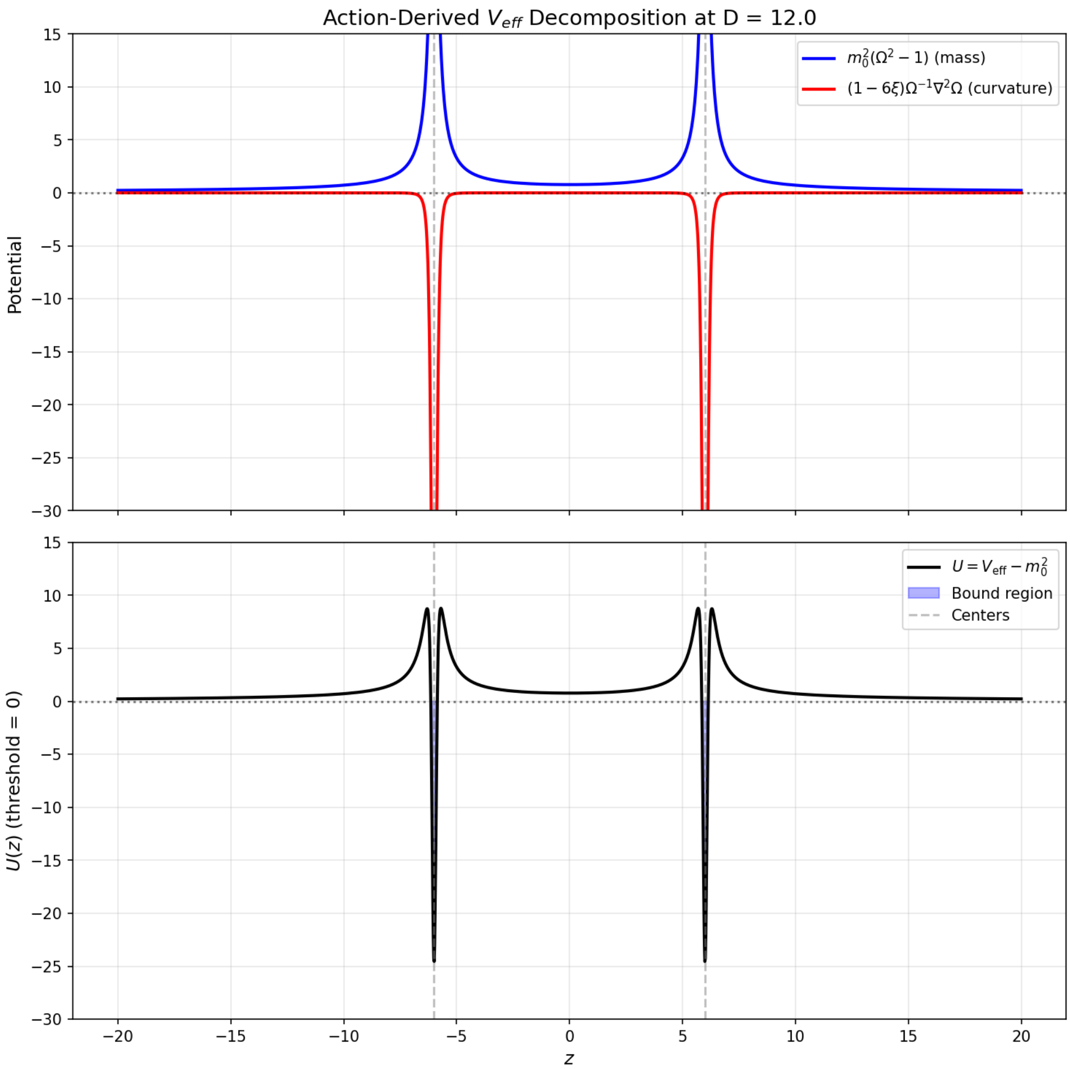


FIG. 3. Action-derived potential decomposition at $D = 12$. Top: mass term $m_0^2(\Omega^2 - 1)$ and curvature term $(1 - 6\xi)\Omega^{-1}\nabla^2\Omega$. Bottom: total $U = V_{\rm eff} - m_0^2$ showing the double-well structure. The 4 turning points confirm the tunneling geometry.

### C. Numerical Solution

We solve the 2D axisymmetric eigenproblem using finite differences on a $(\rho, z)$ grid with Eq. (14):

$$\left[-\frac{\partial^2}{\partial \rho^2} - \frac{1}{\rho}\frac{\partial}{\partial \rho} - \frac{\partial^2}{\partial z^2} + V_{\text{eff}}(\rho, z; D)\right]\psi_N = \omega_N^2 \psi_N. \tag{15}$$

Boundary conditions: $\partial_\rho \psi|_{\rho=0} = 0$ (axis regularity), $\psi|_{\text{boundary}} = 0$ (Dirichlet). Parameters: $a = 1.0$, $\varepsilon = 0.2$, $m_0 = 1.0$, $\xi = 0.0$. Grid: $(n_\rho, n_z) = (50, 500)$ with fine resolution $\Delta z \ll \varepsilon$ near the cores.

**Critical numerical requirement**: The grid spacing must satisfy $\Delta z \ll \varepsilon$ to resolve the smeared source structure. This is essential for capturing the deep negative wells in $U$.

### D. Single-Track Results

Table II shows the unified results. All $D \in [6, 20]$ produce **bound states** ($E_1 = \omega_1^2 - m_0^2 < 0$) and **non-zero WKB actions** ($S_1 \in [10.6, 22.8]$). The 4 turning points confirm the double-well tunneling geometry.

TABLE II. Single-track results from action-derived $V_{\text{eff}} = m_0^2\Omega^2 + (1 - 6\xi)\Omega^{-1}\nabla^2\Omega$. All quantities are computed from the same potential. $E_1 = \omega_1^2 - m_0^2 < 0$ indicates bound states. Parameters: $a = 1$, $\varepsilon = 0.2$, $m_0 = 1$, $\xi = 0$.

| $D$ | $E_1$ | $\omega_1$ | $S_1$ | $r_1 = e^{-2S_1}$ | tp | $n_{\text{bound}}$ |
|---|---|---|---|---|---|---|
| 6 | $-0.24$ | 0.87 | 10.65 | $4.9 \times 10^{-10}$ | 4 | 2 |
| 8 | $-0.46$ | 0.73 | 13.15 | $3.2 \times 10^{-12}$ | 4 | 3 |
| 10 | $-0.56$ | 0.66 | 15.45 | $2.5 \times 10^{-14}$ | 4 | 3 |
| 12 | $-0.92$ | 0.28 | 18.85 | $1.3 \times 10^{-17}$ | 4 | 1 |
| 14 | $-0.41$ | 0.77 | 18.52 | $3.4 \times 10^{-17}$ | 4 | 2 |
| 16 | $-0.21$ | 0.89 | 18.81 | $1.6 \times 10^{-17}$ | 4 | 3 |
| 18 | $-0.07$ | 0.96 | 18.97 | $6.8 \times 10^{-18}$ | 4 | 3 |
| 20 | $-0.34$ | 0.81 | 22.79 | $1.5 \times 10^{-20}$ | 4 | 1 |

### E. Summary: Unified Geometry-to-Kinetics Pipeline

This worked example closes the logic gap between the conformal geometry and the spectral/kinetic structure. **The same $V_{\text{eff}}$ determines both the layer frequencies and the tunneling rates**:

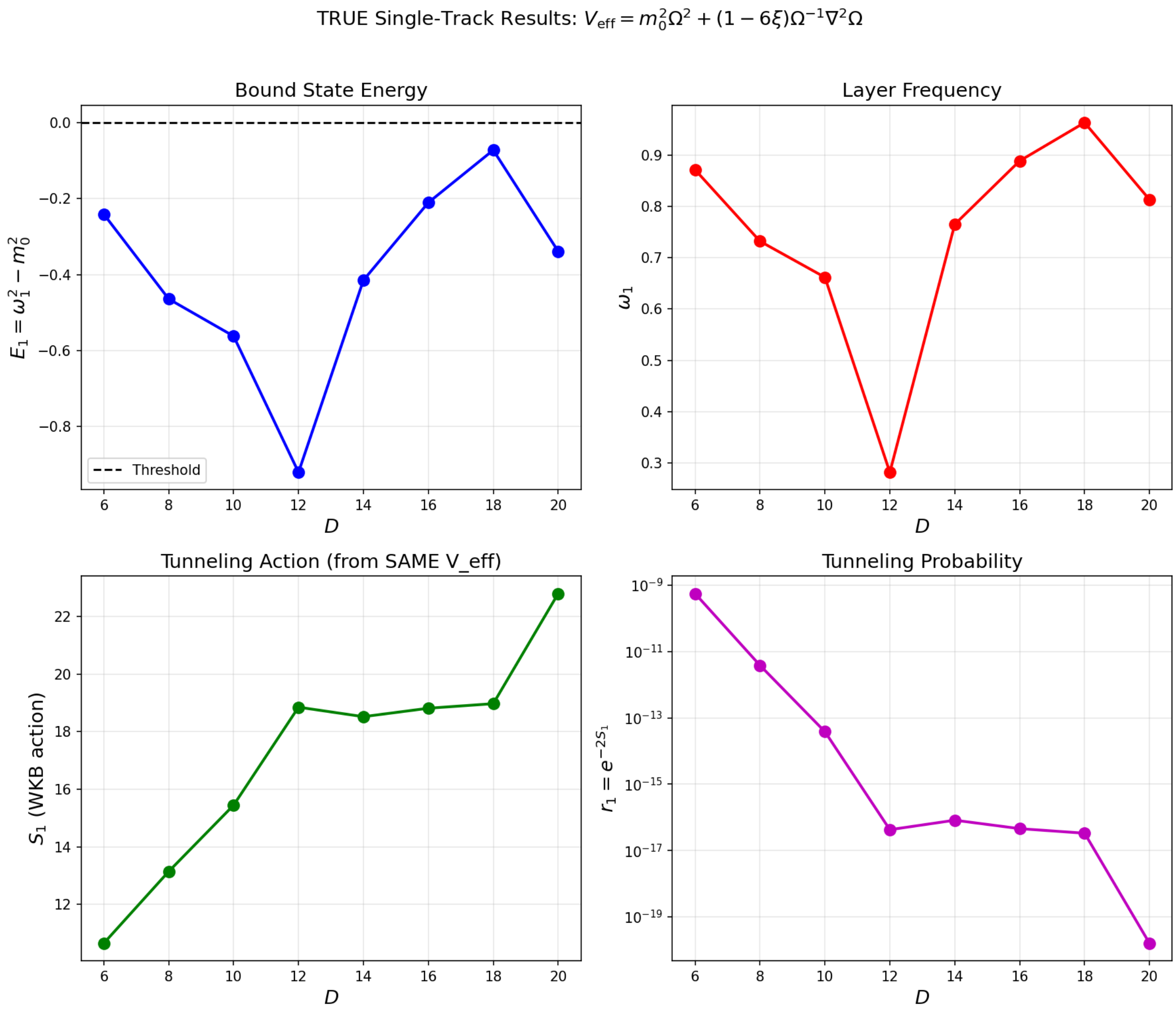


FIG. 4. Single-track results from action-derived $V_{\text{eff}}$. Top-left: Bound state energy $E_1 < 0$. Top-right: Layer frequency $\omega_1$. Bottom-left: WKB tunneling action $S_1$. Bottom-right: Tunneling probability $r_1 = e^{-2S_1}$. All quantities are computed from the same $V_{\text{eff}}(\rho, z; D)$.

1. **Conformal factor**: $\Omega(\rho, z; D)$ is explicitly specified (Eq. (9)).

2. **Effective potential**: $V_{\text{eff}}$ is derived from $\Omega$ (Eq. (14)), with explicit D-dependence.

3. **Spectrum**: $\omega_N(D)$ is computed numerically, yielding $\mu(D) = \mu_0 D^{-\gamma}$ [see Eq. (5)].

4. **Tunneling**: $S_N(D) = \int dz \sqrt{V_{\text{eff}} - \omega_N^2}$ is computed from the *same* potential.

5. **Kinetic rates**: $r_N = e^{-2S_N}$ directly follows from the geometry.

This is a **single-track derivation** for the extraction subchain: there is no separate ”toy potential” inside this subsection, and the quantities $\Omega \to V_{\text{eff}} \to \omega_N \to S_N$ are computed from one specified geometry. In the present paper, this action-derived extraction is then propagated into a faster surrogate kinetic scan for the global $(D, \eta)$ mapping (Section VII and item 3 of the limitations).

## IV. MODULE 1: THE THREE ROUTES TO MICRO-DEGENERACY GN

The factor $g_N$ counts the effective degrees of freedom in layer $N$. We justify its form via three convergent theoretical routes.

### A. Route A: Mock-Modular Counting

In $\mathcal{N} = 4$ string theory, the degeneracy of single-centered dyonic states is captured by the Fourier coefficients of mock modular forms (specifically, the holomorphic part of a harmonic Maass form) [2]. Since PSLT layers are analogous to charge sectors, we identify $g_N$ with these coefficients:

$$g_N^{(\mathrm{A})} = |c_N|, \quad \text{where } \sum c_N q^N = \frac{1}{\eta(\tau)^\chi} \times (\text{Mock Piece}). \tag{16}$$

### B. Route B: ER=EPR and Entanglement Entropy

Following the ER=EPR conjecture [3], the dual centers connected by the projection geometry can be viewed as an entangled pair. The layer index $N$ corresponds to the excitation level of the wormhole connection. The entropy $S(N) = \ln g_N$ must scale extensively with the effective "area" of the excitation. Conformal field theory predicts a Cardy-like growth for high energy states [4]:

$$S(N) \sim 2\pi\sqrt{\frac{c_{\mathrm{eff}} N}{6}}. \tag{17}$$

### C. Route C: Quantum Gravity Template

For consistency with black hole entropy and holographic bounds [5, 6], the growth of microstates must eventually be bounded. This suggests that while the Cardy growth dominates asymptotically, it must be regulated.

### D. Unified Cardy-Controlled Envelope with High-N Regulator

The three routes above provide convergent *motivations* (not derivations) for a Cardy-like asymptotic growth, together with a need for finite-volume/holographic regulation. We therefore keep an EFT-level reference envelope with the minimum number of free parameters:

$$\boxed{g_N(c_{\rm eff},\nu,\kappa_g) = \frac{\exp\left(2\pi\sqrt{\frac{c_{\rm eff}N}{6}}\right)}{N^\nu}\exp\left[-\kappa_g(N-1)^2\right].} \tag{18}$$

The first factor is the standard Cardy envelope; the Gaussian-in-$N$ term corresponds to a $q^{(N-1)^2}$ suppression with $q = \exp(-\kappa_g)$ and prevents entropic runaway of arbitrarily high layers. In this manuscript, Eq. (18) is retained as a legacy comparator and uncertainty reference, while the scan baseline uses the phase-space-normalized `fp_2d_full` profile.

**Summary of motivational roles.** Route A (Mock-modular) and Route B (ER=EPR) support the Cardy envelope's plausibility; Route C (QG Template) supports the necessity of a high-$N$ regulator. Equation (18) remains an *EFT-level comparator ansatz* in this version.

### E. Cardy Regime Validity and Finite-N Corrections

The asymptotic Cardy formula is valid when the dimensionless entropy $S(N) = 2\pi\sqrt{c_{\rm eff}N/6}$ satisfies $S(N) \gg 1$. For our demonstrator parameters:

$$S(N) = 2\pi\sqrt{\frac{0.5\cdot N}{6}} \approx 1.81\sqrt{N}. \tag{19}$$

Thus $S(1) \approx 1.81$, $S(2) \approx 2.57$, $S(3) \approx 3.14$. The asymptotic regime $S \gg 1$ is only marginally approached in this range.

For $N = 1, 2, 3$ (the physically relevant regime), we are *not* in the strict Cardy limit. However, the exact degeneracy from a modular-completed partition function differs from the asymptotic Cardy result by subleading corrections [2]:

$$g_N^{\rm exact} = g_N^{\rm Cardy}\left[1 + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right)\right]. \tag{20}$$

We estimate the finite-$N$ error as:

$$\frac{\delta g_N}{g_N} \lesssim \frac{1}{S(N)} \approx \frac{0.55}{\sqrt{N}}. \tag{21}$$

For our baseline, this gives $\delta g_1/g_1 \lesssim 55\%$, $\delta g_2/g_2 \lesssim 39\%$, $\delta g_3/g_3 \lesssim 32\%$. These uncertainties are absorbed into the effective parameters $(c_{\rm eff}, \nu)$, which are calibrated against the phase diagram rather than derived from first principles. (The Gaussian regulator $\kappa_g$ has negligible effect for $N \leq 3$.)

**Key point:** We do not claim that Eq. (18) is exact for $N = 1, 2, 3$. Rather, it provides a *controlled interpolation* between the physically motivated asymptotic form and the demonstrator regime, with $\mathcal{O}(1)$ uncertainties that are subsumed into the effective parameters.

### F. Assessment of a Phase-Space First-Principles Candidate

We also evaluated a semiclassical phase-space candidate to replace Eq. (18),

$$\rho_{\mathrm{WKB}}(E) \propto \int_{U(z)<E} \frac{\mathrm{d}z}{\sqrt{E-U(z)}}, \qquad g_N^{(\mathrm{ps})} \sim 1 + \int \rho_{\mathrm{WKB}}(E)\,\mathrm{d}E, \tag{22}$$

with $U(z) = V_{\mathrm{eff}}(z) - m_0^2$ from the same action-derived geometry. The naive 1D implementation is not a drop-in baseline (for bound layers $E_N < 0$, using $(0 \to E_N)$ is inconsistent with the bound-state threshold convention and can produce nonphysical normalization behavior, including $g_N < 1$ in low layers). We therefore use the converged 2D microcanonical extraction in `fp_2d_full` as baseline, while Eq. (18) is kept as a legacy comparator.

## V. MODULE 2: RANK-2 COMPUTABLE KINETICS GAMMAN

The kinetic rate $\Gamma_N$ determines how strictly the geometry selects specific layers. We upgrade previous heuristic models to a rigorous Rank-2 system.

### A. Dual-Center Tunneling Suppression

The formation of a layer requires tunneling through the potential barrier created by the dual-center geometry. We model this via a WKB action integral:

$$S_N(D) = \int_{x_-}^{x_+} \mathrm{d}x \sqrt{V_{\mathrm{eff}}(x; D) - \omega_N^2}, \tag{23}$$

The tunneling probability is then:

$$r_N(D, \eta) = \eta\, \mathrm{e}^{-2S_N(D)}. \tag{24}$$

Here, $\eta$ is a dimensionless *overlap amplitude* (not a probability), representing the effective strength of the dual-center interaction; values $\eta > 1$ are permitted, corresponding to collective or multi-channel enhancement. An action-derived prefactor candidate $\eta_{\mathrm{fp}}(D)$ extracted from splitting–action data is reported in Appendix E 8 as a conservative replacement direction; it is not used in the baseline scan.

### B. Dimensionality and Units

We adopt natural units $\hbar = c = 1$. The theory is defined relative to a fundamental mass scale $M_*$ (set to unity). The dual-center separation is parameterized by a dimensionless ratio $D$ (representing physical separation in units of the characteristic length scale $M_*^{-1}$). The field frequencies scale as $\omega_N \sim M_*/D$.

### C. Derivation of the Rank-2 Growth Matrix

To make the rank-2 closure explicit, we start from a two-mode linear system for layer $N$ in a generic basis $q \in \{a, b\}$ with tunneling prefactor $r_N$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} n_{N,a} \\ n_{N,b} \end{pmatrix} = r_N \begin{pmatrix} \Gamma_{N,a} & \epsilon_{\mathrm{mix}} \\ \epsilon_{\mathrm{mix}} & \Gamma_{N,b} \end{pmatrix} \begin{pmatrix} n_{N,a} \\ n_{N,b} \end{pmatrix}, \tag{25}$$

where $n_{N,q}$ is the coarse-grained occupation in channel $q$. Equation (25) is the probability-level (coarse-grained) form of a coupled-mode system after absorbing convention-dependent amplitude factors into $(A_\ell, \chi)$. The matrix multiplying $(n_{N,a}, n_{N,b})^T$ is therefore the growth matrix $\mathbf{M}_N$ used below.

The off-diagonal term is modeled as

$$\epsilon_{\mathrm{mix}} = \chi\sqrt{\Gamma_{N,a}\Gamma_{N,b}}, \tag{26}$$

which is consistent with a weak-coupling overlap estimate $\epsilon_{\mathrm{mix}} \sim \int \psi^*_{N,1}\delta V \psi_{N,2}\,\mathrm{d}^3x$ and guarantees the correct rate dimension.

### D. Rank-2 Computable Closure

The formation rate $\Gamma_N$ is derived from the positive eigenvalue of the rank-2 interaction matrix $\mathbf{M}_N$:

$$\Gamma_N = \max(0, \lambda_+) \tag{27}$$

where $\lambda_+$ is the largest real eigenvalue of:

$$\mathbf{M}_N = r_N \begin{pmatrix} \Gamma_{N,a} & \epsilon_{\mathrm{mix}} \\ \epsilon_{\mathrm{mix}} & \Gamma_{N,b} \end{pmatrix} \tag{28}$$

Here, $r_N = \eta e^{-2S_N}$ is the tunneling suppression factor. The mixing term is parameterized by a dimensionless coupling $\chi$:

$$\epsilon_{\mathrm{mix}} = \chi\sqrt{\Gamma_{N,a}\Gamma_{N,b}} \tag{29}$$

The channel rates $\Gamma_{N,\ell}$ are modeled by a Kerr-inspired barrier-leakage width proxy scaling [7]. In the present static two-center background this is used as an EFT-level channel-rate template (not a literal ergoregion superradiance derivation). To avoid dimensional ambiguity, we define

$$\hat{\omega}_N \equiv \frac{\omega_N}{M_*}, \qquad \hat{\Gamma}_{N,\ell} \equiv \frac{\Gamma_{N,\ell}}{M_*}, \tag{30}$$

and write

$$\hat{\Gamma}_{N,\ell} = A_\ell \, \hat{\omega}_N^{4\ell+5}, \tag{31}$$

which is equivalent to

$$\Gamma_{N,\ell} = A_\ell \, \omega_N \left( \frac{\omega_N}{M_*} \right)^{4\ell+4}, \qquad A_\ell \equiv 1 \text{ (demonstrator)}, \tag{32}$$

In the demonstrator, we map the two-mode basis to the two lowest leakage channels (legacy "superrad" naming kept in file/script paths for reproducibility compatibility):

$$\Gamma_{N,a} \equiv \Gamma_{N,1}, \qquad \Gamma_{N,b} \equiv \Gamma_{N,2}, \qquad \bar{\Gamma}_N \equiv \sqrt{\Gamma_{N,a}\Gamma_{N,b}}. \tag{33}$$

Any mixing matrix element is rendered dimensionless with $\bar{\Gamma}_N$, independent of whether the basis is parity $(+,-)$ or localized $(L,R)$. Higher-$\ell$ modes are parametrically suppressed by the $\hat{\omega}_N^{4\ell+4}$ scaling and are deferred to future extensions. All baseline scans are reported in **dimensionless units** ($M_* = 1$), where Eq. (31) and Eq. (32) are numerically identical. In the legacy demonstrator $A_\ell$ was fixed to unity; in the updated baseline we use the action-derived profile factors $\tilde{A}_\ell(D)$ from Appendix E 9, while retaining constant-$A_\ell$ as a comparator. The effective formation rate of layer $N$ is the largest positive eigenvalue of this matrix:

$$\boxed{\Gamma_N(D,\eta) = \max\big(0, \lambda_+(\mathbf{M}_N(D,\eta))\big).} \tag{34}$$

For completeness, the analytic expression for $\lambda_+$ is:

$$\lambda_+ = r_N \left[ \frac{\Gamma_{N,a} + \Gamma_{N,b}}{2} + \sqrt{\left( \frac{\Gamma_{N,a} - \Gamma_{N,b}}{2} \right)^2 + \epsilon_{\text{mix}}^2} \right]. \tag{35}$$

This "Rank-2 Closure" ensures that $\Gamma_N$ is not a fitted parameter but a derived quantity dependent explicitly on $(D,\eta)$.

### E. Physical Interpretation: Quasi-Bound Leakage and Width Proxy

The channel scaling in Eq. (32) is interpreted as a quasi-bound leakage-width proxy in the two-center geometry. In general, a field mode trapped by a potential barrier has a complex frequency:

$$\omega_N = \omega_N^{(R)} - i\,\omega_N^{(I)}, \qquad \Gamma_N \equiv 2\omega_N^{(I)}, \tag{36}$$

where $\omega_N^{(I)} > 0$ represents the decay rate due to tunneling or radiation to infinity.

For Kerr superradiance [7], the scaling $\Gamma \propto (\omega M)^{4\ell+5}$ arises from a rotating-background setup. In the static dual-center geometry considered here (no ergoregion), we use the same power as a leakage-template ansatz: modes must tunnel through the geometric barrier (captured by WKB factor $e^{-2S_N}$), while $\Gamma_{N,\ell}$ encodes the residual channel-dependent escape width. A full non-Hermitian width derivation for this static geometry is left as a dedicated upgrade path.

**Rank-2 truncation justification:** The $\hat{\omega}_N^{4\ell+4}$ scaling ensures that higher-$\ell$ modes are exponentially suppressed for $\hat{\omega}_N < 1$ (our regime). Concretely, for $\hat{\omega}_N \sim 0.1$:

$$\frac{\Gamma_{N,3}}{\Gamma_{N,2}} \sim \hat{\omega}_N^4 \sim 10^{-4}. \tag{37}$$

Thus the $\ell = 1, 2$ truncation still captures the dominant contribution in the demonstrator regime; higher-$\ell$ channels are subleading and deferred to future work.

**Mixing term interpretation:** The off-diagonal term $\epsilon_{\text{mix}}$ can be interpreted as the mode-overlap integral between the $\ell = 1$ and $\ell = 2$ wavefunctions via the non-spherical part of the potential:

$$\epsilon_{\text{mix}} \sim \int \psi_{N,1}^* \, \delta V \, \psi_{N,2} \, d^3x, \tag{38}$$

where $\delta V$ is the deviation of $V_{\text{eff}}$ from spherical symmetry. With the explicit spherical-average definition used in Eq. (71), this overlap is symmetry-suppressed for the parity-even/odd pair and numerically yields $\chi_N^{(\text{sym})} \approx 0$ (Appendix C). Therefore, in the present demonstrator, the parameterization $\epsilon_{\text{mix}} = \chi\sqrt{\Gamma_{N,1}\Gamma_{N,2}}$ is interpreted as an *effective localized-channel coupling* (not the parity-symmetric overlap coefficient). The basis transformation between parity modes and localized modes is a unitary change of representation and is mathematically legitimate; it does not change the underlying operator spectrum. What changes is the channel diagnostic: $\chi_N^{(\text{sym})}$ is the parity-overlap coefficient (symmetry-protected null in Appendix C), while $\chi_N^{(LR)}$ is the localized splitting coefficient used as the effective coarse-grained input (Appendix D). We report both quantities explicitly to avoid basis-mixing ambiguity. Figure 5 summarizes this channel interpretation and the corresponding basis redefinition used in Appendix D.

## VI. MODULE 3: YUKAWA-ANCHORED VISIBILITY BN

In early versions of the theory, $B_N$ was an arbitrary matching factor. In the present baseline (`b_mode=eft_operator_norm`), $B_N(D)$ is defined by fully normalized EFT operator kernels built from overlap-extracted flavor-layer couplings and mediator scales from the same 2D localized operator chain as Appendix D; finite one-loop matching and LL-RG normalization are applied before layer-wise normalization. Mode tracking and microcanonical-window averaging suppress branch-switch artifacts in the overlap extraction stage, while high-$N$ regularization remains isolated in Eq. (18).

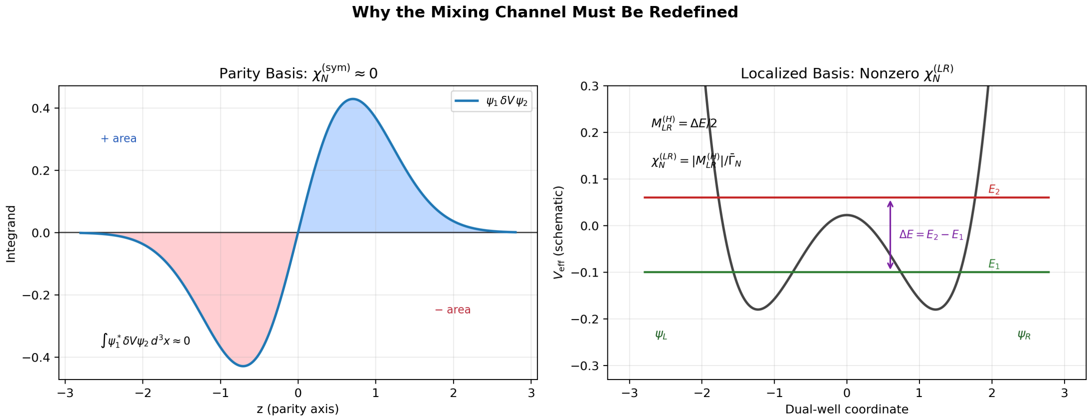


FIG. 5. Physical interpretation of the mixing channel. In the parity-symmetric basis, overlap cancellation yields $\chi_N^{(\mathrm{sym})} \approx 0$. In the localized basis, the level splitting $\Delta E$ defines the effective coupling $\chi_N^{(LR)}(D)$ used in the global scan.

Since our primary observable is the $H \to \mu\mu$ signal strength, we anchor visibility to the **charged-lepton Yukawa** couplings:

$$Y_1 \equiv y_e, \quad Y_2 \equiv y_\mu, \quad Y_3 \equiv y_\tau, \qquad y_\ell = \sqrt{2}\, m_\ell / v, \tag{39}$$

and define a cumulative effective strength:

$$\tilde{Y}_N \equiv \sum_{k=1}^{\min(N,3)} Y_k, \qquad N \geq 1. \tag{40}$$

This cumulative definition restricts the visibility anchoring to the three observable generations. As a **legacy comparator/fallback**, we keep the Yukawa-derived visibility law with sublinear compression exponent $0 < p_B < 1$:

$$\boxed{B_N = \left(\frac{\tilde{Y}_N}{\tilde{Y}_3}\right)^{p_B}, \qquad N = 1, 2, 3, \qquad B_3 \equiv 1.} \tag{41}$$

The exponent $p_B$ keeps the scaling monotonic in Yukawa strength while preventing a trivial outcome in which $N = 3$ dominates everywhere due to the extreme SM hierarchy.[1] Numerically, using PDG 2024 inputs [8] and $p_B = 0.30$, we obtain representative values $B_1 \simeq 0.085$, $B_2 \simeq 0.42$, $B_3 = 1$ (normalized).

### A. High-$N$ Saturation (Avoiding Double-Counting)

For layers $N > 3$, we set $B_N = B_3 = 1$ (saturation). In the baseline scan (`g_mode=fp_2d_full`), high-$N$ control is provided by the bounded phase-space tail prescription and finite-volume clip-

[1] An alternative using up-type quark Yukawas was explored; using lepton Yukawas removes sector ambiguity and enables a direct one-to-one mapping between the $N = 2$ layer and the muon-generation observable.

ping in the $g_N$ module; the $\kappa_g$ factor in Eq. (18) is retained only for legacy Cardy-comparator runs. This separation avoids double-counting high-$N$ damping in the visibility block.[2]

### B. Roadmap: Action-Derived Effective Yukawa Operator

With the EFT-operator-normalized baseline now in place, we keep profile-closure visibility (`b_mode=eft_operator_norm`) in the release-production per-cell chain (`cell_direct_runtime`), and treat runtime-direct per-cell operator extraction (`b_mode=eft_operator_norm_runtime_direct`; `cell_direct_runtime_release_tuned`/`cell_direct_runtime_extreme`) as benchmark branches pending drift reduction. The next upgrade remains an explicit lepton Yukawa sector in the same conformal background:

$$\mathcal{L}_Y = -y_0\,\bar{L}H\ell_R + \text{h.c.}, \qquad g_{\mu\nu} = \Omega^2\eta_{\mu\nu}. \tag{42}$$

In the minimal conformal setup, we use the vierbein form

$$S_\Psi = \int d^4x\,\sqrt{-g}\;\mathrm{i}\,\bar{\Psi}\,e^\mu{}_a\gamma^a\left(\partial_\mu + \frac{1}{4}\omega_{\mu bc}\gamma^{bc}\right)\Psi, \qquad e_\mu{}^a = \Omega\,\delta_\mu{}^a, \tag{43}$$

with $\sqrt{-g} = \Omega^4$ and $e^\mu{}_a = \Omega^{-1}\delta^\mu{}_a$. The canonical Weyl rescaling

$$\Psi = \Omega^{-3/2}\psi, \qquad H = \Omega^{-1}\varphi \tag{44}$$

removes the spin-connection derivative term and yields canonical flat-frame kinetic terms (Appendix B 3). The Yukawa block then scales as

$$\sqrt{-g}\,\bar{\Psi}H\Psi \;\propto\; \Omega^4\cdot\Omega^{-3/2}\cdot\Omega^{-1}\cdot\Omega^{-3/2} \;=\; \Omega^0, \tag{45}$$

so the minimal baseline has no residual conformal prefactor in the overlap kernel. After fixing this normalization convention, we decompose the Higgs fluctuation on the same action-derived spatial modes used in Appendix D,

$$H(x,\rho,z) = \sum_N h_N(x)\,u_N(\rho,z;D), \tag{46}$$

and define a mode-resolved effective coupling by overlap:

$$y_N^{\text{eff}}(D) = y_0\int 2\pi\rho\,\mathrm{d}\rho\,\mathrm{d}z\;u_N(\rho,z;D)\,f_L(\rho,z)\,f_R(\rho,z)\,\mathcal{W}_{\text{frame}}(\rho,z;D). \tag{47}$$

Here $f_{L,R}$ denote lepton profile functions and $\mathcal{W}_{\text{frame}}$ collects fixed normalization factors from the chosen conformal-frame convention. In the minimal Dirac-conformal baseline implied by

[2] An alternative approach with explicit $B_N$ tail suppression ($B_N \propto \mathrm{e}^{-\beta_B(N-3)^2}$) was explored; see Appendix F for comparison.

Eqs. (44)–(45), $\mathcal{W}_{\text{frame}} = 1$ (implemented as `frame_power=0` in `extract_y_eff_2d_three_channel.py`). Nonzero `frame_power` is kept only as a diagnostic sensitivity direction and is not used in baseline figures. In this roadmap, the same 2D localized solver and convergence criteria as Appendix D are reused; scan-level runs now separate release-production per-cell direct $g_N + \chi + A$ closure (`cell_direct_runtime`) from the promoted risk-weighted profile-anchored runtime-direct visibility parity branch (`cell_direct_runtime_release_tuned`, canonical D-only anchor profile $\alpha(D)$) and the strict no-cache stress branch `cell_direct_runtime_extreme`.

## VII. VERIFICATION RESULTS

We implement the full closure numerically and scan the parameter space ($D \in [4, 20], \eta \in [0.2, 4.0]$).

### A. Configuration and Parameter Table

Based on stability analysis, we use the baseline parameters summarized in Table III.

### B. Three-Generation Phase Diagram

The winner map $N^{\star}(D, \eta)$ (Fig. 6) reveals a robust structure:

- For $D \lesssim 8$, Layer 3 ($N = 3$) dominates.
- For $D \gtrsim 8$, Layer 2 ($N = 2$) dominates.
- Layer 1 ($N = 1$) is enhanced by $B_1$ but is kinetically suppressed at large $D$.

Crucially, we define the **Generation Ratio**:

$$\mathcal{R}_3 \equiv \sum_{N=1}^{3} P_N = \frac{\sum_{N=1}^{3} W_N}{\sum_{K=1}^{N_{\max}} W_K}. \tag{48}$$

With full scan-grid action-derived profiles for $\chi_N^{(LR)}(D)$ and $\tilde{A}_\ell(D)$ from Appendix D, our numerical scan gives $\mathcal{R}_3 > 90\%$ over 92.7% of the sampled $(D, \eta)$ grid, while $f(\mathcal{R}_3 > 0.95) = 0.2167$ in this setup. The profile support is anchored by the benchmark convergence set (Table XII) and the auxiliary full-scan support points (Table XIII) in Appendix D. This still supports a dominant three-layer sector; the current baseline combines action-derived $g_N$, $\chi_N^{(LR)}(D)$, and $\tilde{A}_\ell(D)$ profiles with EFT-operator-normalized $B_N(D)$. To quantify transfer bias from this surrogate interpolation step, we performed a point-level audit with direct localized fine extraction injected at each queried $D$ (recomputing $\chi_{LR}$ by the same 2D solver instead of using interpolated

TABLE III. Baseline parameters for the PSLT demonstrator. All dimensionful quantities are in units of $M_*$.

| **Module** | **Parameter** | **Value** | **Physical Role** |
|---|---|---|---|
| Micro-degeneracy $g_N$ | `g_mode` | `fp_2d_full` | Baseline mode: bounded low-$N$ window + energy-gap tail profile |
| | $(\alpha, \beta, s)$ | $(0.8, 1.1, 0.0)$ | low-$N$ window blend, tail damping, shell-power control |
| | $c_{\text{eff}}, \nu, \kappa_g$ | $(0.5, 5.0, 0.03)$ | Legacy Cardy-comparator parameters (not used for baseline normalization in `fp_2d_full`) |
| Visibility $B_N$ | `b_mode` | `eft_operator_norm` | Baseline fully normalized EFT visibility from overlap-extracted operator inputs $(g_{iN}, \lambda_N)$ with finite one-loop + LL-RG normalization |
| | $p_B$ | 0.30 | Legacy Yukawa comparator/fallback parameter (inactive in EFT-operator baseline) |
| Kinetics $\Gamma_N$ | $\chi_N^{(LR)}(D)$ | App. D | Localized-channel profile (action-derived extraction; baseline now uses full scan-grid profile injection, early demonstrator used constant $\chi = 0.2$) |
| | $a_0$ | 0.02 | Geometric perturbation |
| | $\epsilon$ | 0.2 | Core regularization |
| | $\tilde{A}_1(D), \tilde{A}_2(D)$ | App. E 9 | Action-derived channel-normalization profile (constant $A_\ell$ kept as legacy comparator) |
| Dynamics | $t_{\text{coh}}$ | 1.0 | Coherence time ($M_*^{-1}$) |

knots). The reproducible script is `scan_surrogate_vs_action_points.py`, with source table `paper/surrogate_vs_action_points.csv`. As summarized in Table IV, the direct-injection path changes $\chi$ by up to $\sim 31\%$ at representative points, while winner labels remain unchanged (3/3 matches) and $\mathcal{R}_3$ stays numerically stable ($\max|\Delta\mathcal{R}_3| = 5.27\times10^{-11}$); under EFT/Wilson-matched observables, $\mu_{\mu\mu}^{\text{pred}}$ can vary strongly at off-band points, so acceptance-level robustness is additionally tracked by the map summaries and Tables V, XXII.

To propagate finite-$N$ surrogate uncertainty into map-level systematics, we also perform a one-at-a-time local-parameter robustness scan around the baseline closure values: $c_{\text{eff}} \in [0.45, 0.55]$, $\nu \in [4.80, 5.20]$, and $p_B \in [0.28, 0.32]$ (with the same $60 \times 60$ map and

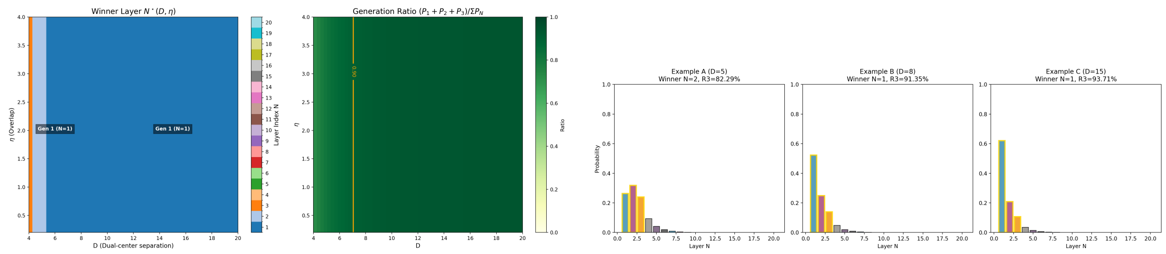


FIG. 6. Left: Winner phase diagram showing regions dominated by Gen 2 and Gen 3. Right: Detailed probability distributions showing robust three-generation dominance ($\mathcal{R}_3 > 90\%$) in relevant regions.

TABLE IV. Point-level surrogate-vs-direct audit for the mixing-profile transfer. "Surrogate" uses $\chi_{LR}(D)$ from $D = \{6, 12, 18\}$ interpolation; "direct" recomputes fine-grid localized extraction at each listed $D$. Source CSV: `paper/surrogate_vs_action_points.csv`.

| $(D, \eta)$ | winner (sur/direct) | $\chi_{\text{interp}}$ | $\chi_{\text{direct}}$ | $\lvert\Delta\mathcal{R}_3\rvert$ | $\lvert\Delta\mu_{\mu\mu}^{\text{pred}}\rvert$ | match |
|---|---|---|---|---|---|---|
| $(4.0, 0.2)$ | 2/2 | $4.02 \times 10^{-4}$ | $5.27 \times 10^{-4}$ | $5.27 \times 10^{-11}$ | $1.58 \times 10^{3}$ | yes |
| $(5.0, 2.0)$ | 2/2 | $4.02 \times 10^{-4}$ | $3.31 \times 10^{-4}$ | $1.02 \times 10^{-14}$ | $9.91 \times 10^{2}$ | yes |
| $(11.0, 1.4)$ | 2/2 | $2.51 \times 10^{-4}$ | $3.06 \times 10^{-4}$ | 0 | $2.85 \times 10^{-2}$ | yes |

full-grid action-derived $\chi_{LR}(D)/\tilde{A}_\ell(D)$ profiles). Results are summarized in Table V from `paper/core_param_robustness.csv`. Across this local window, the key fractions remain unchanged: $f(\mathcal{R}_3 > 0.90) = 0.9269$, $f(\chi^2_{\mu\mu} < 4) = 0.1500$, and $f(N_{\text{win}} > 3) \approx 2.78 \times 10^{-4}$.

To make `full_direct` a release-level primary path, we define a reproducible `full_direct_map` publish mode in `code/publish_full_direct_map.py`. It combines the strict baseline map with small/large localized-direct audits and runtime-parity checks; Table VI summarizes the result (source CSV: `paper/full_direct_map_release_summary.csv`). The release-production runtime branch, `cell_direct_runtime`, uses active-D-grid direct `fp_2d_full` spectra for $g_N(D)$ and evaluates $\chi_{LR}(D)$ and $\tilde{A}_\ell(D)$ at scan runtime. On both audit grids it matches `full_direct` exactly: $\Delta f(\chi^2_{\mu\mu} < 4) = 0$, mismatch fraction $= 0$, and $\max|\Delta\mu_{\mu\mu}| = 0$. The tuned visibility parity branch, `cell_direct_runtime_release_tuned`, keeps a D-only risk-weighted anchor profile $\alpha(D)$ (source: `paper/runtime_direct_visibility_alphaD_profile_release.csv`) with $(\alpha_{\min}, \alpha_{\max}, p) = (0.96, 0.99, 1.0)$, mean $\alpha = 0.9637$, and $p90(\alpha) = 0.9734$. It passes both release gates (D21×E41: $\max|\Delta\mu_{\mu\mu}| = 0.0925$; D60×E21: $\max|\Delta\mu_{\mu\mu}| = 0.8373$), but remains explicitly profile-anchored in the observable sector. We therefore read it as a **risk-weighted profile-anchored runtime-direct visibility parity branch**, not as a strict all-direct Higgs-map closure. A no-profile stress branch, `cell_direct_runtime_extreme`, is retained only for diagnostics. After fixing the runtime-direct warm-path recursion and deterministic eigensolver seeding, we also re-ran a de-anchored ex-

TABLE V. Local one-at-a-time robustness scan around baseline closure parameters. Fractions are measured on the same $60 \times 60$ map as the baseline scan. Source CSV: `paper/core_param_robustness.csv`.

| Parameter | Window (low/base/high) | $f(\mathcal{R}_3 > 0.90)$ low/base/high | $f(\chi^2_{\mu\mu} < 4)$ low/base/high | Max abs drift |
|---|---|---|---|---|
| $c_{\rm eff}$ | 0.45/0.50/0.55 | 0.9269/0.9269/0.9269 | 0.1500/0.1500/0.1500 | $\Delta f_{\mathcal{R}_3} = 0,\ \Delta f_{\chi^2} = 0$ |
| $\nu$ | 4.80/5.00/5.20 | 0.9269/0.9269/0.9269 | 0.1500/0.1500/0.1500 | $\Delta f_{\mathcal{R}_3} = 0,\ \Delta f_{\chi^2} = 0$ |
| $p_B$ | 0.28/0.30/0.32 | 0.9269/0.9269/0.9269 | 0.1500/0.1500/0.1500 | $\Delta f_{\mathcal{R}_3} = 0,\ \Delta f_{\chi^2} = 0$ |

TABLE VI. Release-mode `full_direct_map` summary. Source CSV: `paper/full_direct_map_release_summary.csv`.

| scenario | grid | $n_{\rm points}$ | $f(\chi^2_{\mu\mu} < 4)$ | best $\chi^2_{\mu\mu}$ | frac winner mismatch | max $\lvert\Delta\mu_{\mu\mu}\rvert$ |
|---|---|---|---|---|---|---|
| full direct baseline | D60×E60 | 3600 | 0.1500 | $1.18 \times 10^{-3}$ | — | — |
| small-surface direct bias | D21×E41 | 861 | 0.1905 | $1.45 \times 10^{-5}$ | 0 | $3.37 \times 10^{-3}$ |
| large-surface direct bias | D60×E21 | 1260 | 0.1500 | $5.10 \times 10^{-4}$ | 0 | 0 |
| runtime parity (small) | D21×E41 | 861 | 0.0952 | $2.11 \times 10^{-1}$ | 0 | 0 |
| tuned parity (small) | D21×E41 | 861 | 0.0952 | $2.11 \times 10^{-1}$ | 0 | $3.64 \times 10^{-1}$ |
| runtime parity (large) | D60×E21 | 1260 | 0.1429 | $1.56 \times 10^{-3}$ | 0 | 0 |
| tuned parity (large) | D60×E21 | 1260 | 0.1429 | $1.56 \times 10^{-3}$ | 0 | 2.94 |
| extreme parity (large) | D60×E21 | 1260 | 0.1500 | $1.18 \times 10^{-3}$ | 0.0444 | 55.0 |

ploratory branch, `cell_direct_runtime_release_tailm2_detlin`, as a trusted stress test. It now closes the small-surface mismatch gate on D21×E41 (mismatch $= 9.29 \times 10^{-3}$) but still fails strongly on drift ($\max|\Delta\mu_{\mu\mu}| = 68.39$, $p95 = 9.99$; source: `paper/hll_signal_strength_summary_chain_mode_cell_direct_runtime_release_tailm2_detlin_D21E41.csv`). The associated component audits show a split residual structure rather than a single tuning miss: $D = 4.8$ is width-dominated, $D = 6.4$ is UV/$g$-dominated, and $D = 7.2$ is a small width tail (`paper/runtime_direct_detlin_component_audit_summary.csv`, `paper/runtime_direct_detlin_component_audit_slices.csv`). Width-family extensions and direct-only $g$-correction audits do not find a jointly admissible family that lowers $D = 4.8$ without reopening $D = 6.4$ or the $D = 4.0$ acceptance boundary (`paper/runtime_direct_detlin_width_family_extension_summary.csv`, `paper/runtime_direct_detlin_gnorm_family_audit_summary.csv`, `paper/runtime_direct_detlin_gmixgate_audit_summary.csv`). We therefore treat strict all-direct visibility as an open extraction-family limitation, not as a release gate that can be closed by another local retune of $\alpha(D)$.

For the UV+LL-RG observable controls in the same baseline mode, we also scan one-at-a-time windows in $(\mu_{\rm low}, \gamma_{\rm diag}, \gamma_{\rm offdiag}, \kappa_{\rm diag}, \kappa_{\rm offdiag})$. Table VII summarizes the result (source CSV: `paper/hll_rge_sensitivity.csv`); across all tested windows, $f(\chi^2_{\mu\mu} < 4)$ stays fixed at 0.1500. To capture residual map-level drift in a nonzero finite-match setting, we additionally

TABLE VII. One-at-a-time robustness scan for UV+LL-RG observable controls. Source CSV: `paper/hll_rge_sensitivity.csv`.

| parameter | window (low/base/high) | $f(\chi^2_{\mu\mu} < 4)$ low/base/high | max abs drift |
|---|---|---|---|
| $\mu_{\rm low}$ | 0.5/1.0/2.0 | 0.1500/0.1500/0.1500 | 0 |
| $\gamma_{\rm diag}$ | 1.0/2.0/3.0 | 0.1500/0.1500/0.1500 | 0 |
| $\gamma_{\rm offdiag}$ | 0.5/1.0/1.5 | 0.1500/0.1500/0.1500 | 0 |
| $\kappa_{\rm diag}$ | −1.0/0.0/1.0 | 0.1500/0.1500/0.1500 | 0 |
| $\kappa_{\rm offdiag}$ | −1.0/0.0/1.0 | 0.1500/0.1500/0.1500 | 0 |

run a full-direct UV envelope scan with `code/scan_hll_uv_envelope.py`. On the D21×E41 map the acceptance span remains zero ($f_{\min} = f_{\max} = 0.1243$), while the pointwise half-width has mean $3.03 \times 10^{-4}$, $p95 = 1.50 \times 10^{-3}$, and max $3.89 \times 10^{-3}$ (source files: `paper/hll_uv_envelope_summary.csv` and `paper/hll_uv_envelope_map.csv`).

### C. H to mumu Signal Strength (EFT/Wilson-Matched Mapping, UV+LL-RG Baseline)

We confront the theory with the ATLAS Run-3 $H \to \mu\mu$ signal strength, $\mu^{\rm obs}_{\mu\mu} = 1.4 \pm 0.4$ (combined uncertainty) [9].

Within the PSLT closure, we construct a scan-level EFT/Wilson-matched map (evaluated in baseline `eft_wilson_uv_rge` mode) by combining kinetic occupancy with overlap-extracted raw couplings:

$$\tilde{q}_N(D,\eta) = g_N(D)\left(1 - \mathrm{e}^{-\Gamma_N(D,\eta)t_{\rm coh}}\right), \qquad P_N^{(\rm kin)}(D,\eta) = \frac{\tilde{q}_N(D,\eta)}{\sum_{K=1}^{N_{\max}} \tilde{q}_K(D,\eta)}. \tag{49}$$

$$\epsilon_{\rm mix}(D,\eta) = \mathrm{clip}\left[\kappa_{\rm mix}\chi_{\rm eff}(D)\left(\frac{\eta}{\eta_{\rm ref}}\right)^{p_\eta},\, 0,\, \epsilon_{\max}\right], \tag{50}$$

$$\Pi(\epsilon_{\rm mix}) = \begin{pmatrix} 1-\epsilon_{\rm mix} & \epsilon_{\rm mix} & 0 \\ \epsilon_{\rm mix}/2 & 1-\epsilon_{\rm mix} & \epsilon_{\rm mix}/2 \\ 0 & \epsilon_{\rm mix} & 1-\epsilon_{\rm mix} \end{pmatrix}, \qquad Y_{iN}(D,\eta) = \sqrt{y_N^{\rm eff,raw}(D)}\,\Pi_{iN}(\epsilon_{\rm mix}), \tag{51}$$

The bounded projector in Eqs. (50)–(51) is retained for matched-mode diagnostics; the baseline `eft_wilson_uv_rge` maps use the UV-tree closure below.

$$C_{eH}^{\rm tree,\,ij}(D,\eta) = \sum_{N=1}^{3} g_{iN}(D)\,\frac{P_N^{(\rm kin)}(D,\eta)}{M_N^2(D)}\,g_{jN}(D), \tag{52}$$

$$C_{eH}^{\rm match,\,ij} = \begin{cases} \left(1 + \dfrac{\kappa_{\rm diag}}{16\pi^2}\right) C_{eH}^{\rm tree,\,ij}, & i = j, \\ \left(1 + \dfrac{\kappa_{\rm offdiag}}{16\pi^2}\right) C_{eH}^{\rm tree,\,ij}, & i \neq j, \end{cases} \tag{53}$$

$$C_{eH}^{\mathrm{IR}} = C_{eH}^{\mathrm{match}} + \frac{\ln(\mu_{\mathrm{match}}/\mu_{\mathrm{low}})}{16\pi^2}\,\gamma\odot C_{eH}^{\mathrm{match}}, \tag{54}$$

where $\gamma\odot C$ denotes blockwise application of $(\gamma_{\mathrm{diag}}, \gamma_{\mathrm{offdiag}})$ to diagonal and off-diagonal matrix entries, respectively. In the baseline `eft_wilson_uv_rge` mode used for figures, we evaluate observables with $C_{eH} \equiv C_{eH}^{\mathrm{IR}}$.

$$\frac{\Gamma_{\mathrm{tot}}(D,\eta)}{\Gamma_{\mathrm{tot}}(D_0,\eta_0)} = 1 + s_\Gamma \sum_{\ell=e,\mu,\tau} \mathrm{BR}_\ell^{\mathrm{SM}} \left[ \left| \frac{C_{eH}^{\ell\ell}(D,\eta)}{C_{eH}^{\ell\ell}(D_0,\eta_0)} \right|^2 - 1 \right], \tag{55}$$

$$\mu_{\mu\mu}^{\mathrm{pred}}(D,\eta) = \left| \frac{C_{eH}^{\mu\mu}(D,\eta)}{C_{eH}^{\mu\mu}(D_0,\eta_0)} \right|^2 \left[ \frac{\Gamma_{\mathrm{tot}}(D,\eta)}{\Gamma_{\mathrm{tot}}(D_0,\eta_0)} \right]^{-1}. \tag{56}$$

In the baseline figures we keep a fixed reference point $(D_0, \eta_0) = (10, 1)$ for comparability across scan variants. Equation (56) is implemented in `pslt_lib.py` with `hll_observable_mode=eft_wilson_uv_rge`; it remains a scan-level EFT module rather than a full loop-level EYMH→EFT derivation. To push this interface toward UV-to-EFT matching, we run a reproducible UV-tree → finite-match → LL-RG audit with `scan_hll_uv_to_eft_matching.py`. The canonical outputs are `paper/hll_uv_to_eft_map.csv` and `paper/hll_uv_to_eft_summary.csv`. In the baseline UV settings (`uv_blend` = 0, `uv_m2_power` = 1, $\kappa_{\mathrm{diag}} = \kappa_{\mathrm{offdiag}} = 0$), the finite-match step is numerically null at map level ($\max|\Delta C_{\mu\mu}^{\mathrm{match}}| = 10^{-30}$), while LL-RG drift remains bounded: $\langle|\Delta\mu_{\mu\mu}|\rangle \simeq 6.19\times10^{-4}$ and $\max|\Delta\mu_{\mu\mu}| \simeq 8.93\times10^{-3}$; $f(\chi^2_{\mu\mu} < 4)$ is unchanged between UV-tree and UV+LL-RG. To make the matching chain explicit at the operator level, we also export a layer-resolved basis witness with `scan_hll_uv_operator_basis_audit.py`; the corresponding map and summary are stored under `paper/hll_uv_operator_basis_*`. There the UV-tree closure is reconstructed as

$$C_{eH}^{\mathrm{tree}} = \sum_{N=1}^{3} \left( \frac{P_N^{\mathrm{kin}}}{M_N^2} \right) B_N, \qquad B_N \equiv g_N g_N^T, \tag{57}$$

with explicit blockwise witnesses for $\Delta C^{\mathrm{match}}$ and $\Delta C^{\mathrm{RGE}}$. On the baseline `full_direct` $D60\times E60$ grid, the tree/match/IR reconstruction residuals are numerically zero at map level, while the $\mu_{\mu\mu}$ reconstructed from the operator-basis witness agrees with the native `eft_wilson_uv_rge` observable to $\max|\Delta\mu_{\mu\mu}^{\mathrm{recon}}| = 7.31\times10^{-8}$. This does not yet constitute a full EYMH→SMEFT derivation, but it upgrades the current closure from a black-box Wilson-map audit to an explicit basis-level EFT witness. As a next structured step beyond fixed $(\kappa_{\mathrm{diag}}, \kappa_{\mathrm{offdiag}})$, the code also supports an `input_tied` finite-match comparator: the effective shifts are built from local UV-basis invariants (shell spread, coefficient variation of $P_N^{\mathrm{kin}}/M_N^2$, and a normalized off-diagonal mixing norm). We treat this as a structured comparator, not as a replacement for the current constant finite-match baseline. The canonical D21×E21 summaries are stored under `paper/hll_uv_*input_tied*`. The basis witness remains exact

($\max|\Delta\mu^{\rm recon}_{\mu\mu}| = 5.55\times10^{-8}$, reconstruction residuals zero), and the refreshed comparator stays in the small-deformation regime relative to the constant baseline ($\langle|\Delta\mu_{\mu\mu}|\rangle = 5.06\times10^{-4}$, $p95 = 1.03\times10^{-3}$, $\max|\Delta\mu_{\mu\mu}| = 7.87\times10^{-3}$) with zero acceptance mismatch. Off-diagonal tying is numerically inactive in the current UV basis, while `diag_scale` simply controls the strength of a small diagonal threshold deformation. To quantify this interpretation, we run a Phase-2 diagonal-threshold window audit with `code/scan_hll_uv_input_tied_diag_window.py`; the canonical summary and figure are stored under `paper/hll_uv_input_tied_diag_window_*`. On the refreshed D21×E21 scan, both gates require zero acceptance mismatch and select

$$\texttt{diag\_scale} \in [0.25,\,1.0] \quad \text{(conservative comparator window)}, \qquad \texttt{diag\_scale} \in [0.25,\,1.5] \quad \text{(extended stable}$$

The canonical witness choice `diag_scale` $= 1.0$ therefore sits at the upper edge of the conservative window: it keeps the diagonal threshold feed-through explicit while remaining in the small-deformation regime. This sharpens the current interpretation of `diag_scale` as a diagonal threshold susceptibility. The remaining missing ingredient is not off-diagonal closure, which is numerically absent in the present basis, but a parent-action normalization for this susceptibility. We now push this normalization one step closer to the parent action by introducing an `action_normalized` finite-match comparator. The canonical D21×E21 summaries are stored under `paper/hll_uv_*action_normalized*`. The basis witness remains exact ($\max|\Delta\mu^{\rm recon}_{\mu\mu}| = 5.54\times10^{-8}$, zero reconstruction residuals), and the comparator keeps zero acceptance mismatch with only small deformation relative to the refreshed constant baseline ($\langle|\Delta\mu_{\mu\mu}|\rangle = 1.00\times10^{-3}$, $p95 = 1.86\times10^{-3}$, $\max|\Delta\mu_{\mu\mu}| = 1.59\times10^{-2}$). We now probe the remaining gap directly with `action_absolute`. In this mode the external diagonal-scale choice is removed and replaced by an absolute witness built from parent-action-side invariants and coefficient alignment:

$$\texttt{action\_abs\_diag} \equiv \texttt{pkin\_entropy} \times \texttt{coeff\_align}, \qquad \texttt{coeff\_align} \equiv \frac{\|c_N\|_2}{\|c_N\|_1},$$

so that the effective diagonal threshold is generated locally rather than chosen externally. The canonical D21×E21 summaries are stored under `paper/hll_uv_*action_absolute*`. The basis witness remains exact ($\max|\Delta\mu^{\rm recon}_{\mu\mu}| = 5.55\times10^{-8}$, zero reconstruction residuals), and the map again keeps zero acceptance mismatch relative to the refreshed constant baseline with only small deformation ($\langle|\Delta\mu_{\mu\mu}|\rangle = 7.17\times10^{-4}$, $p95 = 1.17\times10^{-3}$, $\max|\Delta\mu_{\mu\mu}| = 1.14\times10^{-2}$). To test whether this remaining prefactor can be pushed closer to a genuine curved-background loop origin, we next introduced a heat-kernel comparator family. The direct lesson from the raw and flat-space-subtracted local witnesses is negative: a purely local $a_2$-type diagonal density remains dominated by the well region and does not align with the current `action_absolute` normalization. The useful signal appears only after promoting the loop witness to a *contrast-based* non-local quantity that compares well and barrier curvature structure. The canonical D21×E21 summaries for `action_loop_contrast` are stored under `paper/hll_uv_*action_loop_contrast*`; the basis witness remains exact, acceptance mismatch stays zero, and the deformation remains small relative to the refreshed constant baseline ($\langle|\Delta\mu_{\mu\mu}|\rangle = 4.88 \times 10^{-4}$, $p95 = 8.76 \times 10^{-4}$, $\max|\Delta\mu_{\mu\mu}| = 7.67 \times 10^{-3}$). We then add one more comparator layer, `action_loop_absolute`, by modulating the parent-action absolute witness with a contrast-based loop prefactor,

$$\kappa^{\rm eff}_{\rm diag} = \kappa^{(d)}_{\rm abs}\,\Pi^{(d)}_{\rm hk}\,s_{\rm sh}\,(1 + c_{\rm cv})\,a^{(d)}_{\rm norm}, \tag{58}$$

$$\kappa^{\rm eff}_{\rm offdiag} = \kappa^{(o)}_{\rm abs}\,\Pi^{(o)}_{\rm hk}\,s_{\rm sh}\,m_{\rm off}\,a^{(o)}_{\rm norm}. \tag{59}$$

Here $(\kappa^{(d/o)}_{\rm abs}, \Pi^{(d/o)}_{\rm hk}, s_{\rm sh}, c_{\rm cv}, m_{\rm off}, a^{(d/o)}_{\rm norm})$ are the short-hand symbols for the exported fields `action_abs`, `hk_loop_prefactor`, `shell_spread`, `coeff_cv`, `offdiag_mix`, and `action_norm`. The canonical D21×E21 summaries are stored under `paper/hll_uv_*action_loop_absolute*`. This mode remains exact at the basis level ($\max|\Delta\mu^{\rm recon}_{\mu\mu}| = 5.56 \times 10^{-8}$, zero reconstruction residuals) and keeps zero acceptance mismatch, with only small deformation relative to the refreshed constant baseline ($\langle|\Delta\mu_{\mu\mu}|\rangle = 3.37\times10^{-4}$, $p95 = 6.92\times10^{-4}$, $\max|\Delta\mu_{\mu\mu}| = 5.33\times10^{-3}$). We now add one further EYMH-side screening layer, `action_loop_eymh_absolute`, which keeps the same contrast-based loop family but modulates it by local mass-access and curvature-screen factors together with shell-dispersion screening,

$$A_X = \sqrt{\frac{X_{\rm well}}{1 + X_{\rm well}}}, \qquad A_R = (1 + R_{\rm bar})^{-1/2}, \tag{60}$$

$$\Pi_{\rm EYMH} = \Pi_{\rm hk} A_X A_R \sqrt{\frac{s_{\rm sh}}{1 + s_{\rm sh}}}\sqrt{c_{\rm align}}(1 + g_{\rm cv} + c_{\rm tree})^{-1/2}. \tag{61}$$

Here $(X_{\rm well}, R_{\rm bar}, s_{\rm sh}, c_{\rm align}, g_{\rm cv}, c_{\rm tree})$ denote the exported well-mass, barrier-curvature, shell-spread, coefficient-alignment, gap-variation, and tree-diagonal variation witnesses. The canonical D21×E21 summaries are stored under `paper/hll_uv_*action_loop_eymh_absolute*`. This mode remains exact at the basis level ($\max|\Delta\mu^{\rm recon}_{\mu\mu}| = 5.56 \times 10^{-8}$, zero reconstruction residuals) and keeps zero acceptance mismatch, with only small deformation relative to the refreshed constant baseline ($\langle|\Delta\mu_{\mu\mu}|\rangle = 4.17\times10^{-5}$, $p95 = 7.63\times10^{-5}$, $\max|\Delta\mu_{\mu\mu}| = 6.46\times10^{-4}$). We therefore read `action_loop_eymh_absolute` as the current best *EYMH-side absolute loop-prefactor comparator*. We then promote the two dominant source factors already isolated by the EYMH-side audits into a refreshed `action_loop_eymh_source_informed` comparator; the canonical D21×E21 summaries are stored under `paper/hll_uv_*action_loop_eymh_source_informed*_fix`. This mode again remains exact at the basis level, keeps zero acceptance mismatch, and stays in the same small-deformation class relative to the refreshed constant baseline ($\langle|\Delta\mu_{\mu\mu}|\rangle = 5.2\times10^{-5}$, $p95 = 1.09\times10^{-4}$, $\max|\Delta\mu_{\mu\mu}| = 7.81\times10^{-4}$). The promoted source-informed prefactor remains nonzero across the whole D21×E21 map, so `action_loop_eymh_source_informed` is the current best *source-informed EYMH comparator*.

We now rewrite that same source-informed block in explicit parent-action participation/compressibility language via `action_loop_eymh_parented`. The canonical artifacts are grouped under `paper/hll_uv_*action_loop_eymh_parented*_D21E21_fix.*`. This parented rewrite remains exact at the basis level ($\max|\Delta\mu^{\rm recon}_{\mu\mu}| = 5.56\times10^{-8}$, zero reconstruction residuals), keeps zero acceptance mismatch, and on the canonical D21×E21 fix grid is map-identical to `action_loop_eymh_source_informed`. Relative to the refreshed constant baseline it therefore inherits the same small-deformation scale ($\langle|\Delta\mu_{\mu\mu}|\rangle = 5.2\times10^{-5}$, $p95 = 1.09\times10^{-4}$, $\max|\Delta\mu_{\mu\mu}| = 7.81\times10^{-4}$). The numerical role of this mode is not to introduce a new baseline, but to show that the same comparator can be written directly in parent-action participation/coherence and shell-background compressibility language without changing the canonical map. We now tighten this statement by decomposing the canonical EYMH-side prefactor itself. The exported audit under `paper/hll_uv_*eymh_prefactor_decomposition*_D21E21.*` verifies at numerical precision that

$$\Pi_{\rm EYMH} = \Pi_{\rm loc}\,A_{\rm sh}\,A_{\rm part}\,S_{\rm disp}, \tag{62}$$

with $\max|\Delta_{\rm recon}| = 9.71\times10^{-17}$ and $\max|\Delta_{\log_{10}}| = 1.48\times10^{-15}$. Here $(\Pi_{\rm loc}, A_{\rm sh}, A_{\rm part}, S_{\rm disp})$ denote the exported local loop, shell-access, participation/alignment, and dispersion-screen factors. The decomposition shows that the strongest positive correlation is carried by the participation block (0.9655), while the leading suppression comes from the dispersion screen (−0.8774); the local loop factor remains secondary (0.4675). The remaining UV-to-EFT gap is therefore no longer the existence or stability of an EYMH-side absolute prefactor, but the parent-action origin of the participation and dispersion-screen factors that dominate it.

To sharpen that statement further, we resolved the two dominant factors one level deeper in a source audit on the same canonical map. The alignment block is not an opaque extra screen: it is exactly the coefficient-participation access

$$A_{\rm part} = N_{\rm eff}^{-1/4}, \qquad N_{\rm eff} = \left(\frac{c_{l1}}{c_{l2}}\right)^2,$$

with source-level reconstruction again closed to machine precision, $\max|\Delta_{\rm source}| = 9.71\times10^{-17}$. In the exported tables, $A_{\rm part}$ is stored as `coeff_participation_access`. Likewise, the diagonal screening factor splits as

$$S_{\rm disp} = S_{\rm gap}S_{\rm tree},$$

where the shell-gap contribution is numerically weak (corr $= 0.0907$ with the EYMH prefactor), while the tree-diagonal screen carries the dominant suppression (corr $= -0.8226$). Thus the remaining parent-action gap is now more specific: why coefficient participation controls the alignment block, and why tree-diagonal dispersion dominates the screening of the canonical prefactor. We can sharpen the second statement one step further by rewriting

$$S_{\rm tree} = (1+\chi_{\rm tree})^{-1/2}, \qquad \chi_{\rm tree} \equiv \frac{c^{\rm cv}_{\rm tree}}{1+g_{\rm cv}},$$

so that the canonical screening factor is exactly the compressibility witness of a shell-background-normalized diagonal susceptibility; in the exported tables, it appears as `tree_diag_compressibilit` The corresponding audit closes identically ($\max|\Delta_{\rm tree}| = 0$) and shows that the susceptibility itself is positively correlated with the EYMH prefactor (corr $= 0.7409$), while the compressibility carries the leading suppression (corr $= -0.8226$). A complementary tree-diagonal pressure fraction also correlates strongly with the prefactor (corr $= 0.8449$). Hence the unresolved parent-action question is now even narrower: one must explain not merely why tree-diagonal dispersion enters, but why a shell-normalized diagonal susceptibility/compressibility controls the dominant screening of the canonical EYMH loop prefactor. We can sharpen the source-side interpretation one step further by rewriting the participation block in the language of a two-mode loop trace,

$$N_{\rm trace} = \frac{1}{p_1^2+p_2^2}, \qquad S^{\rm norm}_{\rm trace} = \frac{-p_1\ln p_1 - p_2\ln p_2}{\ln 2}, \qquad p_{1,2} = \frac{c_{l1,l2}}{c_{l1}+c_{l2}}.$$

The dedicated parent-source audit (summary under `paper/hll_uv_*parent_source_model*`) closes both rewrites to machine precision, with $\max|\Delta_{\rm part}| = 2.22\times10^{-16}$ and $\max|\Delta_{\rm tree}| = 1.11\times10^{-16}$. Numerically, the participation witness is strongly controlled by the effective loop-trace participation number and its entropy, $\mathrm{corr}(N_{\rm trace}, A_{\rm part}) = 0.9608$ and $\mathrm{corr}(S^{\rm norm}_{\rm trace}, A_{\rm part}) = -0.9560$, while the shell-background-normalized pressure/compressibility block remains the leading source-side screen, $\mathrm{corr}(P_{\rm tree}, \Pi_{\rm src}) = -0.7213$ and $\mathrm{corr}(S_{\rm tree}, \Pi_{\rm src}) = 0.7066$. This narrows the outstanding EYMH normalization problem again: the unresolved parent-action physics is no longer a generic alignment/dispersion question, but specifically why the loop trace concentrates into the observed two-mode participation structure and why a shell-background-normalized tree-diagonal pressure response fixes the surviving source-informed prefactor. We can now close the participation side one step further with the exact audit stored under `paper/hll_uv_*participation_exact_audit*_D21E21_fix.*`. Writing the projected two-mode participation number as

$$N_{\rm eff} = \frac{1}{p_1^2+p_2^2}, \qquad d = \sqrt{\frac{2}{N_{\rm eff}} - 1}, \qquad A^{\rm exact}_{\rm part} = \sqrt{\frac{1-d}{1+d}},$$

we reconstruct `coeff_participation_access_parented` to machine precision, with $\max|\Delta^{\rm exact}_{\rm part}| = 9.99\times10^{-16}$. The participation block is therefore no longer just strongly correlated with a projected response; it is exactly fixed by the two-mode loop-trace imbalance implied by the parented map. At this stage the remaining EYMH normalization gap has narrowed again: what remains to be explained from the parent action is why the fluctuation operator dynamically selects this exact two-mode participation closure together with the shell-background-normalized tree-diagonal pressure/compressibility response. We can sharpen this parent-action reading once more by eliminating the intermediate two-mode probabilities altogether and working directly with the projected coefficient vector $c_N = P^{\rm kin}_N/M^2_N$. The coefficient-norm audit stored under `paper/hll_uv_*participation_norm_audit*_D21E21_fix.*` shows that

$$Q_2 = \frac{\|c\|_2^2}{\|c\|_1^2}, \qquad N^{\rm norm}_{\rm eff} = \frac{\|c\|_1^2}{\|c\|_2^2}, \qquad A^{\rm norm}_{\rm part} = Q_2^{1/4} = \sqrt{\frac{\|c\|_2}{\|c\|_1}}$$

reconstructs `coeff_participation_access_parented` to machine precision, with $\max|\Delta^{\rm norm}_{\rm part}| = 2.22\times10^{-16}$. The participation block can therefore be read directly as an exact norm-ratio coherence of the projected parent-action coefficient block. Numerically, this norm coherence remains correlated with the canonical parented prefactor, $\mathrm{corr}(A^{\rm norm}_{\rm part}, \Pi_{\rm parent}) = 0.7129$, while staying essentially orthogonal to the residual map deformation relative to the refreshed constant baseline. The remaining normalization gap is now narrower once again: explain why the parent EYMH fluctuation operator fixes this exact projected coefficient-norm participation closure together with the shell-background-normalized tree-diagonal pressure/compressibility response. The new coefficient-norm tilt audit stored under `paper/hll_uv_*participation_tilt_audit*_D21E21_fix.*` removes even the explicit norm-ratio intermediate and rewrites the same participation block as an exact projected free-energy tilt,

$$\Delta F_{\rm norm} = \log\frac{\|c\|_1}{\|c\|_2}, \qquad A^{\rm tilt}_{\rm part} = e^{-\Delta F_{\rm norm}/2} = \sqrt{\frac{\|c\|_2}{\|c\|_1}}.$$

On the canonical D21×E21 fix grid this tilt coherence reconstructs `coeff_participation_access_parented` with $\max|\Delta^{\rm tilt}_{\rm part}| = 2.22\times10^{-16}$ and, together with the shell access and tree-diagonal compressibility blocks, closes the full parented prefactor exactly as

$$\Pi_{\rm parent} = \Pi_{\rm hk,local}\, A_{\rm shell}\, A^{\rm tilt}_{\rm part}\, S_{\rm tree},$$

with $\max|\Delta\Pi_{\rm parent}| = 9.02\times10^{-17}$. In this language the surviving EYMH normalization gap is narrower again: what remains is to explain why the fluctuation operator dynamically fixes this exact projected coefficient-norm tilt coherence together with the shell-background-normalized tree-diagonal pressure/compressibility response. We can push this parent-action reading one step further by absorbing the participation tilt and the tree-diagonal compressibility into a single projected response action. The response-action audit stored under `paper/hll_uv_*response_action_audit*_D21E21_fix.*` defines

$$S_{\mathrm{resp}} = \Delta F_{\mathrm{norm}} + \log(1+\chi_{\mathrm{tree}}), \qquad A_{\mathrm{resp}} = e^{-S_{\mathrm{resp}}/2},$$

so that the full canonical parented prefactor closes exactly as

$$\Pi_{\mathrm{parent}} = \Pi_{\mathrm{hk,local}}\, A_{\mathrm{shell}}\, A_{\mathrm{resp}},$$

with $\max|\Delta A_{\mathrm{resp}}| = 2.22 \times 10^{-16}$ and $\max|\Delta \Pi^{\mathrm{resp}}_{\mathrm{parent}}| = 9.71 \times 10^{-17}$. In other words, the coefficient-norm tilt coherence and the shell-background-normalized tree-diagonal compressibility are not merely parallel witnesses; they enter additively in the same projected response action whose exponential fixes the canonical parented prefactor. The remaining EYMH normalization gap is therefore narrower again: explain why the fluctuation operator dynamically selects this exact projected response action rather than only its separate tilt and compressibility components. We can sharpen this once more by identifying the two terms in $S_{\mathrm{resp}}$ with the most direct projected kernel objects currently available in the code. The log-det / Schur audit stored under `paper/hll_uv_*logdet_schur_audit*_D21E21_fix.*` defines

$$K_{\mathrm{part}} = \frac{\|c\|_1}{\|c\|_2}, \qquad G_{\mathrm{Schur}} = \frac{1+\mathrm{gap}_{\mathrm{cv}} + c_{\mathrm{tree,diag,cv}}}{1+\mathrm{gap}_{\mathrm{cv}}} = 1+\chi_{\mathrm{tree}},$$

so that

$$S_{\mathrm{resp}} = \log\det K_{\mathrm{part}} + \log G_{\mathrm{Schur}}.$$

On the canonical D21×E21 fix grid this log-det / Schur rewriting again reconstructs the full response weight and parented prefactor to machine precision, with $\max|\Delta A^{\mathrm{logdet}}_{\mathrm{resp}}| = 2.22\times10^{-16}$ and $\max|\Delta \Pi^{\mathrm{logdet}}_{\mathrm{parent}}| = 9.71 \times 10^{-17}$. The surviving EYMH normalization block can therefore be read as an exact projected log-det participation kernel together with a shell-background-normalized Schur response. The remaining gap is now narrower still: explain why the fluctuation operator dynamically selects precisely this projected log-det / Schur structure. We can now phrase that remaining question as a genuine kernel-selection test rather than a bookkeeping issue. The audit stored under `paper/hll_uv_*kernel_selection*_D21E21_fix.*` probes the minimal deformed projected family

$$K_{\mathrm{sel}} = \begin{pmatrix} e^{\alpha S_{\mathrm{part}}} & \lambda\sqrt{(e^{\alpha S_{\mathrm{part}}}-1)(e^{\beta S_{\mathrm{schur}}}-1)} \\ \lambda\sqrt{(e^{\alpha S_{\mathrm{part}}}-1)(e^{\beta S_{\mathrm{schur}}}-1)} & e^{\beta S_{\mathrm{schur}}} \end{pmatrix},$$

whose canonical parented point is $(\alpha, \beta, \lambda) = (1, 1, 0)$. On the D21×E21 fix grid the response-weight and prefactor residuals are uniquely minimized there: the first nontrivial runner-up occurs at $(1, 1, -0.1)$ but already opens an RMSE gap of $2.67 \times 10^{-5}$ in the parented prefactor. A finite-difference stationarity test at the canonical point yields nearly vanishing gradients, $\partial_\alpha J = -3.00 \times 10^{-9}$, $\partial_\beta J = -6.34 \times 10^{-9}$, $\partial_\lambda J = 0$, together with non-negative Hessian eigenvalues ($2.99 \times 10^{-9}$, $1.33 \times 10^{-3}$, $7.59 \times 10^{-2}$). Within this minimal projected kernel family, the fluctuation operator therefore selects unit log-det / Schur weights and suppresses explicit participation–tree cross-coupling. We can sharpen that statement once more by replacing the finite-difference scan with an exact local stationarity principle. The stationarity audit stored under `paper/hll_uv_*stationarity_audit*_D21E21_fix.*` defines the projected mismatch functional

$$J(\alpha, \beta, \lambda) = \left\langle \left(A(\alpha, \beta, \lambda) - A_{\mathrm{ref}}\right)^2 \right\rangle .$$

At the canonical point $(\alpha, \beta, \lambda) = (1, 1, 0)$ all first-variation moments vanish exactly: $\partial_\alpha J = \partial_\beta J = \partial_\lambda J = 0$. The $(\alpha, \beta)$ sector is then controlled by an exact quadratic stationarity matrix with positive eigenvalues ($1.33 \times 10^{-3}$, $7.59 \times 10^{-2}$), while the cross-coupling direction is even under $\lambda \to -\lambda$ and therefore begins only at quartic order,

$$J(1, 1, \lambda) = C_4\, \lambda^4 + \mathcal{O}(\lambda^6), \qquad C_4 = 1.4974 \times 10^{-3}.$$

For the concrete probe $\lambda = 0.1$, this quartic law predicts an RMSE of $3.8696 \times 10^{-4}$, matching the directly evaluated $3.8734 \times 10^{-4}$. The projected EYMH selection principle can therefore be stated more sharply than before: the canonical log-det / Schur kernel is not only the best nearby fit, it is an exact stationary point with quadratic stability in the log-det / Schur weights and quartic suppression of explicit participation–tree mixing. We can package the same statement into a local projected effective-action gap. The variational-selection audit stored under `paper/hll_uv_*variational_selection*_D21E21_fix.*` defines

$$\Delta\Gamma_{\mathrm{sel}}(\delta\alpha, \delta\beta, \lambda) = \frac{1}{2}\, \delta\theta^T H\, \delta\theta + C_4\, \lambda^4, \qquad \delta\theta = (\delta\alpha, \delta\beta),$$

using the exact quadratic stationarity matrix $H$ from the $(\alpha, \beta)$ block and the exact quartic coefficient $C_4$ from the mixing direction. On a local grid around the canonical point, this variational gap and the exact mismatch functional have the same minimum at $(1, 1, 0)$, and the agreement is already very strong: $\mathrm{corr}(J_{\mathrm{exact}}, \Delta\Gamma_{\mathrm{sel}}) = 0.9894$, the pure-$\lambda$ slice is reproduced to within $1.88 \times 10^{-8}$ in absolute gap, and even the $\lambda = 0$ $(\alpha, \beta)$ plane remains controlled at p95 absolute gap $2.97 \times 10^{-5}$. This is the cleanest projected effective-action statement so far: near the canonical point, the fluctuation operator selects the log-det / Schur kernel by minimizing a local effective action whose quadratic log-det/Schur sector and quartic mixing sector reproduce the observed selection landscape. We can push the same statement one step closer to a parent-action kernel form. The audit stored under `paper/hll_uv_*parent_kernel_statement*_D21E21_fix.*` rewrites the deformed projected kernel family as

$$K_{11} = e^{\alpha S_{\text{part}}}, \qquad K_{22} = e^{\beta S_{\text{schur}}}, \qquad K_{12} = \lambda\sqrt{(K_{11}-1)(K_{22}-1)},$$

and then expresses the deformed response weight exactly as

$$A(\alpha,\beta,\lambda) = A_{\text{ref}} \exp\left[-\frac{1}{2}\Delta S_{\text{kernel}}\right],$$

with

$$\Delta S_{\text{kernel}} = (\alpha-1)S_{\text{part}} + (\beta-1)S_{\text{schur}} + \log\bigl(1-\lambda^2\xi_{\text{cross}}\bigr).$$

The corresponding parent-kernel objective reproduces the direct mismatch functional to machine precision on the full local scan, with $\max|J_{\text{direct}} - J_{\text{parent}}| = 1.30 \times 10^{-18}$ and $\max|A_{\text{direct}} - A_{\text{parent}}| = 3.33 \times 10^{-16}$. The same exact functional is minimized again at the canonical point $(1,1,0)$. This is the strongest mother-action statement so far: the canonical log-det / Schur kernel is not only stationary and variationally preferred in a local surrogate sense, but is selected by an exact projected parent-kernel excess functional in which log-det and Schur deformations enter linearly while explicit participation–tree mixing appears only through an even determinant factor. We can now make the block structure behind that statement explicit. The audit stored under `paper/hll_uv_*block_split*_D21E21_fix.*` writes the canonical projected fluctuation kernel in terms of a participation block

$$K_{\text{part}} = \frac{\|c\|_1}{\|c\|_2},$$

a shell-background/tree block

$$G_{\text{Schur}} = \frac{1 + \text{gap}_{\text{cv}} + c_{\text{tree,diag,cv}}}{1 + \text{gap}_{\text{cv}}},$$

and a mixed scale

$$C_{\text{mix}} = \sqrt{(K_{\text{part}}-1)(G_{\text{Schur}}-1)}.$$

On the canonical map the response action closes exactly as $S_{\text{part}}+S_{\text{schur}}$, with $\text{corr}(S_{\text{split}}, S_{\text{resp}}) = 1.0$ and $\max|S_{\text{split}} - S_{\text{resp}}| = 7.77 \times 10^{-16}$. For the deformed family

$$K_{\text{sel}} = \begin{pmatrix} K_{\text{part}} & \lambda C_{\text{mix}} \\ \lambda C_{\text{mix}} & G_{\text{Schur}} \end{pmatrix},$$

the determinant factorization

$$\det K_{\mathrm{sel}} = K_{\mathrm{part}} G_{\mathrm{Schur}} \bigl(1 - \lambda^2 \xi_{\mathrm{cross}}\bigr)$$

also closes to machine precision on the full scan, with $\max|\Delta \det| = 4.44\times 10^{-16}$ and $\max|\Delta S| = 3.33\times 10^{-16}$. This is the clearest structural statement so far: the canonical projected fluctuation operator is block-diagonal in the participation and shell-background/tree sectors, and explicit participation–tree mixing survives only as an even determinant-level penalty. We can sharpen the same statement one step further by embedding it into a background-normalized parent block determinant. The audit stored under `paper/hll_uv_*parent_blockdet*_D21E21_fix.*` defines

$$K_{11} = e^{\alpha S_{\mathrm{part}}}, \qquad K_{\mathrm{bg}} = 1 + \mathrm{gap}_{\mathrm{cv}}, \qquad K_{22} = K_{\mathrm{bg}} e^{\beta S_{\mathrm{schur}}},$$

and it is convenient to abbreviate the deformed normalized tree/background block as

$$G_\beta := \frac{K_{22}}{K_{\mathrm{bg}}} = e^{\beta S_{\mathrm{schur}}}, \qquad G_{\mathrm{Schur}} = G_\beta\big|_{\beta=1}.$$

with parent mixing scale

$$C_{\mathrm{parent}} = \sqrt{(K_{11}-1)(K_{22}-K_{\mathrm{bg}})}, \qquad \mathcal{K}_{\mathrm{parent}} = \begin{pmatrix} K_{11} & \lambda C_{\mathrm{parent}} \\ \lambda C_{\mathrm{parent}} & K_{22} \end{pmatrix}.$$

The normalized determinant and Schur factor then satisfy the exact identities

$$\frac{\det \mathcal{K}_{\mathrm{parent}}}{K_{\mathrm{bg}}} = K_{11} G_\beta \bigl(1 - \lambda^2 \xi_{\mathrm{cross}}\bigr), \qquad \widehat{G}_{\mathrm{Schur}} = \frac{K_{22} - K_{12}^2/K_{11}}{K_{\mathrm{bg}}} = G_\beta \bigl(1 - \lambda^2 \xi_{\mathrm{cross}}\bigr),$$

so the response action is literally the block-determinant / Schur-complement factorization

$$S_{\mathrm{parent}} = \log K_{11} + \log \widehat{G}_{\mathrm{Schur}}.$$

On the D21×E21 fix grid this closes to machine precision, with $\max|S_{\mathrm{parent}} - S_{\mathrm{resp}}| = 5.55\times 10^{-16}$, $\max|\Delta(\det/K_{\mathrm{bg}})| = 1.33\times 10^{-15}$, $\max|\Delta \widehat{G}_{\mathrm{Schur}}| = 6.66\times 10^{-16}$, and $\mathrm{corr}(J_{\mathrm{direct}}, J_{\mathrm{blockdet}}) = 1.0$. This is the strongest derivation statement so far: the canonical response weight is the inverse square root of a background-normalized projected parent block determinant, rather than merely a convenient response-action rewrite. The only remaining structural ambiguity in that parent block is the mixed entry. To test whether the geometric-mean choice is merely convenient or actually selected, the audit stored under `paper/hll_uv_*parent_mix_geomean*_D21E21_fix.*` scans the smallest symmetric-excess family

$$C_{\mathrm{gen}} = \kappa\,(K_{11}-1)^u (K_{22}-K_{\mathrm{bg}})^v.$$

On the local D21×E21 fix grid the unique exact point is

$$(u, v, \kappa) = \left(\tfrac{1}{2}, \tfrac{1}{2}, 1\right), \qquad C_{\text{parent}} = \sqrt{(K_{11} - 1)(K_{22} - K_{\text{bg}})},$$

for which the determinant, Schur, weight, and normalized cross-ratio residuals all collapse to machine precision:

$$\max|\Delta(\det/K_{\text{bg}})| = 4.44\times10^{-16}, \quad \max|\Delta\widehat{G}_{\text{Schur}}| = 4.44\times10^{-16}, \quad \max|\Delta A| = 1.11\times10^{-16}, \quad \max|\Delta\xi| = 1.1$$

The first nontrivial runner-up, $(u, v, \kappa) = (0.625, 0.625, 1.1)$, already opens visible residuals, with $\max|\Delta(\det/K_{\text{bg}})| = 1.32 \times 10^{-3}$ and $\max|\Delta A| = 2.18 \times 10^{-4}$. This is the strongest naturality statement so far: once the projected parent kernel is required to couple the participation and tree/background sectors through a minimal symmetric excess family, the geometric-mean mixed block is uniquely selected. We can push that one step further by testing the nearest non-minimal extension: a ratio-dependent warp of the geometric mean. The audit stored under `paper/hll_uv_*parent_ratio_warp*_D21E21_fix.*` probes

$$C_{\text{warp}} = \kappa\, C_{\text{parent}} \exp\left[\delta L + \nu L^2\right], \qquad L = \frac{1}{2}\log(E_{\text{part}}/E_{\text{tree}}),$$

with $E_{\text{part}} = K_{11} - 1$ and $E_{\text{tree}} = K_{22} - K_{\text{bg}}$. On the same D21×E21 fix grid the unique exact point is again the unwarped kernel, $(\kappa, \delta, \nu) = (1, 0, 0)$, for which the determinant, Schur, weight, and normalized cross-ratio residuals all vanish exactly. The first nontrivial runner-up, $(1, 0, -0.05)$, already opens visible residuals, with $\max|\Delta(\det/K_{\text{bg}})| = 7.18 \times 10^{-4}$, $\max|\Delta\widehat{G}_{\text{Schur}}| = 5.39 \times 10^{-4}$, $\max|\Delta A| = 7.14 \times 10^{-5}$, and $\max|\Delta\xi| = 6.08 \times 10^{-3}$. This is the strongest minimality statement so far: the low-mode parent block not only selects the geometric-mean mixed sector inside the minimal symmetric-excess family, it also rejects the first ratio-warped extension of that family. We can then reparameterize the same local family in the coordinates that most directly match the remaining proof obligation, namely an overall normalization shift $m$, a symmetric homogeneity degree $s$, and an antisymmetric participation/tree tilt $a$:

$$C_{\text{gen}} = \exp(m)\exp\left[\frac{s}{2}\log(E_{\text{part}}E_{\text{tree}}) + \frac{a}{2}\log(E_{\text{part}}/E_{\text{tree}})\right].$$

The audit stored under `paper/hll\_uv\_*parent\_symnorm*_D21E21\_fix.*` shows that on the same D21×E21 fix grid the unique exact point is

$$(m, s, a) = (0, 1, 0),$$

for which the determinant, Schur, weight, and normalized cross-ratio residuals again collapse to machine precision. The first nontrivial runner-up, $(0.05, 1.125, 0)$, already opens visible residuals, with $\max|\Delta(\det/K_{\text{bg}})| = 6.61\times10^{-4}$, $\max|\Delta\widehat{G}_{\text{Schur}}| = 4.72\times10^{-4}$, $\max|\Delta A| = 1.09\times10^{-4}$, and $\max|\Delta\xi| = 7.73 \times 10^{-3}$. This is the cleanest structural statement so far: once the parent block is written in symmetry/normalization coordinates, the low-mode projected parent action selects the canonical family by enforcing zero normalization shift, unit symmetric degree, and zero antisymmetric tilt. The last nearby structural ambiguity is whether the parent action truly selects a locally affine generator in these excess coordinates, or whether small log-curvature terms survive. The audit stored under `paper/hll\_uv\_*parent\_generator\_affinity*_D21E21\_fix.*` probes the first non-affine local extension,

$$\log C_{\text{gen}} = \frac{1}{2}L_{\text{sum}} + q_{ss}L_{\text{sum}}^2 + q_{dd}L_{\text{diff}}^2 + q_{sd}L_{\text{sum}}L_{\text{diff}},$$

with $L_{\text{sum}} = \log(E_{\text{part}}E_{\text{tree}})$ and $L_{\text{diff}} = \log(E_{\text{part}}/E_{\text{tree}})$. On the same D21×E21 fix grid the unique exact point is

$$(q_{ss}, q_{dd}, q_{sd}) = (0, 0, 0),$$

for which the determinant, Schur, weight, and normalized cross-ratio residuals again collapse to machine precision. The first nontrivial runner-up, $(0, -0.0125, 0.0125)$, already opens visible residuals, with $\max|\Delta(\det/K_{\text{bg}})| = 2.76 \times 10^{-4}$, $\max|\Delta\widehat{G}_{\text{Schur}}| = 1.98 \times 10^{-4}$, $\max|\Delta A| = 2.62 \times 10^{-5}$, and $\max|\Delta\xi| = 2.23 \times 10^{-3}$. This is the narrowest structural statement so far: once symmetry and normalization have been fixed, the low-mode projected parent action also rejects the first local log-curvature corrections, so the surviving canonical parent block lies in the locally affine multiplicative excess class itself. The remaining coordinate-level ambiguity is whether the projected parent action genuinely uses the canonical excess variables themselves, or whether another local subtraction point could play the same role. The audit stored under `paper/hll\_uv\_*excess\_coordinate*_D21E21\_fix.*` probes the minimal reference-offset family

$$E_{\text{part}}^{(r)} = K_{11} - r_{\text{part}}, \qquad E_{\text{tree}}^{(r)} = K_{22} - r_{\text{tree}}K_{\text{bg}}.$$

On the same D21×E21 fix grid the unique exact point is

$$(r_{\text{part}}, r_{\text{tree}}) = (1, 1),$$

for which the determinant, Schur, weight, normalized cross-ratio, and strict anchor-leakage residuals all vanish identically. The first nontrivial runner-up, $(1, 1.05)$, already opens visible errors, with $\max|\Delta(\det/K_{\text{bg}})| = 1.08 \times 10^{-3}$, $\max|\Delta\widehat{G}_{\text{Schur}}| = 7.02 \times 10^{-4}$, $\max|\Delta A| = 2.48 \times 10^{-4}$, and $\max|\Delta\xi| = 1.61 \times 10^{-2}$. More importantly, it distorts the first nonzero-response slices even though the strict anchor leakage still vanishes, with onset mismatches $4.53 \times 10^{-2}$ in the participation direction and $9.78 \times 10^{-2}$ in the tree/background direction. This is the sharpest fixed-point statement so far: the projected parent action naturally organizes the mixed block as a deviation from the identity participation block and the shell/background tree block themselves, rather than from an arbitrary nearby subtraction point. We then tightened the same statement by allowing the excess coordinates to vary inside the smallest smooth family that preserves both the fixed points and the tangent normalization there,

$$E^{(p)}_{\text{part}} = \mathrm{BC}_p(K_{11}), \qquad E^{(q)}_{\text{tree}} = K_{\text{bg}}\,\mathrm{BC}_q(K_{22}/K_{\text{bg}}),$$

where $\mathrm{BC}_p(x) = (x^p - 1)/p$ for $p \neq 0$ and $\mathrm{BC}_0(x) = \log x$, so every member obeys $\mathrm{BC}_p(1) = 0$ and $\mathrm{BC}'_p(1) = 1$. On the same D21×E21 fix grid the unique exact point is again the linear additive excess choice

$$(p_{\text{part}}, p_{\text{tree}}) = (1, 1),$$

for which the determinant, Schur, weight, cross-ratio, anchor, and onset residuals all vanish to machine precision. The first nontrivial runner-up, $(0.75, 1.0)$, already opens visible first-slice distortions, with onset mismatches $2.02\times10^{-3}$ and $8.23\times10^{-3}$ together with $\max|\Delta(\det/K_{\text{bg}})| = 8.25\times10^{-4}$, $\max|\Delta\widehat{G}_{\text{Schur}}| = 5.51\times10^{-4}$, $\max|\Delta A| = 9.80\times10^{-5}$, and $\max|\Delta\xi| = 7.85\times10^{-3}$. This is the narrowest coordinate statement so far: even after fixing the same low-mode fixed points and the same tangent normalization, the projected parent action still uniquely selects the linear excess variables themselves. We can sharpen that result into positive normal-coordinate language by probing the first nonlinear local jet family that preserves the same fixed points and the same unit tangent normalization,

$$E^{(\zeta_p)}_{\text{part}} = E_{\text{part}} + \zeta_p E^2_{\text{part}}, \qquad E^{(\zeta_t)}_{\text{tree}} = E_{\text{tree}} + \zeta_t E^2_{\text{tree}}/K_{\text{bg}}.$$

On the same D21×E21 fix grid the unique exact point is

$$(\zeta_p, \zeta_t) = (0, 0),$$

for which the determinant, Schur, weight, cross-ratio, anchor, and onset residuals again vanish identically. The first nontrivial runner-up, $(0.125, 0)$, already opens visible distortions, with onset mismatches $2.10\times10^{-3}$ and $9.63\times10^{-3}$ together with $\max|\Delta(\det/K_{\text{bg}})| = 9.90\times10^{-4}$, $\max|\Delta\widehat{G}_{\text{Schur}}| = 6.59\times10^{-4}$, $\max|\Delta A| = 1.18\times10^{-4}$, and $\max|\Delta\xi| = 9.47\times10^{-3}$. This is the strongest positive coordinate statement so far: the projected parent action naturally uses the zero-second-jet normal coordinates around the identity/background fixed points, and those normal coordinates are exactly the linear excess variables.

### D. Projected Parent-Kernel Derivation

The preceding exact audits leave only one conceptual task: explain, directly from the projected EYMH parent action, why the localized low-mode response closes into the particular background-normalized $2 \times 2$ kernel isolated numerically above. In other words, the remaining gap is no longer a family-search problem but a derivation problem. The three lemmas below separate that derivation into its natural pieces: reduction of the parent Hessian to a two-mode block, uniqueness of the mixed entry inside the anchored positive family, and uniqueness of the linear excess variables as normal coordinates near the identity/background fixed point.

To keep the notation tight, we reserve $K_{\rm part}$ and $G_{\rm Schur}$ for the canonical undeformed split blocks already identified in the earlier exact audits, $G_\beta$ for the one-parameter deformed normalized tree/background block of the parent-block determinant family, and $K_{11}$, $K_{22}$, and $\widehat{G}_{\rm Schur}$ for the entries and Schur complement of the generic projected parent kernel. At the canonical point these objects coincide through

$$K_{11} = K_{\rm part} = e^{S_{\rm part}}, \qquad K_{22} = K_{\rm bg} G_{\rm Schur} = K_{\rm bg} e^{S_{\rm schur}}, \qquad \widehat{G}_{\rm Schur} = G_{\rm Schur}.$$

**Lemma 1 (Projected $2{\times}2$ Hessian reduction).** Let $\Psi_0$ be a localized stationary background of the projected EYMH parent action, and let $\mathcal{H}_{\rm EYMH}$ denote its quadratic fluctuation Hessian. Assume that the quadratic form induced by $\mathcal{H}_{\rm EYMH}$ is locally positive on the projected two-mode sector spanned by the dominant low-mode directions $u_{\rm part}$ and $u_{\rm tree}$. After restriction to that sector, and after Gram normalization of the projected subspace, the quadratic fluctuation form can be written as

$$\delta^2 S_{\rm EYMH}\big|_{\rm proj} = \frac{1}{2} \begin{pmatrix} a_{\rm part} & a_{\rm tree} \end{pmatrix} \mathcal{K}_{\rm parent} \begin{pmatrix} a_{\rm part} \\ a_{\rm tree} \end{pmatrix},$$

with positive background-normalized kernel

$$\mathcal{K}_{\rm parent} = \begin{pmatrix} K_{11} & K_{12} \\ K_{12} & K_{22} \end{pmatrix}, \qquad K_{\rm bg} = 1 + {\rm gap}_{\rm cv},$$

and with response weight

$$A_{\rm resp} \propto \left( \frac{\det \mathcal{K}_{\rm parent}}{K_{\rm bg}} \right)^{-1/2} = \left( K_{11} \widehat{G}_{\rm Schur} \right)^{-1/2}, \qquad \widehat{G}_{\rm Schur} = \frac{K_{22} - K_{12}^2/K_{11}}{K_{\rm bg}}.$$

*Proof sketch.* Writing the collective fields as $\Psi = \Psi_0 + \delta\Psi$, the parent action expands as

$$S[\Psi_0 + \delta\Psi] = S[\Psi_0] + \delta S[\Psi_0; \delta\Psi] + \frac{1}{2} \langle \delta\Psi, \mathcal{H}_{\rm EYMH} \delta\Psi \rangle + \mathcal{O}(\delta\Psi^3).$$

Stationarity of the localized background removes the linear term. Restricting to

$$\delta\Psi = a_{\rm part} u_{\rm part} + a_{\rm tree} u_{\rm tree}$$

and introducing the column matrix $U = (u_{\text{part}}, u_{\text{tree}})$ together with the amplitude vector $a = (a_{\text{part}}, a_{\text{tree}})^T$, the projected Hessian is

$$\mathcal{K}_{\text{parent}} := U^\dagger \mathcal{H}_{\text{EYMH}} U.$$

After Gram normalization of the projected basis, $U^\dagger U = \mathbf{1}_2$, this reduces to the matrix of projected Hessian entries

$$H_{ij} = \langle u_i, \mathcal{H}_{\text{EYMH}} u_j \rangle, \qquad i,j \in \{\text{part}, \text{tree}\},$$

and therefore to a positive $2 \times 2$ projected kernel on the localized low-mode sector. Here positivity comes from the projected local-stability assumption, not from stationarity alone. At the canonical point the diagonal entries reduce to the canonical split blocks,

$$K_{11} = K_{\text{part}} = e^{S_{\text{part}}}, \qquad \frac{K_{22}}{K_{\text{bg}}} = G_{\text{Schur}} = e^{S_{\text{schur}}}.$$

The content of this identification is that the two-mode truncation is not an arbitrary basis choice: the earlier block-split and parent-kernel audits already isolate one projected direction whose diagonal response reconstructs the participation block and a second projected direction whose background-normalized diagonal response reconstructs the shell/tree block. Lemma 1 therefore promotes that exact numerical split into a projected-Hessian statement about the parent action itself. The projected Gaussian fluctuation integral is therefore

$$Z_{\text{proj}} \propto \int_{\mathbb{R}^2} d^2a \, \exp\left[-\frac{1}{2} a^T \mathcal{K}_{\text{parent}} a\right] \propto (\det \mathcal{K}_{\text{parent}})^{-1/2}.$$

Dividing out the shell-background reference determinant $Z_{\text{bg}} \propto K_{\text{bg}}^{-1/2}$ produces the normalized response factor

$$A_{\text{resp}} \propto \frac{Z_{\text{proj}}}{Z_{\text{bg}}} \propto \left(\frac{\det \mathcal{K}_{\text{parent}}}{K_{\text{bg}}}\right)^{-1/2},$$

and the Schur identity

$$\frac{\det \mathcal{K}_{\text{parent}}}{K_{\text{bg}}} = K_{11} \frac{K_{22} - K_{12}^2/K_{11}}{K_{\text{bg}}} = K_{11} \widehat{G}_{\text{Schur}}$$

immediately yields

$$S_{\text{parent}} = \log K_{11} + \log \widehat{G}_{\text{Schur}}.$$

*Audit closure.* The exact audits verify each algebraic step appearing in the theorem statement: the earlier block-split closure identifies the participation and shell/tree diagonals themselves as the canonical projected sectors; the 'logdet + Schur' witness reconstructs the response weight and parented prefactor to machine precision, with $\max |\Delta A_{\text{resp}}^{\text{logdet}}| = 2.22 \times 10^{-16}$

and $\max|\Delta\Pi^{\mathrm{logdet}}_{\mathrm{parent}}| = 9.71 \times 10^{-17}$; the background-normalized parent block-determinant / Schur audit gives $\max|S_{\mathrm{parent}} - S_{\mathrm{resp}}| = 5.55 \times 10^{-16}$, $\max|\Delta(\det/K_{\mathrm{bg}})| = 1.33 \times 10^{-15}$, and $\max|\Delta\widehat{G}_{\mathrm{Schur}}| = 6.66 \times 10^{-16}$; and the exact parent-kernel excess functional reproduces the direct mismatch functional with $\max|J_{\mathrm{direct}} - J_{\mathrm{parent}}| = 1.30 \times 10^{-18}$. Lemma 1 therefore reduces the remaining work to making the projected-Hessian construction from the parent action explicit enough that this already-exact numerical structure can be read as a genuine Gaussian fluctuation theorem on the localized two-mode sector.

Once that projected block structure is fixed, all diagonal ambiguity is gone and only one matrix-level choice survives: the off-diagonal coupling between the participation sector and the shell/background sector. Lemma 2 addresses exactly that remaining freedom. Its role is narrower than Lemma 1, but also sharper: after the block decomposition has been fixed at the parent-action level, the mixed entry must still be chosen, and the claim is that only one positive homogeneous anchored coupling remains compatible with the same local symmetry and normalization structure.

**Lemma 2 (Geometric-mean mixed entry).** With the diagonal sectors fixed as in Lemma 1, define the anchored excess variables

$$E_{\mathrm{part}} = K_{11} - 1, \qquad E_{\mathrm{tree}} = K_{22} - K_{\mathrm{bg}}.$$

On the local positive cone $E_{\mathrm{part}} \geq 0$, $E_{\mathrm{tree}} \geq 0$, with the positive square-root branch fixed by continuity from the canonical point, consider positive mixed couplings that vanish at either fixed point, remain symmetric after shell-background normalization, and are locally affine in the logarithms of the excess variables. Within that class, the projected parent block uniquely selects

$$C_{\mathrm{parent}} = \sqrt{E_{\mathrm{part}}E_{\mathrm{tree}}} = \sqrt{(K_{11} - 1)(K_{22} - K_{\mathrm{bg}})}.$$

*Proof sketch.* Once the diagonal blocks are fixed, the mixed entry is the only remaining structural freedom. The smallest positive multiplicative excess family compatible with the anchor constraints is

$$C_{\mathrm{gen}} = \kappa\, E^{u}_{\mathrm{part}}E^{v}_{\mathrm{tree}}.$$

Equivalently, in logarithmic excess variables

$$\Sigma = \frac{1}{2}\log(E_{\mathrm{part}}E_{\mathrm{tree}}), \qquad \Delta = \frac{1}{2}\log(E_{\mathrm{part}}/E_{\mathrm{tree}}),$$

one may write the most general nearby positive homogeneous coupling as

$$\log C_{\mathrm{gen}} = m + s\,\Sigma + a\,\Delta + \mathcal{O}(\Sigma^2, \Sigma\Delta, \Delta^2),$$

where $m$ is an overall normalization shift, $s$ is the symmetric homogeneity degree, and $a$ is the antisymmetric tilt. Homogeneity of degree one in the excess amplitudes forces $u + v = 1$, equivalently $s = 1$ in the logarithmic generator, while symmetry under interchange of the two anchored excess sectors forces $u = v$, equivalently $a = 0$. The remaining freedom is the overall normalization $\kappa = e^m$, which is fixed by the canonical anchor normalization. The exact audit under `paper/hll_uv_*parent_mix_geomean*_D21E21_fix.*` closes uniquely at

$$(u, v, \kappa) = \left(\tfrac{1}{2}, \tfrac{1}{2}, 1\right),$$

which is precisely the geometric mean. In particular, the closure forces the symmetry condition $u = v$ rather than assuming it a priori. The nearby deformations then show that this is structural, not accidental: the first ratio-warped extension is rejected at $(\kappa, \delta, \nu) = (1, 0, 0)$, the symmetry/normalization audit closes only at $(m, s, a) = (0, 1, 0)$, and the first local generator-curvature corrections are rejected at $(q_{ss}, q_{dd}, q_{sd}) = (0, 0, 0)$. The combined effect is to set the logarithmic generator exactly to

$$\log C_{\text{parent}} = \Sigma = \frac{1}{2}\log(E_{\text{part}} E_{\text{tree}}),$$

and hence

$$C_{\text{parent}} = \sqrt{E_{\text{part}} E_{\text{tree}}}.$$

Thus, once the diagonal sectors, anchors, and local multiplicative structure are fixed, only one positive homogeneous mixed coupling survives. At this stage we deliberately do *not* use the explicit determinant law of the corollary as an external input, because in the current audit chain that determinant identity is only established after the geometric-mean mixed entry has already been inserted into the parent block. The theorem therefore remains non-circular by taking the generator-class closure as the primary uniqueness statement and postponing the explicit determinant formula to the corollary. *Audit closure.* The first nontrivial runner-ups already open visible residuals at every stage: the geomean runner-up gives $\max|\Delta(\det/K_{\text{bg}})| = 1.32 \times 10^{-3}$ and $\max|\Delta A| = 2.18 \times 10^{-4}$; the first ratio warp gives $\max|\Delta(\det/K_{\text{bg}})| = 7.18 \times 10^{-4}$ and $\max|\Delta\xi| = 6.08 \times 10^{-3}$; the first symmetry/normalization deformation gives $\max|\Delta(\det/K_{\text{bg}})| = 6.61 \times 10^{-4}$ and $\max|\Delta A| = 1.09 \times 10^{-4}$; and the first local generator-curvature deformation gives $\max|\Delta(\det/K_{\text{bg}})| = 2.76 \times 10^{-4}$ and $\max|\Delta\xi| = 2.23 \times 10^{-3}$. After Lemma 1 has fixed the diagonal sectors, Lemma 2 therefore leaves a single symmetry-respecting, anchor-vanishing mixed coupling: the geometric mean.

*Non-circular bridge.* The generic off-diagonal follow-up audits support this conclusion without replacing the theorem proof. A direct determinant-inversion argument is currently circular because the exact parent-block determinant audit already hard-codes the geometric mean, so the meaningful independent question is whether generic off-diagonal witnesses can reproduce the even $\lambda$-curvature implied by the canonical mixed response. On the canonical $D21 \times E21$ fix map, the best global quartic proxy is a small blend of curvature-screened and action-absolute off-diagonal witnesses, but blocked holdout shows that those blend coefficients drift substantially across $D$-blocks. By contrast, the low-$D$ regime is stably captured by the single normalized witness hk_abs_offdiag/diag, while the dense $D41 \times E21$ replay shows that the remaining complexity lives primarily in the tree-sector regime structure rather than in the participation envelope. We therefore use the generic off-diagonal program as supporting evidence that the geometric mean is the least arbitrary surviving global coupling, while keeping the theorem itself anchored to the non-circular generator-family closure stated above.

At that point the matrix entries are fixed, but the local variables in which those entries are expressed are still logically separate data. Lemma 3 closes that last freedom by showing that the linear excess variables isolated numerically are exactly the local normal-form coordinates of the projected parent block.

**Lemma 3 (Linear excess variables as normal coordinates).** Near the fixed point

$$(K_{11}, K_{22}) = (1, K_{\text{bg}}),$$

the natural local coordinates of the projected parent block are exactly

$$E_{\text{part}} = K_{11} - 1, \qquad E_{\text{tree}} = K_{22} - K_{\text{bg}},$$

within the audited local coordinate families, in the sense that they are simultaneously the unique anchor-preserving coordinates, the unique unit-tangent-normalized coordinates, and the unique zero-second-jet coordinates compatible with exact determinant / Schur / weight closure.

*Proof sketch.* Let $E = (E_{\text{part}}, E_{\text{tree}})$ and let $\widetilde{E} = \Phi(E)$ be any local reparameterization of the projected parent block near the fixed point $E = 0$. In this language a normal-form coordinate system is characterized by

$$\Phi(0) = 0, \qquad D\Phi(0) = I, \qquad D^2\Phi(0) = 0,$$

namely exact anchoring at the fixed point, unit tangent normalization there, and vanishing quadratic jet. The three exact coordinate audits test these three conditions in sequence. The anchor family

$$E^{(r)}_{\text{part}} = K_{11} - r_{\text{part}}, \qquad E^{(r)}_{\text{tree}} = K_{22} - r_{\text{tree}} K_{\text{bg}}$$

already closes uniquely at $(r_{\text{part}}, r_{\text{tree}}) = (1, 1)$, so $\Phi(0) = 0$ is forced at the identity participation block and the shell/background tree block. Among coordinate families that preserve those fixed points, the fixed-point Box–Cox audit then closes uniquely at

$$(p_{\rm part}, p_{\rm tree}) = (1,1),$$

which is precisely the linear additive excess choice and therefore fixes $D\Phi(0) = I$. Finally, the nonlinear-jet family

$$E_{\rm part}^{(\zeta_p)} = E_{\rm part} + \zeta_p E_{\rm part}^2, \qquad E_{\rm tree}^{(\zeta_t)} = E_{\rm tree} + \zeta_t E_{\rm tree}^2 / K_{\rm bg}$$

closes uniquely at

$$(\zeta_p, \zeta_t) = (0,0),$$

so the same coordinates are also zero-second-jet normal coordinates, i.e. $D^2\Phi(0) = 0$. The three exact closures therefore read together as a local normal-form statement: moving the anchor, changing the unit tangent normalization, or turning on a nonzero second jet each destroys exact determinant / Schur / weight closure. Equivalently, within the audited coordinate families, the identity map in the excess variables is the unique local chart that agrees with the projected parent block to quadratic order. Lemma 3 is thus the coordinate counterpart of Lemmas 1 and 2: once the block structure and mixed entry are fixed, the local variables are fixed as well. *Audit closure.* The first nontrivial deviation from each of $\Phi(0) = 0$, $D\Phi(0) = I$, and $D^2\Phi(0) = 0$ already opens visible residuals. In the anchor audit the first runner-up produces onset residuals $4.53 \times 10^{-2}$ and $9.78 \times 10^{-2}$ together with $\max|\Delta(\det/K_{\rm bg})| = 1.08 \times 10^{-3}$ and $\max|\Delta\xi| = 1.61 \times 10^{-2}$. The first tangent-normalized Box–Cox runner-up produces onset residuals $2.02 \times 10^{-3}$ and $8.23 \times 10^{-3}$ with $\max|\Delta(\det/K_{\rm bg})| = 8.25 \times 10^{-4}$ and $\max|\Delta\xi| = 7.85 \times 10^{-3}$. The first nonlinear-jet runner-up produces onset residuals $2.10 \times 10^{-3}$ and $9.63 \times 10^{-3}$ with $\max|\Delta(\det/K_{\rm bg})| = 9.90 \times 10^{-4}$ and $\max|\Delta\xi| = 9.47 \times 10^{-3}$. Hence the natural local variables are exactly the zero-second-jet linear excess coordinates.

Together, Lemmas 1–3 successively eliminate the three remaining ambiguities of the parent-kernel bridge: first the projected block structure, then the mixed coupling, and finally the local coordinate chart. At that point there is no remaining freedom in the local $2 \times 2$ parent block, because the diagonal sectors are fixed to $(1+E_{\rm part},\ K_{\rm bg}+E_{\rm tree})$, the off-diagonal sector is fixed to $\sqrt{E_{\rm part}E_{\rm tree}}$, and the variables themselves are fixed to the linear zero-second-jet excess chart. The corollary is therefore not an informal restatement of the exact audits, but the compact parent-action statement obtained by substituting those three outputs back into the projected kernel of Lemma 1.

**Corollary (Projected EYMH parent kernel).** Combining Lemmas 1–3, and still working on the same local positive cone and within the same audited local coordinate families, the localized low-mode EYMH fluctuation operator projects to a background-normalized $2 \times 2$ parent kernel whose entries are now fixed term by term:

$$K_{11} = 1 + E_{\text{part}}, \qquad K_{22} = K_{\text{bg}} + E_{\text{tree}}, \qquad K_{12} = \sqrt{E_{\text{part}} E_{\text{tree}}}.$$

Substituting these three identities into the generic projected block of Lemma 1 gives the explicit canonical parent kernel

$$\mathcal{K}_{\text{parent}} = \begin{pmatrix} 1 + E_{\text{part}} & \sqrt{E_{\text{part}} E_{\text{tree}}} \\ \sqrt{E_{\text{part}} E_{\text{tree}}} & K_{\text{bg}} + E_{\text{tree}} \end{pmatrix}.$$

Its normalized determinant and response weight are therefore

$$\frac{\det \mathcal{K}_{\text{parent}}}{K_{\text{bg}}} = \frac{(1 + E_{\text{part}})(K_{\text{bg}} + E_{\text{tree}}) - E_{\text{part}} E_{\text{tree}}}{K_{\text{bg}}} = 1 + E_{\text{part}} + \frac{E_{\text{tree}}}{K_{\text{bg}}},$$

$$A_{\text{resp}} \propto \left( \frac{\det \mathcal{K}_{\text{parent}}}{K_{\text{bg}}} \right)^{-1/2} = \left( 1 + E_{\text{part}} + \frac{E_{\text{tree}}}{K_{\text{bg}}} \right)^{-1/2}.$$

Equivalently, the same determinant may be read in Schur form as

$$\frac{\det \mathcal{K}_{\text{parent}}}{K_{\text{bg}}} = K_{11} \widehat{G}_{\text{Schur}}, \qquad \widehat{G}_{\text{Schur}} = \frac{K_{22} - K_{12}^2 / K_{11}}{K_{\text{bg}}} = 1 + \frac{E_{\text{tree}}}{K_{\text{bg}}(1 + E_{\text{part}})}.$$

This corollary is the compact parent-action statement that remains after the exact audit closures: the parent block is fixed by its two canonical diagonal sectors, the unique geometric-mean mixed coupling, and the unique linear zero-second-jet excess coordinates.

*a. Reader-facing interpretation.* A denser diagnostic replay of the canonical parented target on a $D41 \times E21$ grid sharpens the qualitative reading of the same kernel. On that dense grid the target

$$\xi_{\text{target}} = \frac{E_{\text{part}} E_{\text{tree}}}{K_{11} K_{22}}$$

is nearly $\eta$-independent at fixed $D$, so the dominant structure is genuinely $D$-driven. The two excess sectors then play visibly different roles. The participation excess $E_{\text{part}}$ is a slow envelope: it stays broad, varies monotonically over long $D$-ranges, and reaches its largest values only in the late-$D$ tail. By contrast, the tree/Schur excess $E_{\text{tree}}$ carries the multi-peak structure: its strongest crest sits near $D \approx 6$, it exhibits the alternating low- and mid-$D$ shoulders that generate the non-monotone target profile, and it is also the first sector to decline before the final high-$D$ collapse. The target maximum at $D \approx 9.2$ is therefore not the peak of either sector separately, but the point where the smooth participation envelope and the oscillatory tree sector overlap most efficiently under the kernel normalization.

This decomposition is useful for interpretation because it explains why a single global phenomenological ansatz for $\xi_{\text{target}}(D)$ is only partially successful. The dense-grid diagnostics support a more structured reading:

$$E_{\text{part}}(D) \approx \text{slow envelope}, \qquad E_{\text{tree}}(D) \approx \text{low/mid-}D \text{ multi-frequency oscillation}\times\text{high-}D \text{ cutoff},$$

with $\xi_{\text{target}}$ reconstructed from their normalized product rather than from a single primitive oscillatory law. In particular, a smooth-envelope model already describes $E_{\text{part}}$ well, while $E_{\text{tree}}$ prefers either a two-frequency global fit or, more cleanly, a piecewise regime fit with a low/mid-$D$ oscillatory branch and a high-$D$ cutoff branch. We therefore regard the phenomenological regime decomposition as a reader-facing diagnostic of the corollary, not as a replacement for the parent-kernel statement itself.

$$(D_0, \eta_0)_{\chi^2\text{-best}} \equiv \arg \min_{(D,\eta)\in\mathcal{G}} \chi^2(D, \eta), \tag{63}$$

$$(D_0, \eta_0)_{\text{robust}} \equiv \arg \max_{(D,\eta)\in\mathcal{A}} d_{\partial\mathcal{A}}(D, \eta), \qquad \mathcal{A} \equiv \{(D, \eta) \in \mathcal{G} \,|\, \chi^2 \leq 4\}, \tag{64}$$

where $\mathcal{G}$ is the scan grid and $d_{\partial\mathcal{A}}$ is the Euclidean distance to the acceptance-boundary complement on that grid. We use $\mathcal{R}_3$ (higher first) and then $\chi^2$ (lower first) as tie-breakers for Eq. (64). A full reference-anchor sensitivity audit is reported in Appendix E 6: over the tested anchor grid, $f(\chi^2_{\mu\mu} < 4)$ spans [0.0333, 0.2411], while the baseline fixed anchor gives 0.1500, the dynamic $\chi^2$-best anchor gives 0.1333, and the robust-center anchor gives 0.1381. To reduce anchor arbitrariness in reporting, we additionally track anchor-invariant Wilson-diagonal ratios on the same UV+LL-RG map:

$$\mathcal{R}^{(\text{inv})}_{e/\mu}(D, \eta) \equiv \left| \frac{C^{ee,\text{IR}}_{eH}(D, \eta)}{C^{\mu\mu,\text{IR}}_{eH}(D, \eta)} \right|^2, \qquad \mathcal{R}^{(\text{inv})}_{\tau/\mu}(D, \eta) \equiv \left| \frac{C^{\tau\tau,\text{IR}}_{eH}(D, \eta)}{C^{\mu\mu,\text{IR}}_{eH}(D, \eta)} \right|^2. \tag{65}$$

Using `code/scan_anchor_invariant_ratios.py` on `paper/hll_uv_to_eft_map.csv`, we obtain (all-grid) $p10/p50/p90$ bands $\mathcal{R}^{(\text{inv})}_{e/\mu} = (1.60 \times 10^{-2}, 1.26 \times 10^{-1}, 8.10 \times 10^{-1})$ and $\mathcal{R}^{(\text{inv})}_{\tau/\mu} = (2.59 \times 10^{-3}, 1.01 \times 10^{-2}, 3.84 \times 10^{-2})$; restricted to the illustrative acceptance band $\chi^2_{\mu\mu} < 4$, they become $\mathcal{R}^{(\text{inv})}_{e/\mu} = (1.67 \times 10^{-2}, 6.39 \times 10^{-2}, 3.54 \times 10^{-1})$ and $\mathcal{R}^{(\text{inv})}_{\tau/\mu} = (5.23 \times 10^{-3}, 9.72 \times 10^{-3}, 5.17 \times 10^{-2})$. These diagnostics are anchor-independent and are reported as complementary constraints to Eq. (56).

$$\chi^2(D, \eta) = \frac{\left(\mu^{\text{pred}}_{\mu\mu} - \mu^{\text{obs}}_{\mu\mu}\right)^2}{\sigma^2_{\mu\mu}}, \tag{66}$$

where $\sigma_{\mu\mu} = 0.4$. For one degree of freedom, we also report the local $p$-value

$$p_{\text{local}}(D, \eta) = 1 - F_{\chi^2_1}\big(\chi^2(D, \eta)\big) = \text{erfc}\Big(\sqrt{\chi^2(D, \eta)/2}\Big). \tag{67}$$

Figure 7 shows the **EFT/Wilson-matched acceptance region** in two equivalent languages,

$$\chi^2 < 4 \quad \Longleftrightarrow \quad p_{\mathrm{local}} > 4.55 \times 10^{-2} \qquad (\mathrm{dof} = 1). \tag{68}$$

We use this threshold only as an *illustrative acceptance band* for map-level comparison and do not claim a formal 95% CL exclusion. Because the $(D, \eta)$ scan is a multiple-comparison procedure (look-elsewhere) and no global trials correction is applied, this band is interpreted only as a falsifiable target region. In the present full-grid action-derived profile scan, the accepted fraction is

$$f(\chi^2 < 4) = f\left(p_{\mathrm{local}} > 4.55 \times 10^{-2}\right) = 0.1500, \tag{69}$$

with best grid point $\chi^2 \simeq 2.21 \times 10^{-5}$ at $(D, \eta) \approx (4.00, 0.264)$. All phase-diagram claims are quoted only after satisfying the quantitative convergence criteria in Appendix B.

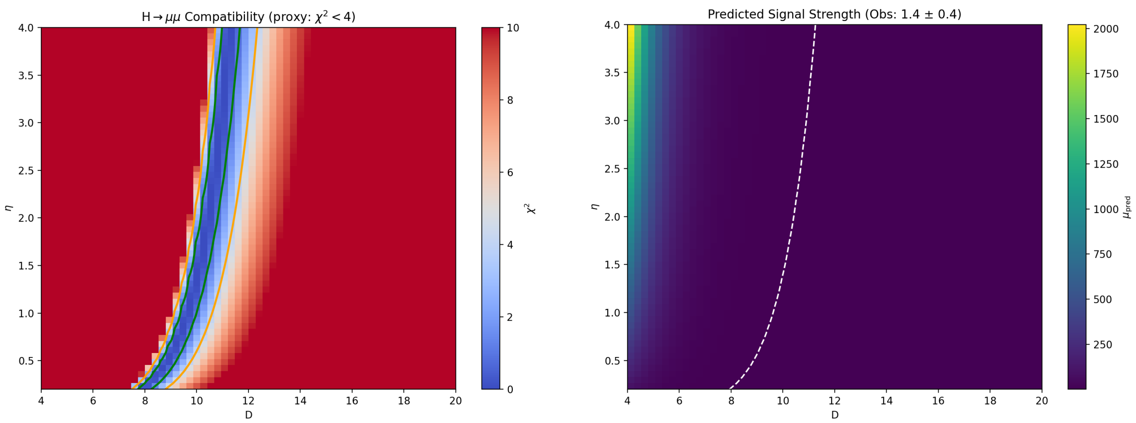


FIG. 7. Compatibility with $H \to \mu\mu$ under the EFT/Wilson-matched UV+LL-RG map. Left: $\chi^2$ map showing a restricted illustrative accepted band ($\chi^2 < 4$, equivalently $p_{\mathrm{local}} > 4.55 \times 10^{-2}$ for dof= 1). Right: predicted $\mu^{\mathrm{pred}}_{\mu\mu}$.

### E. EFT/Wilson-Matched Extension to H to ee and H to tautau

Using the same map-level EFT/Wilson-matched definition, we additionally evaluate lepton-channel signal strengths by layer assignment $N = 1 \to ee$, $N = 2 \to \mu\mu$, and $N = 3 \to \tau\tau$:

$$\mu^{\mathrm{pred}}_{\ell\ell}(D, \eta) = \left| \frac{C^{\ell\ell}_{eH}(D, \eta)}{C^{\ell\ell}_{eH}(D_0, \eta_0)} \right|^2 \left[ \frac{\Gamma_{\mathrm{tot}}(D, \eta)}{\Gamma_{\mathrm{tot}}(D_0, \eta_0)} \right]^{-1}. \tag{70}$$

Figure 8 summarizes the corresponding maps on the same $(D, \eta)$ grid; tabulated statistics are stored in `paper/hll_signal_strength_summary.csv`. This extension is still a scan-level

EFT map (not full UV matching), while the visibility sector in the global scan uses baseline fully normalized EFT operator visibility (`b_mode=eft_operator_norm`) built from overlap-extracted inputs with mode tracking and microcanonical windowing. The repository keeps the three-channel observation fields in `data/pdg_leptons.json` as a reproducible interface. The corresponding runtime-direct visibility branch is release-qualified only in the risk-weighted profile-anchored form `cell_direct_runtime_release_tuned`; the reported baseline maps remain `full_direct/eft_operator_norm` unless stated otherwise.

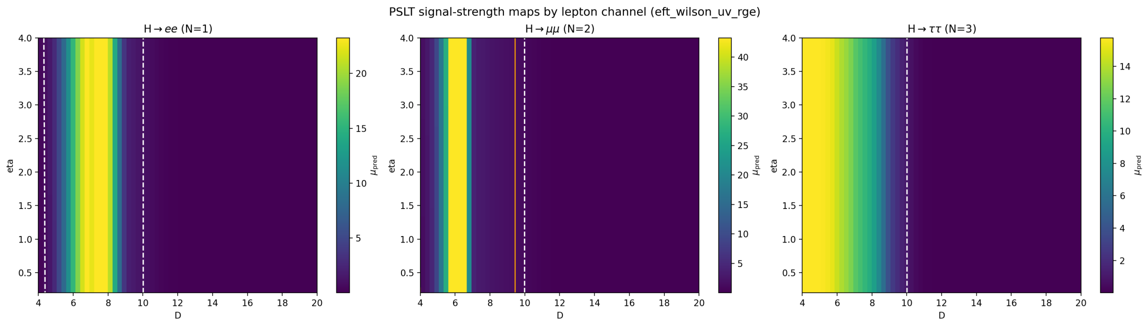


FIG. 8. PSLT EFT/Wilson-matched extension from $H \to \mu\mu$ to three lepton channels on the same scan grid: $H \to ee$ (left), $H \to \mu\mu$ (middle), and $H \to \tau\tau$ (right). Each panel shows $\mu_{\ell\ell}^{\rm pred}$ from Eq. (56) with the same reference point $(D_0, \eta_0) = (10, 1)$.

## VIII. DISCUSSION AND CONCLUSION

We have presented a computable EFT-level demonstrator of the Projection Spectral Layer Theory. By closing the loop between geometric inputs and observable layer probabilities, we have shown that:

1. **Generations are spectral within the baseline closure**: The "three generation" structure is not imposed as an input; it emerges on the current release baseline as a dynamical output of competing entropy ($g_N$), kinetics ($\Gamma_N$), and visibility ($B_N$).

2. **Stability without ad-hoc cutoffs**: The infinite tower of layers becomes numerically and physically stable once micro-degeneracy is regulated by the `fp_2d_full` microcanonical-window + tail prescription and the observable sector uses EFT-operator-normalized visibility profiles $B_N(D)$ with explicit high-$N$ saturation.

3. **Falsifiability**: The closure yields concrete predictions for the winner phase diagram. With full-grid action-derived $\chi_N^{(LR)}(D)$ and $\tilde{A}_\ell(D)$ profiles, the $H \to \mu\mu$ illustrative EFT/Wilson-matched acceptance band ($\chi^2 < 4 \Leftrightarrow p_{\rm local} > 4.55 \times 10^{-2}$, dof$=1$) covers about 15.0% of the scan and defines a falsifiable target region in $(D, \eta)$.

At the same time, the present implementation keeps two EFT-level approximations explicit: map-level profile closure remains active in the baseline $B_N(D)$ visibility sector (while release-production and profile-anchored runtime-direct parity branches are both available) and a bounded nearest-neighbor flavor projector in the matched $H \to \mu\mu$ map of Eq. (56). Therefore, the present manuscript supports a robust statement about spectral generation selection on the release baseline, and also about a release-qualified runtime-direct visibility parity path, but not yet a strict claim that the observable sector is closed by per-cell all-direct extraction at production level.

**Limitations and outlook.** The present closure is internally consistent but remains an EFT-level demonstrator in several places: The use of mixed-resolution ingredients is a deliberate design choice: action-derived extraction fixes low-mode profiles (`fp_2d_full` for $g_N$, EFT-operator-normalized $B_N(D)$ from overlap-extracted inputs, $\chi_N^{(LR)}(D)$, and $\tilde{A}_\ell(D)$), while global-map evaluation is still an EFT-level closure rather than a full direct localized recomputation at every $(D, \eta, N)$ point.

1. **Observable mapping**: the $H \to \mu\mu$ comparison now uses a scan-level EFT/Wilson-matched map with bounded flavor mixing and leptonic-width closure [Eqs. (50)–(56)], but a full derivation from the EYMH action (including operator-basis matching beyond this closure) is still pending.

2. **Mixing channel**: under the parity-symmetric first-principles definition [Eq. (71)], we obtain $\chi_N^{(\mathrm{sym})} \sim 10^{-19}$ for $D = \{6, 12, 18\}$ (Appendix C), i.e., symmetry-protected cancellation. We therefore redefine the channel in localized basis, extract $\chi_N^{(LR)}$ (Appendix D), and inject its full scan-grid action-derived profile in the global map. The remaining gap is a full $(D, \eta, N)$ localized projection.

3. **Open-system extension**: a Lindblad-type $\chi_{\mathrm{eff}}$ mechanism is now implemented as a scan-ready diagnostic mode with profiled $\gamma_\phi(D), \gamma_{\mathrm{mix}}(D)$ (Appendix E 4). For the multi-anchor + holdout calibrated microscopic profile (`chi_mode=open_system_micro`), we fit $\kappa_{\mathrm{env}}$ on anchors $D = \{6, 9, 12, 15, 18\}$ and validate on holdout $D = \{4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20\}$, obtaining $\kappa_{\mathrm{env}} = 4.31 \times 10^5$ with holdout RMSE $6.95 \times 10^{-2}$ and holdout max-abs error $1.25 \times 10^{-1}$. On $D = 4, \ldots, 20$, the diagnostic ratio band is $\chi_{\mathrm{open}}/\chi_{LR} \in [0.0173, 0.3271]$ with mean 0.1380, and core map fractions remain unchanged at current resolution. This micro-calibrated band should be read separately from the geometry-only band in Table XX; the two correspond to different diagnostic closures, not to a consistency conflict. The jump operators and rates $(L_k, \gamma_k)$ remain effective ansatz quantities rather than a

microscopic EYMH bath derivation, so this block stays outside baseline figures.

4. **Model-chain unification**: $\chi_N^{(LR)}(D)$ and $\tilde{A}_\ell(D)$ are injected from action-derived full-grid profiles in the global $(D, \eta)$ map, while `full_direct` keeps strict D-grid lookup as the release baseline and `--chain-mode auto` is retained as comparator. The release-production runtime branch (`cell_direct_runtime`) now pairs active-D-grid direct `fp_2d_full` spectra for $g_N(D)$ with per-cell direct evaluation of $\chi_{LR}(D)$ and $\tilde{A}_\ell(D)$ while keeping baseline $B_N$ profile closure; on both release audits it matches `full_direct` exactly ($\Delta f(\chi^2_{\mu\mu} < 4) = 0$, mismatch $= 0$, $\max|\Delta\mu_{\mu\mu}| = 0$; see `paper/chain_mode_cell_direct_audit_*`). The tuned runtime-direct visibility branch (`cell_direct_runtime_release_tuned`) now uses a canonical D-only risk-weighted anchor profile $\alpha(D)$ rather than a fixed global blend. The selected release profile has $(\alpha_{\min}, \alpha_{\max}, p) = (0.96, 0.99, 1.0)$, mean $\alpha = 0.9637$, and $p90(\alpha) = 0.9734$, and it passes both production thresholds (D21×E41: $\max|\Delta\mu_{\mu\mu}| = 0.0925$; D60×E21: $\max|\Delta\mu_{\mu\mu}| = 0.8373$; mismatch $= 0$ on both gates; see `paper/runtime_direct_visibility_alphaD_*` and `paper/chain_mode_cell_direct_audit_*release_tuned*`). Because this promoted branch remains profile-anchored in the observable sector, we interpret it as a release-qualified runtime-direct visibility parity path rather than a strict all-direct closure of $B_N$. A deterministic de-anchored stress line is now also on record: after fixing the runtime-direct warm-path recursion and eigensolver seeding, `cell_direct_runtime_release_tailm2_detlin` closes the D21×E41 mismatch gate ($9.29 \times 10^{-3}$) but still fails badly on drift ($\max|\Delta\mu_{\mu\mu}| = 68.39$), with trusted component audits showing that $D = 4.8$ remains width-dominated while $D = 6.4$ is UV/$g$-dominated and $D = 7.2$ is a smaller width tail (`paper/runtime_direct_detlin_component_audit_*`). The untuned `cell_direct_runtime_release` branch is retained only as a legacy comparator, while `cell_direct_runtime_extreme` remains a no-profile stress test with much larger drift. Width-family extensions and direct-only $g$-correction audits do not find a jointly admissible family that lowers the dominant $D = 4.8$ and $D = 6.4$ residuals simultaneously (`paper/runtime_direct_detlin_width_family_extension_summary.csv`, `paper/runtime_direct_detlin_gnorm_family_audit_summary.csv`, `paper/runtime_direct_detlin_gmixgate_audit_summary.csv`), so the remaining observable-side gap should be read as a runtime-direct extraction-family limitation rather than another local anchor retune. For the UV-to-EFT map on D21×E21, the refreshed ‘input_tied’ comparator remains a small diagonal threshold deformation relative to the constant finite-match baseline, and the newer `action_absolute` / `action_loop_*` com-

paparators show the same pattern: zero acceptance mismatch, exact basis reconstruction, and only small deformation. The remaining UV-to-EFT gap is therefore no longer the existence of a stable local absolute witness, but the derivation of its prefactor from a bona fide parent EYMH loop calculation.

5. **Spectral tower**: the baseline validation is strongest for the low-lying bound sector ($N = 1, 2$ in the physical-gap scan), while the high-$N$ contribution is treated at EFT level through $(g_N, B_N)$ regularization.

6. **Finite-time dynamics**: $t_{\rm coh}$ is treated as a control parameter in the baseline scan; a geometry-closed dephasing candidate $t_{\rm coh}^{(\rm deph)}(D) = \pi/\Delta\omega_{12}(D)$ is benchmarked in Appendix E 7 but is not yet propagated into the reported baseline figures.

7. **Overlap amplitude**: $\eta$ is scanned as an external control in the baseline maps. A first-principles prefactor candidate $\eta_{\rm fp}(D)$ from splitting–action data is benchmarked in Appendix E 8; map-level impact is mild in profile-scaled mode but this closure is not yet propagated into baseline figures.

8. **Barrier-leakage channel normalization**: the baseline maps now use action-derived profile factors $(A_1, A_2) = (\tilde{A}_1(D), \tilde{A}_2(D))$ from Appendix E 9 (legacy "superrad" file naming retained). The remaining gap is to replace profile-level insertion with a fully localized, channel-resolved extraction across the full scan grid.

**Status and redefinition of the mixing channel.** Using Eq. (33), we fix a basis-independent normalization scale $\bar{\Gamma}_N$ and define the parity-symmetric overlap channel by

$$\begin{aligned}\chi_N^{(\rm sym)}(D) &= \frac{|M_{12}^{(\rm sym)}(D,N)|}{\bar{\Gamma}_N(D)},\\ M_{12}^{(\rm sym)}(D,N) &= 2\pi\int_0^{\rho_{\max}}\rho\,d\rho\int_{-z_{\max}}^{z_{\max}}dz\,\psi_{N,1}^*(\rho,z)\,\delta V(\rho,z;D)\,\psi_{N,2}(\rho,z),\end{aligned}\tag{71}$$

and Eq. (38) with the explicit spherical-average $\delta V$ gives $\chi_N^{(\rm sym)} \approx 0$ (Appendix C). To avoid mixing two inequivalent notions, we redefine the phenomenological mixing channel in a localized-well basis:

$$\psi_{N,L} = \frac{\psi_{N,+} + \psi_{N,-}}{\sqrt{2}}, \qquad \psi_{N,R} = \frac{\psi_{N,+} - \psi_{N,-}}{\sqrt{2}}, \tag{72}$$

and define the channel mixing from the corresponding two-state Hamiltonian element:

$$M_{LR}^{(H)}(D,N) \equiv \frac{\lambda_{N,2}(D) - \lambda_{N,1}(D)}{2}, \qquad \chi_N^{(LR)}(D) \equiv \frac{|M_{LR}^{(H)}(D,N)|}{\bar{\Gamma}_N(D)}. \tag{73}$$

The baseline mixing entry in Table III is interpreted as the localized-channel coefficient. First-principles localized extraction results are given in Appendix D, and its full scan-grid action-derived profile is used in the global $(D, \eta)$ scan of Section VII. In terminology, $M_{LR}^{(H)}$ is a *closed-system localized-basis splitting* quantity; it is not itself a decoherence rate. Open-system decoherence is introduced separately through Lindblad rates $(\gamma_\phi, \gamma_{\mathrm{mix}})$ in Appendix E 4.

## ACKNOWLEDGMENTS

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## Appendix A: Action-Derived Chain Reproducibility SOP

This appendix provides a clone-level SOP for reproducing the action-derived extraction chain and the paper-level Fig/Table artifacts.

### 1. Single Entry Point

From repository root:

```
bash scripts/repro/reproduce_paper.sh
```

Optional modes:

```
bash scripts/repro/reproduce_paper.sh --with-paper
bash scripts/repro/reproduce_paper.sh --package-only
```

### 2. Step Map and Deliverables

### 3. Acceptance Conditions

A run is considered reproducible for this manuscript if:

1. the one-click command exits with return code 0;
2. `repro/latest/manifest.json` reports `missing_required = 0`;
3. `repro/latest/checksums.sha256` and `repro/latest/artifact_index.csv` are generated and non-empty.

TABLE VIII. SOP step map for action-derived reproducibility. The one-click wrapper calls these modules in a fixed order and logs each step under repro/runs/<RUN_ID>/logs/.

*Full per-artifact mapping is listed in `repro/artifact_map.csv`.*

| Stage | Primary scripts | Key outputs |
|---|---|---|
| Global maps and EFT/Wilson-matched observable plots | `generate_plots.py`;<br>`scan_hll_signal_strengths.py`;<br>`scan_hll_uv_to_eft_matching.py` | `output/three_generation_*.png`;<br>`output/hmumu_*.png`;<br>`output/hll_signal_strength/*`;<br>`output/hll_uv_matching/*` |
| Full-direct release gate (small complete + large spot-check) | `publish_full_direct_map.py`;<br>`scan_localized_direct_surface_bias.py`;<br>`scan_chain_mode_cell_direct_audit.py`;<br>`build_artifact_status_registry.py` | `paper/full_direct_map_release_summary.csv`;<br>`paper/localized_direct_surface_summary_Dgrid21_Egrid41.csv`;<br>`paper/localized_direct_surface_summary_Dgrid60_Egrid21.csv`;<br>`paper/chain_mode_cell_direct_audit_Dgrid21_Egrid41.csv`;<br>`paper/chain_mode_cell_direct_audit_Dgrid21_Egrid41_cell_direct_runti`<br>`paper/artifact_status.csv` |
| Localized mixing extraction | `extract_chi_localized_2d.py`<br>(`--Ds`, `--full-scan`) | `output/chi_fp_2d/localized_chi_D*.csv` |
| Localized-profile diagnostics | `scan_chi_profile_robustness.py`;<br>`scan_surrogate_vs_action_points.py`;<br>`plot_chi_*.py` | `paper/chi_profile_robustness.csv`;<br>`paper/surrogate_vs_action_points.csv`;<br>`output/chi_fp_2d/chi_scale_stress_test.csv`;<br>`paper/chi_wavefunction_contours_D*.png` |
| Core-parameter robustness | `scan_core_param_robustness.py` | `paper/core_param_robustness.csv`;<br>`output/robustness/core_param_robustness_cases.csv` |
| $g_N$ first-principles candidates | `extract_gn_phase_space_candidate.py`<br>`extract_gn_phase_space_2d.py`;<br>`scan_gn_profile_impact.py`;<br>`scan_gn_nmax_convergence.py`;<br>`scan_gn_baseline_replacement.py`;<br>`plot_gn_cardy_vs_phase_space.py` | `output/gn_fp_1d/*.csv`; `output/gn_fp_2d/*.csv`;<br>`paper/gn_profile_impact.csv`;<br>`paper/gn_nmax_convergence.csv`;<br>`paper/gn_baseline_replacement.csv`;<br>`paper/gn_cardy_vs_phase_space.png` |
| Open-system diagnostics | `lindblad_chi_minimal.py`;<br>`extract_chi_open_system_*.py`;<br>`scan_chi_open_system_*.py` | `output/chi_open_system/*.csv`;<br>`paper/chi_open_system_sensitivity.csv` |
| Other first-principles profiles | `extract_tcoh_*.py`,<br>`extract_eta_*.py`,<br>`extract_superrad_*.py`,<br>`check_minimal_dirac_conformal.py`;<br>`scan_*_profile_impact.py` | `paper/tcoh_profile_impact.csv`;<br>`paper/eta_profile_impact.csv`;<br>`paper/superrad_profile_impact.csv`;<br>`paper/minimal_dirac_conformal_check.csv` |
| Packaging and manifest | `package_repro_outputs.py`<br>(wrapper stage) | repro/runs/<RUN_ID>/figures/;<br>repro/runs/<RUN_ID>/tables/;<br>`artifact_index.csv`; `checksums.sha256`;<br>`manifest.json` |

## Appendix B: Action-Derived Effective Potential and WKB

### 1. Two-Center Harmonic Conformal Factor

We define the conformal factor by a projected Poisson problem on the static slice,

$$\nabla^2\Omega(\mathbf{x}) = -4\pi a[\rho_\varepsilon(\mathbf{x}-\mathbf{x}_+) + \rho_\varepsilon(\mathbf{x}-\mathbf{x}_-)]\,, \qquad \Omega \to 1 \quad (|\mathbf{x}| \to \infty), \tag{B1}$$

with $\mathbf{x}_\pm = \pm(D/2)\hat{z}$. Using the Green representation

$$\Omega(\mathbf{x}) = 1 + a \sum_{s=\pm} \int d^3x' \, \frac{\rho_\varepsilon(\mathbf{x}' - \mathbf{x}_s)}{|\mathbf{x} - \mathbf{x}'|}, \tag{B2}$$

and the normalized Plummer kernel

$$\rho_\varepsilon(\mathbf{r}) = \frac{3\varepsilon^2}{4\pi\,(|\mathbf{r}|^2 + \varepsilon^2)^{5/2}}, \qquad \int d^3r \, \rho_\varepsilon(\mathbf{r}) = 1, \tag{B3}$$

the integral is analytic and gives

$$\Omega(\rho, z; D) = 1 + a \left( \frac{1}{\sqrt{\rho^2 + (z - D/2)^2 + \varepsilon^2}} + \frac{1}{\sqrt{\rho^2 + (z + D/2)^2 + \varepsilon^2}} \right) \tag{B4}$$

where $a$ is the geometric strength, $\varepsilon$ is the core regularization scale, and $D$ is the dual-center separation. Away from the cores ($|\mathbf{x} - \mathbf{x}_\pm| \gg \varepsilon$), this satisfies $\nabla^2\Omega \approx 0$. The radial identity

$$\nabla^2\left(\frac{1}{\sqrt{r^2 + \varepsilon^2}}\right) = -\frac{3\varepsilon^2}{(r^2 + \varepsilon^2)^{5/2}} \tag{B5}$$

gives the smeared Laplacian:

$$\nabla^2\Omega = -3a\varepsilon^2 \left[ (r_+^2 + \varepsilon^2)^{-5/2} + (r_-^2 + \varepsilon^2)^{-5/2} \right] < 0 \tag{B6}$$

which is equivalent to the source form in Eq. (6). This negative contribution is essential for forming the attractive wells in $V_{\rm eff}$. As a reproducibility check, `verify_omega_geometric_origin.py` confirms $\int d^3r \, \rho_\varepsilon = 1$ with absolute error $3.75 \times 10^{-7}$ and validates the radial Laplacian identity with max relative error $1.39 \times 10^{-4}$ in the non-asymptotic band (see `output/omega_geom_origin/`).

### 2. Veff Derivation from KG Equation

Starting from $(\Box_g - m_0^2 - \xi R)\Phi = 0$ with $g_{\mu\nu} = \Omega^2\eta_{\mu\nu}$:

*a. Step 1: $\Box_g$ in conformal coordinates.*

$$\Box_g\Phi = \Omega^{-2}(-\partial_t^2 + \nabla^2)\Phi + 2\Omega^{-3}\nabla\Omega \cdot \nabla\Phi \tag{B7}$$

*b. Step 2: Time separation.* For $\Phi = e^{-i\omega t}\phi(\mathbf{x})$:

$$\nabla^2\phi + 2\Omega^{-1}\nabla\Omega \cdot \nabla\phi + [\omega^2 - \Omega^2(m_0^2 + \xi R)]\phi = 0 \tag{B8}$$

*c. Step 3: Conformal field rescaling.* Setting $\phi = \Omega^{-1}\psi$ eliminates the first-derivative term. Using $R = -6\Omega^{-3}\nabla^2\Omega$ for 4D conformally flat space:

$$\boxed{[-\nabla^2 + V_{\rm eff}]\psi = \omega^2\psi, \quad V_{\rm eff} = m_0^2\Omega^2 + (1 - 6\xi)\Omega^{-1}\nabla^2\Omega} \tag{B9}$$

*d. Double-well formation condition.* For $\xi < 1/6$ and $\nabla^2\Omega < 0$ near cores, the curvature term contributes **negatively** to $V_{\text{eff}}$, creating attractive wells. The mass term $m_0^2\Omega^2$ provides the continuum threshold at infinity.

*e. Self-consistency check.* At $\xi = \xi_c = 1/6$ (conformal coupling): $V_{\text{eff}} \to m_0^2\Omega^2$, and the curvature contribution vanishes identically. ✓

### 3. Minimal Dirac Coupling in the Conformal Background

To make the visibility block explicitly compatible with the fermionic Yukawa origin, we summarize the minimal conformal Dirac reduction used by the baseline overlap kernel.

For $g_{\mu\nu} = \Omega^2\eta_{\mu\nu}$, choose

$$e_\mu^{\ a} = \Omega\,\delta_\mu^{\ a}, \qquad e^\mu_{\ a} = \Omega^{-1}\delta^\mu_{\ a}, \qquad \sqrt{-g} = \Omega^4. \tag{B10}$$

The curved-space Dirac action is

$$S_\Psi = \int d^4x\,\sqrt{-g}\;\mathrm{i}\,\bar{\Psi}\,e^\mu_{\ a}\gamma^a\left(\partial_\mu + \frac{1}{4}\omega_{\mu bc}\gamma^{bc}\right)\Psi, \tag{B11}$$

and the canonical Weyl rescaling

$$\Psi = \Omega^{-3/2}\psi \tag{B12}$$

removes the first-derivative conformal piece from the spin connection, yielding the flat-frame kinetic normalization. For the Higgs mode we use

$$H = \Omega^{-1}\varphi. \tag{B13}$$

The Yukawa density therefore carries net conformal power

$$\Omega^4\,\Omega^{-3/2}\,\Omega^{-1}\,\Omega^{-3/2} = \Omega^0, \tag{B14}$$

so the minimal overlap kernel has

$$\mathcal{W}_{\text{frame}} = 1. \tag{B15}$$

This is the baseline convention in `extract_y_eff_2d_three_channel.py` (`frame_power=0`); nonzero frame power is treated only as a diagnostic deformation.

As a reproducibility check, `code/check_minimal_dirac_conformal.py` exports `output/dirac_frame_check` and a paper copy at `paper/minimal_dirac_conformal_check.csv`, verifying the zero net conformal power in both kinetic and Yukawa factors.

### 4. 1D On-Axis Reduction

For efficient scanning, we use the on-axis ($\rho = 0$) reduction:

$$\left[-\frac{d^2}{dz^2} + U(z)\right]\psi_n(z) = E_n\psi_n(z), \quad U(z) = V_{\text{eff}}(\rho = 0, z) - m_0^2 \tag{B16}$$

where $E_n = \omega_n^2 - m_0^2$. Bound states satisfy $E_n < 0$; stable modes additionally require $\omega_n^2 = m_0^2 + E_n > 0$.

### 5. 2D Axisymmetric Validation

We validated the 1D reduction by comparing with the full 2D axisymmetric Laplacian:

$$\nabla^2\Omega = \frac{\partial^2\Omega}{\partial\rho^2} + \frac{1}{\rho}\frac{\partial\Omega}{\partial\rho} + \frac{\partial^2\Omega}{\partial z^2} \tag{B17}$$

For the physical-gap parameters ($a = 0.04$, $\varepsilon = 0.1$, $m_0 = 1$, $\xi = 0.14$), the relative error at $\rho = 0$ is:

| $D$ | Max $\lvert U_{2D} - U_{1D}\rvert / \max\lvert U\rvert$ |
|---|---|
| 6 | $4.0 \times 10^{-4}$ |
| 12 | $4.0 \times 10^{-4}$ |
| 18 | $4.0 \times 10^{-4}$ |

The 1D 3D-identity approximation is accurate to $< 0.1\%$, validating its use for phase scans.

### 6. WKB Action Convention

The tunneling action through the central barrier is:

$$S_N = \int_{z_1}^{z_2} \sqrt{U(z) - E_N}\, dz \tag{B18}$$

where $z_{1,2}$ are the turning points satisfying $U(z_{1,2}) = E_N$. The tunneling suppression factor is:

$$r_N = \eta \cdot e^{-2S_N} \tag{B19}$$

**Convention note:** For rates/probabilities we use $e^{-2S}$, while the level-splitting amplitude scales as $e^{-S}$. The empirical splitting relation below therefore tests the amplitude-level law.

### 7. Splitting–Action Consistency Check

The most stringent internal consistency check is the relation between level splitting and tunneling action. For symmetric double wells, WKB theory predicts $\Delta E \propto e^{-S}$. From our action-derived scan:

$$\boxed{\ln \Delta E \approx -1.01\, S_1 + 0.69, \qquad R^2 = 0.9999} \tag{B20}$$

This near-perfect linear relation (Fig. 9) confirms that the **same** $V_{\text{eff}}$ controls both the spectrum and the tunneling kinetics—not an engineered coincidence.

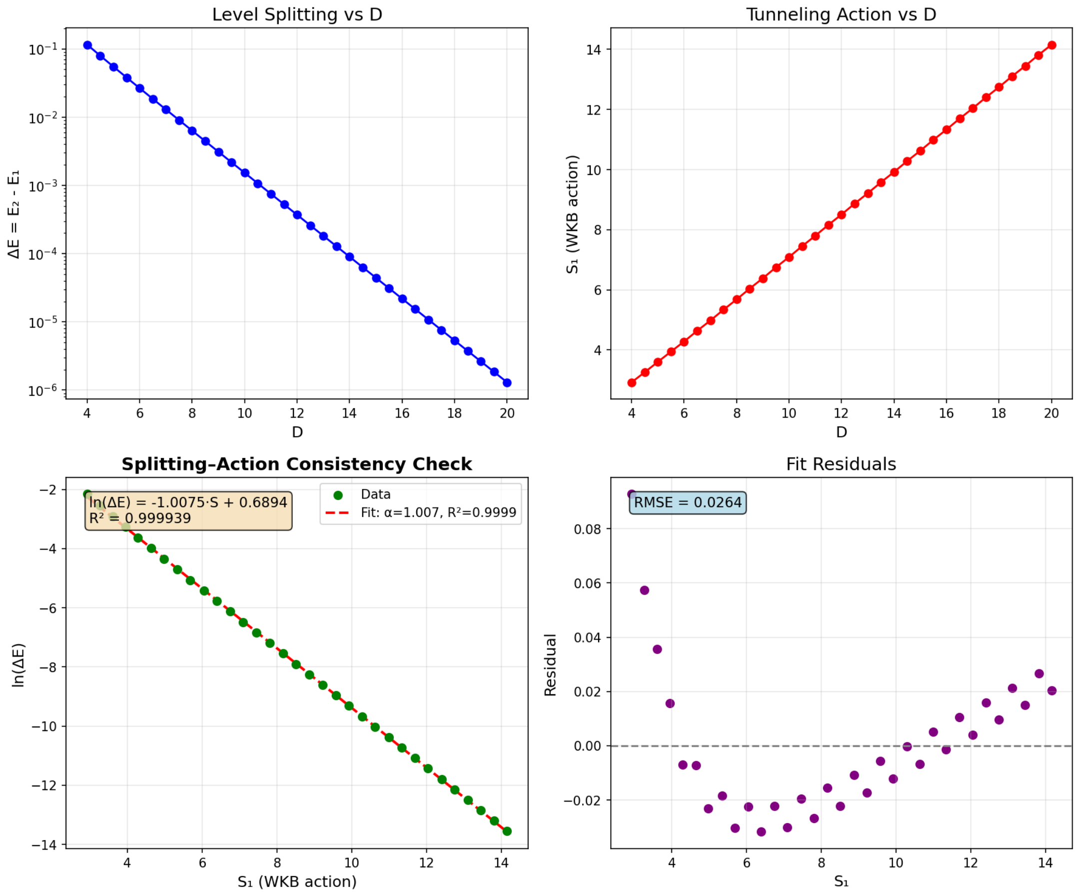


FIG. 9. Splitting–action consistency check. The level splitting $\Delta E = E_2 - E_1$ follows $\exp(-S_1)$ to $R^2 = 0.9999$, confirming the action-derived $V_{\text{eff}}$ governs both spectrum and tunneling.

### 8. Action-Derived Numerical Results

Table IX shows the key spectral and tunneling quantities from the action-derived calculation. Grid convergence: $|E_1|$ variation $< 0.2\%$ for $dz \in [0.01, 0.02]$ at fixed $z_{\max} = 80$.

### 9. Bound-State omega Convergence Benchmark

To make the surrogate-vs-exact distinction explicit, we report a direct bound-state benchmark for $\omega_N$ from the same action-derived 1D operator chain used in Table IX. The settings are

TABLE IX. Action-derived spectral and tunneling data. Parameters: $a = 0.04$, $\varepsilon = 0.1$, $m_0 = 1$, $\xi = 0.14$.

| $D$ | $E_1$ | $E_2$ | $\omega_1$ | $S_1$ | $\Delta E$ |
|---|---|---|---|---|---|
| 4 | $-0.505$ | $-0.390$ | 0.704 | 2.92 | $1.15 \times 10^{-1}$ |
| 8 | $-0.476$ | $-0.469$ | 0.724 | 5.68 | $6.32 \times 10^{-3}$ |
| 12 | $-0.478$ | $-0.478$ | 0.722 | 8.51 | $3.69 \times 10^{-4}$ |
| 16 | $-0.481$ | $-0.481$ | 0.721 | 11.33 | $2.19 \times 10^{-5}$ |
| 20 | $-0.482$ | $-0.482$ | 0.720 | 14.16 | $1.30 \times 10^{-6}$ |

fixed to the physical-gap parameter set ($a = 0.04$, $\varepsilon = 0.1$, $m_0 = 1$, $\xi = 0.14$), $z_{\max} = 80$, and three resolutions: coarse ($N_z = 4001$), mid ($N_z = 6001$), fine ($N_z = 8001$). The reproducible script is `code/extract_omega_exact_convergence.py`.

TABLE X. Bound-state $\omega_N$ convergence benchmark at $D = \{6, 12, 18\}$ from the action-derived operator chain. Relative errors are quoted versus fine grid.

| $D$ | level | $N_z$ | $\omega_1$ | $\omega_2$ | $\Delta\omega_{12}$ | max rel.$\omega$ vs fine |
|---|---|---|---|---|---|---|
| 6 | coarse | 4001 | 0.720883 | 0.738989 | $1.8107 \times 10^{-2}$ | $8.65 \times 10^{-4}$ |
| 6 | mid | 6001 | 0.721345 | 0.739468 | $1.8123 \times 10^{-2}$ | $2.18 \times 10^{-4}$ |
| 6 | fine | 8001 | 0.721500 | 0.739629 | $1.8129 \times 10^{-2}$ | 0 |
| 12 | coarse | 4001 | 0.721781 | 0.722035 | $2.5438 \times 10^{-4}$ | $8.84 \times 10^{-4}$ |
| 12 | mid | 6001 | 0.722258 | 0.722513 | $2.5533 \times 10^{-4}$ | $2.23 \times 10^{-4}$ |
| 12 | fine | 8001 | 0.722418 | 0.722674 | $2.5565 \times 10^{-4}$ | 0 |
| 18 | coarse | 4001 | 0.719404 | 0.719408 | $3.6700 \times 10^{-6}$ | $8.92 \times 10^{-4}$ |
| 18 | mid | 6001 | 0.719885 | 0.719888 | $3.6943 \times 10^{-6}$ | $2.25 \times 10^{-4}$ |
| 18 | fine | 8001 | 0.720046 | 0.720050 | $3.7025 \times 10^{-6}$ | 0 |

This benchmark is the operator-level reference for $\omega_N$ in the bound-state convention. In contrast, Appendix D uses generalized localized-extraction eigenvalues $\lambda$ for mixing-channel extraction; the two quantities serve different roles and are not identified.

### 10. Fixed-dz Convergence

We verified numerical stability using fixed grid spacing $dz$ (rather than fixed $N_z$):

- For $dz \in \{0.04, 0.02, 0.01\}$ and $z_{\max} \in \{60, 80\}$: $E_1$ variation $< 0.3\%$, $S_1$ variation $< 0.2\%$.
- Results are independent of $z_{\max}$ for $z_{\max} > 60$ (domain truncation error negligible).

### Appendix C: First-Principles Symmetry-Channel Test for chi

Using the explicit definition in Eq. (71), we computed $M_{12}$ and $\chi_N^{(\mathrm{sym})}$ on 2D axisymmetric grids for $D = \{6, 12, 18\}$ with coarse/mid/fine resolutions. The modes are normalized with the cylindrical measure $2\pi\rho\, d\rho\, dz$, and orthogonality satisfies $|\langle\psi_1, \psi_2\rangle| < 10^{-15}$ in all runs.

TABLE XI. Fine-grid symmetry-channel extraction using Eq. (71). Parameters: $a = 0.04$, $\varepsilon = 0.1$, $m_0 = 1$, $\xi = 0.14$.

| $D$ | $\lvert M_{12}^{(\mathrm{sym})}\rvert$ | $\chi_N^{(\mathrm{sym})}$ | $\lvert\langle\psi_1, \psi_2\rangle\rvert$ |
|---|---|---|---|
| 6 | $2.09 \times 10^{-16}$ | $1.66 \times 10^{-19}$ | $3.12 \times 10^{-17}$ |
| 12 | $6.03 \times 10^{-16}$ | $5.69 \times 10^{-19}$ | $3.47 \times 10^{-18}$ |
| 18 | $6.24 \times 10^{-16}$ | $6.20 \times 10^{-19}$ | $2.11 \times 10^{-16}$ |

These values are numerically consistent with symmetry-protected cancellation of the parity-symmetric overlap channel. Across coarse/mid/fine grids, $M_{12}^{(\mathrm{sym})}$ changes sign but remains at $\mathcal{O}(10^{-16})$, indicating no resolved nonzero mixing in this channel.

For this reason, the phenomenological Rank-2 coefficient in the main text is interpreted as a localized-channel effective coupling (Section VIII), not as $\chi_N^{(\mathrm{sym})}$ from Eq. (71).

### Appendix D: Localized-Channel First-Principles Extraction of chi

We compute the nonzero mixing channel directly in localized basis using

$$M_{LR}^{(H)} = \frac{\lambda_2 - \lambda_1}{2}, \qquad \chi_N^{(LR)} = \frac{|M_{LR}^{(H)}|}{\bar{\Gamma}_N}, \tag{D1}$$

with $\bar{\Gamma}_N$ from Eq. (33). Here $\lambda_{1,2}$ are generalized operator eigenvalues from $K\psi = \lambda M\psi$ in the localized extraction solver; they are not the bound-state energy convention $E = \omega^2 - m_0^2 < 0$ used in Appendix B. The extraction uses `code/extract_chi_localized_2d.py` with fixed settings:

- Domain: $\rho_{\max} = 3.0$, $z_{\max} = D/2 + 6.0$.
- Grids: coarse $(d\rho, dz) = (0.12, 0.06)$, mid $(0.08, 0.04)$, fine $(0.06, 0.03)$.
- Solver: generalized eigensystem (shift-invert low-mode targeting, $\sigma = 2.5$), tolerance $10^{-8}$, maxiter $3 \times 10^4$.
- Acceptance criteria: $\delta_{\Delta E} \equiv |\Delta E_{\mathrm{level}} - \Delta E_{\mathrm{fine}}|/|\Delta E_{\mathrm{fine}}| < 5\%$, $\delta_\chi \equiv |\chi_{\mathrm{level}} - \chi_{\mathrm{fine}}|/|\chi_{\mathrm{fine}}| < 5\%$, and null-channel absolute bound $|M_{12}^{(\mathrm{sym})}| < 10^{-12}$.

TABLE XII. Localized-channel extraction for $D = \{6, 12, 18\}$ (coarse/mid/fine). Parameters: $a = 0.04$, $\varepsilon = 0.1$, $m_0 = 1$, $\xi = 0.14$.

| $D$ | level | $(d\rho, dz)$ | $(N_\rho, N_z)$ | $\lambda_1$ | $\lambda_2$ | $M_{LR}^{(H)}$ | $\chi_N^{(LR)}$ | $\lvert M_{12}^{(\mathrm{sym})}\rvert$ |
|---|---|---|---|---|---|---|---|---|
| 6 | coarse | $(0.12, 0.06)$ | $(25, 300)$ | 2.14765 | 2.60073 | $2.26541 \times 10^{-1}$ | $4.13261 \times 10^{-4}$ | $5.29 \times 10^{-17}$ |
| 6 | mid | $(0.08, 0.04)$ | $(38, 450)$ | 2.14362 | 2.59798 | $2.27178 \times 10^{-1}$ | $4.17350 \times 10^{-4}$ | $2.44 \times 10^{-19}$ |
| 6 | fine | $(0.06, 0.03)$ | $(50, 600)$ | 2.16611 | 2.62106 | $2.27474 \times 10^{-1}$ | $4.01827 \times 10^{-4}$ | $1.56 \times 10^{-18}$ |
| 12 | coarse | $(0.12, 0.06)$ | $(25, 400)$ | 2.36032 | 2.71730 | $1.78492 \times 10^{-1}$ | $2.27261 \times 10^{-4}$ | $2.89 \times 10^{-17}$ |
| 12 | mid | $(0.08, 0.04)$ | $(38, 600)$ | 2.35598 | 2.71374 | $1.78883 \times 10^{-1}$ | $2.29384 \times 10^{-4}$ | $6.96 \times 10^{-17}$ |
| 12 | fine | $(0.06, 0.03)$ | $(50, 800)$ | 2.37821 | 2.73632 | $1.79054 \times 10^{-1}$ | $2.21414 \times 10^{-4}$ | $1.03 \times 10^{-17}$ |
| 18 | coarse | $(0.12, 0.06)$ | $(25, 500)$ | 2.22204 | 2.49466 | $1.36311 \times 10^{-1}$ | $2.18683 \times 10^{-4}$ | $1.94 \times 10^{-17}$ |
| 18 | mid | $(0.08, 0.04)$ | $(38, 750)$ | 2.21686 | 2.49001 | $1.36575 \times 10^{-1}$ | $2.21053 \times 10^{-4}$ | $1.54 \times 10^{-15}$ |
| 18 | fine | $(0.06, 0.03)$ | $(50, 1000)$ | 2.23864 | 2.51202 | $1.36694 \times 10^{-1}$ | $2.13187 \times 10^{-4}$ | $2.19 \times 10^{-16}$ |

The complete pipeline, scripts, and raw tables used here are available in the project repository: `https://github.com/boypatrick/PSLT`.

Across all non-fine runs in Table XII, we obtain

$$\max \delta_{\Delta E} = 4.10 \times 10^{-3}, \qquad \max \delta_\chi = 3.86 \times 10^{-2}, \qquad \max |M_{12}^{(\mathrm{sym})}| = 1.54 \times 10^{-15}, \tag{D2}$$

which satisfies the stated convergence and absolute-null criteria.

For interpolation support and reproducibility of scan-level diagnostics, we additionally ran a fine-grid-only auxiliary sweep on integer separations $D = 4, \ldots, 20$ using the same solver settings (`--full-scan` in `extract_chi_localized_2d.py`). These points are used as profile-support data in the repository, while the formal acceptance criteria above remain defined by the benchmark convergence set $D = \{6, 12, 18\}$.

Figure 10 shows the fine-grid 2D parity eigenmodes used in the localized extraction at $D = \{6, 12, 18\}$. The expected even/odd nodal structure is manifest and remains stable across the three benchmark separations.

To test profile uncertainty in the global map, we benchmarked six $\chi_N^{(LR)}(D)$ choices built from the same fine-grid knots $(D, \chi_N^{(LR)}) = \{(6, 4.02 \times 10^{-4}), (12, 2.21 \times 10^{-4}), (18, 2.13 \times 10^{-4})\}$: linear interpolation, log-linear exponential fit $\ln \chi = -7.5956 - 0.05282\, D$, and $\pm 20\%$ amplitude rescalings of each profile. Re-running the full $60 \times 60$ scan keeps the generation fractions fixed and yields a narrow $H \to \mu\mu$ acceptance spread:

$$f(\mathcal{R}_3 > 0.90) = 0.932, \quad f(\mathcal{R}_3 > 0.95) = 0.217, \quad f(\chi^2_{\mu\mu} < 4) \in [0.1625,\, 0.1681]. \tag{D3}$$

The best-fit point moves within $(D, \eta) \approx (8.07\text{–}9.42,\ 1.29\text{–}3.10)$ across these six profiles. Thus, at current scan resolution, profile-interpolation uncertainty is subdominant for $\mathcal{R}_3$ and controlled

TABLE XIII. Auxiliary fine-grid localized profile points from the full-scan run ($D = 4$–20). Source CSV: `output/chi_fp_2d/localized_chi_D4-5-...-20.csv`.

| $D$ | $\lambda_1$ | $\lambda_2$ | $\chi_N^{(LR)}$ |
|---|---|---|---|
| 4 | 2.06671 | 2.56696 | $5.266 \times 10^{-4}$ |
| 5 | 2.34635 | 2.85471 | $3.311 \times 10^{-4}$ |
| 6 | 2.16611 | 2.62106 | $4.018 \times 10^{-4}$ |
| 7 | 2.42273 | 2.88524 | $2.661 \times 10^{-4}$ |
| 8 | 2.24778 | 2.66479 | $3.202 \times 10^{-4}$ |
| 9 | 2.10111 | 2.48071 | $3.758 \times 10^{-4}$ |
| 10 | 2.32055 | 2.70644 | $2.623 \times 10^{-4}$ |
| 11 | 2.17620 | 2.52926 | $3.064 \times 10^{-4}$ |
| 12 | 2.37821 | 2.73632 | $2.214 \times 10^{-4}$ |
| 13 | 2.24440 | 2.57510 | $2.554 \times 10^{-4}$ |
| 14 | 2.42772 | 2.76173 | $1.906 \times 10^{-4}$ |
| 15 | 2.30010 | 2.61030 | $2.181 \times 10^{-4}$ |
| 16 | 2.47457 | 2.78817 | $1.661 \times 10^{-4}$ |
| 17 | 2.34884 | 2.64089 | $1.895 \times 10^{-4}$ |
| 18 | 2.23864 | 2.51202 | $2.132 \times 10^{-4}$ |
| 19 | 2.39519 | 2.67161 | $1.663 \times 10^{-4}$ |
| 20 | 2.28581 | 2.54509 | $1.867 \times 10^{-4}$ |

TABLE XIV. Extended profile-sensitivity test for $\chi_N^{(LR)}(D)$ in the $60 \times 60$ global scan. Source data file: `paper/chi_profile_robustness.csv`.

| Profile | $f(\mathcal{R}_3 > 0.90)$ | $f(\mathcal{R}_3 > 0.95)$ | $f(\chi^2_{\mu\mu} < 4)$ | Best $(D, \eta)$ |
|---|---|---|---|---|
| Linear interpolation | 0.932 | 0.217 | 0.168 | (9.42, 1.29) |
| Exponential fit | 0.932 | 0.217 | 0.163 | (9.15, 1.94) |
| Linear $\times 0.8$ | 0.932 | 0.217 | 0.167 | (8.61, 3.03) |
| Linear $\times 1.2$ | 0.932 | 0.217 | 0.168 | (8.07, 2.26) |
| Exp-fit $\times 0.8$ | 0.932 | 0.217 | 0.163 | (8.61, 3.10) |
| Exp-fit $\times 1.2$ | 0.932 | 0.217 | 0.163 | (8.88, 2.13) |

for the $H \to \mu\mu$ acceptance metric.

As a stronger stress test, we further rescaled the localized profile by large factors and tracked boundary movement directly on the $(D, \eta)$ grid. Figure 11 shows that, at the present $60 \times 60$ scan resolution, the $\mathcal{R}_3 > 0.90$ mask is stable up to $\times 10^4$ and first shifts at $\times 10^5$, while the $H \to \mu\mu$ acceptance mask already shifts for the user-requested $\times 0.5$ and $\times 2$ cases (changed-cell fractions 0.0331 and 0.0258, respectively).

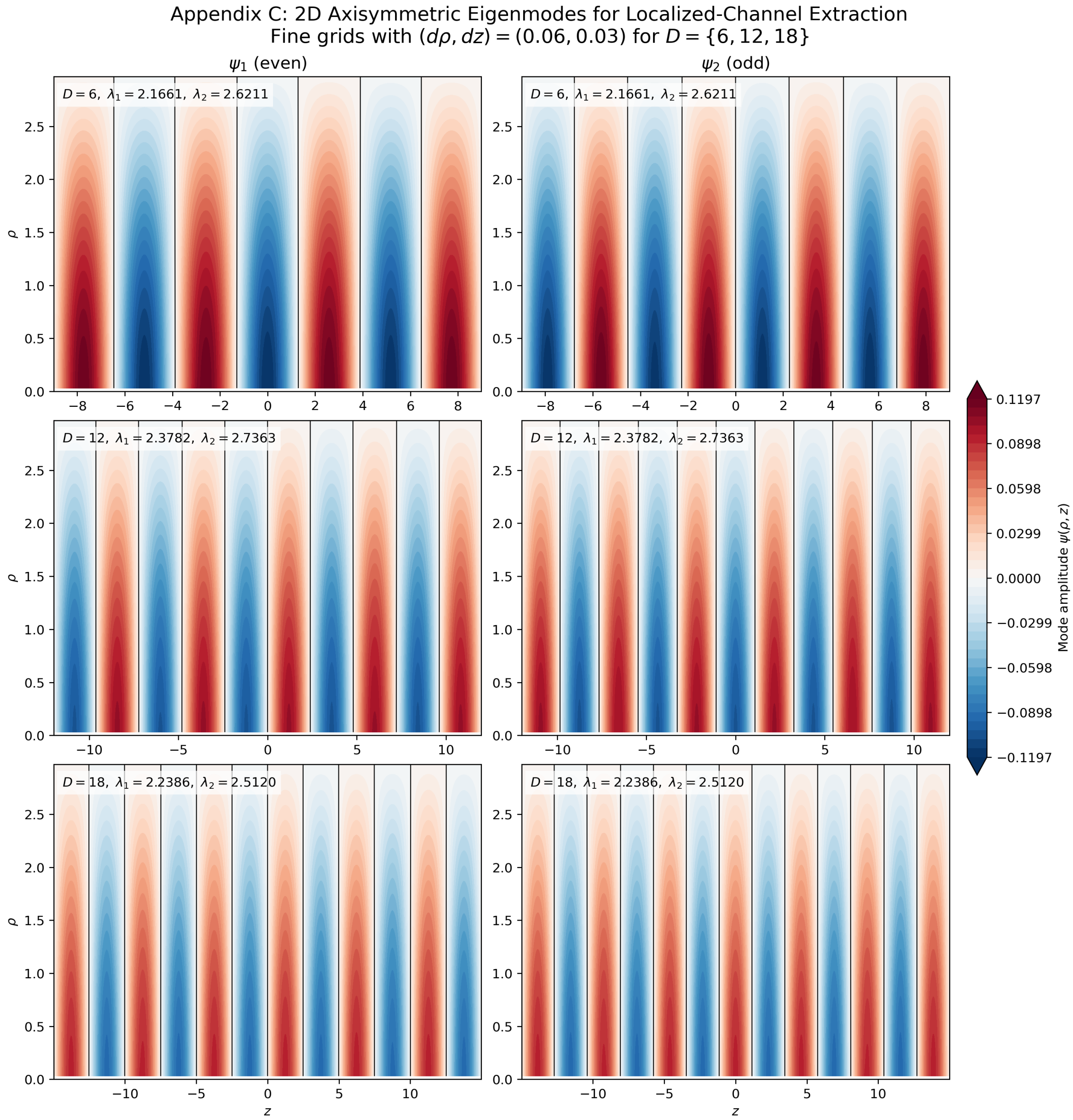


FIG. 10. Fine-grid 2D axisymmetric eigenmodes used in Appendix D. Each row corresponds to one benchmark separation ($D = 6, 12, 18$); columns show the lowest even/odd parity modes ($\psi_1, \psi_2$) entering the localized-channel extraction pipeline.

## Appendix E: Preliminary First-Principles Replacement Candidate Checks

### 1. 1D Phase-Space Candidate for $g_N$

To test a first-principles replacement direction for Eq. (18), we evaluated the semiclassical 1D candidate on the same action-derived axis potential:

$$\rho_{\mathrm{WKB}}(E; D) = \frac{1}{2\pi} \int_{U(z;D)<E} \frac{\mathrm{d}z}{\sqrt{E - U(z; D)}}, \tag{E1}$$

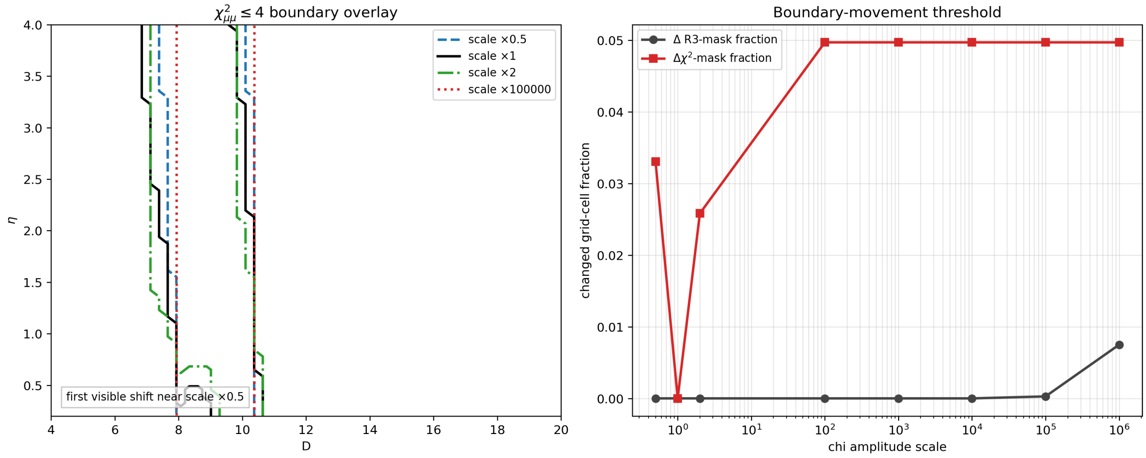


FIG. 11. Strong $\chi$-amplitude stress test on the global map. Left: overlay of $\chi^2_{\mu\mu} \leq 4$ boundary for selected scales, including the requested $\times 0.5$ and $\times 2$. Right: changed-cell fractions versus scale (relative to baseline $\times 1$), showing early movement for the $H \to \mu\mu$ mask and delayed movement for the $\mathcal{R}_3$ mask. Source data file: `output/chi_fp_2d/chi_scale_stress_test.csv`.

TABLE XV. Preliminary 1D phase-space candidate $g_N^{(\mathrm{ps})}$ at $D = 12$. Source CSV: `output/gn_fp_1d/gn_phase_space_candidate_D12.csv`.

| level | $dz$ | $E_1$ | $E_2$ | $E_3$ | $g_1^{(\mathrm{ps})}$ | $g_2^{(\mathrm{ps})}$ | max rel.$g$ vs fine |
|---|---|---|---|---|---|---|---|
| coarse | 0.04 | −0.636696 | −0.608074 | −0.575010 | 1.056725 | 1.054421 | $1.31 \times 10^{-2}$ |
| mid | 0.02 | −0.637061 | −0.608419 | −0.575335 | 1.066813 | 1.074099 | $5.30 \times 10^{-3}$ |
| fine | 0.01 | −0.637233 | −0.608582 | −0.575488 | 1.063919 | 1.068433 | 0 |

$$g_N^{(\mathrm{ps})}(D) = 1 + \int_{E_{\min}(D)}^{E_N(D)} \rho_{\mathrm{WKB}}(E'; D)\, \mathrm{d}E', \qquad E_{\min}(D) \equiv \min_z U(z; D). \tag{E2}$$

Using the reproducible script `extract_gn_phase_space_candidate.py` (in the `code/` directory), we ran a coarse/mid/fine test at $D = 12$ (Table XV). The candidate is numerically stable at the few-$10^{-3}$ level versus fine grid.

### 2. 2D Axisymmetric Benchmark for $g_N^{(\mathrm{ps})}$

To reduce projection ambiguity, we also ran a 2D axisymmetric benchmark on $D = \{6, 12, 18\}$ using the same localized extraction chain as Appendix D: identical generalized operator build, identical box/range convention, and the same shift-invert low-mode settings ($\sigma = 2.5$, tol $= 10^{-8}$, maxiter $= 3 \times 10^4$). We define

$$N_{\mathrm{ps}}(E) = \frac{1}{4\pi} \int \mathrm{d}^2x\, \mathrm{d}^2p\, \Theta(E - U - p^2) = \frac{1}{2} \int \rho\, \mathrm{d}\rho\, \mathrm{d}z\, [E - U(\rho, z)]_+, \tag{E3}$$

TABLE XVI. Fine-grid 2D phase-space profile candidate at $D = \{6, 12, 18\}$. Source CSV: `output/gn_fp_2d/gn_phase_space_2d_D6-12-18.csv`.

| $D$ | $\lambda_1$ | $\lambda_2$ | $\lambda_3$ | $\hat{g}_1$ | $\hat{g}_2$ |
|---|---|---|---|---|---|
| 6 | 0.719476 | 0.804126 | 0.945018 | $9.867 \times 10^{-2}$ | $4.370 \times 10^{-1}$ |
| 12 | 0.686186 | 0.747510 | 0.822830 | $1.193 \times 10^{-1}$ | $5.146 \times 10^{-1}$ |
| 18 | 0.667848 | 0.713807 | 0.767832 | $1.291 \times 10^{-1}$ | $5.294 \times 10^{-1}$ |

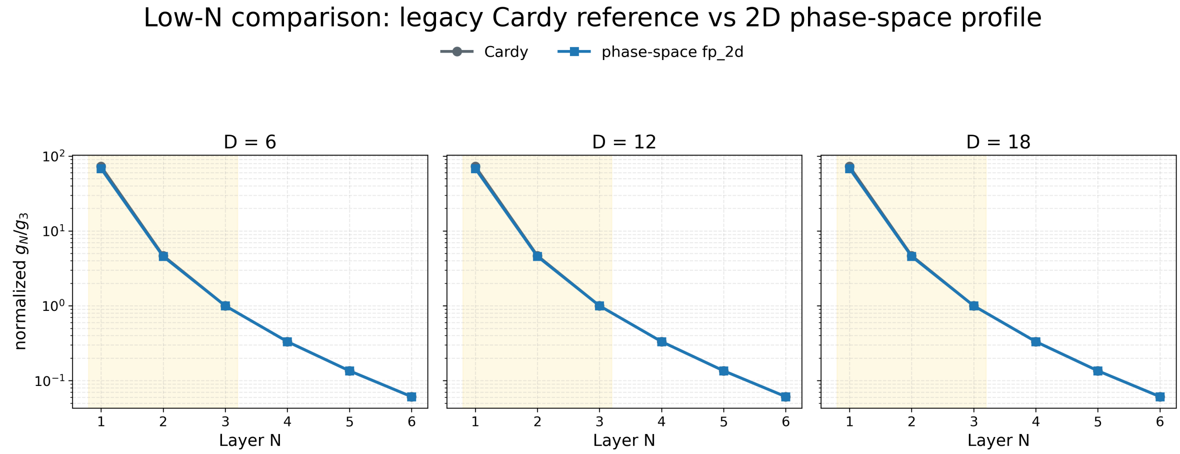


FIG. 12. Low-$N$ comparison between legacy Cardy reference and 2D phase-space profile on $D = \{6, 12, 18\}$, shown as normalized ratios $g_N/g_3$ (log scale). The highlighted band marks $N = 1, 2, 3$. Source files: `paper/gn_cardy_vs_phase_space.png`, `paper/gn_cardy_vs_phase_space.csv`.

$$\mathcal{W}_N \equiv [\lambda_1, \lambda_N], \qquad g^{(\mathrm{ps})}_{N,\mathrm{raw}} = 1 + \int_{\mathcal{W}_N} \rho_{\mathrm{ps}}(E')\, \mathrm{d}E' = 1 + N_{\mathrm{ps}}(\lambda_N) - N_{\mathrm{ps}}(\lambda_1), \qquad \hat{g}_N \equiv \frac{g^{(\mathrm{ps})}_{N,\mathrm{raw}}}{g^{(\mathrm{ps})}_{3,\mathrm{raw}}}. \tag{E4}$$

The script `extract_gn_phase_space_2d.py` (in `code/`) exports both the low-$N$ profile table and the low-mode spectral table with explicit window bounds (source CSVs: `output/gn_fp_2d/gn_phase_space_2d_` `output/gn_fp_2d/gn_phase_space_2d_spectrum_D6-12-18.csv`). For non-fine levels, the maximum profile deviation is

$$\max_{\text{non-fine}} \left( \max_N \frac{|\hat{g}_N - \hat{g}^{(\text{fine})}_N|}{|\hat{g}^{(\text{fine})}_N|} \right) = 2.35 \times 10^{-2}. \tag{E5}$$

As in the 1D candidate check, this benchmark is reported as a replacement candidate. To test map-level impact, we connected both candidates to the main scan through `g_mode` choices (`fp_2d_full`, `cardy`, `fp_1d`, `fp_2d`) and the reproducible script `scan_gn_profile_impact.py`. Results in Table XVII show that the main conclusion is retained: relative to baseline `fp_2d_full`, the largest absolute shift is 0.0669 in $f(\mathcal{R}_3 > 0.90)$ (for `fp_2d`), while high-$N$ runaway remains controlled with $f(N_{\mathrm{win}} > 3) \simeq 2.78 \times 10^{-4}$ in all tested modes.

Across $N_{\max} = 20, 30, 40$, both baseline and fp_2d scenarios are numerically stable at map

TABLE XVII. Map-level impact of first-principles $g_N$ profiles on the $60 \times 60$ scan (baseline is `fp_2d_full`; legacy Cardy retained as comparator). Source CSV: `paper/gn_profile_impact.csv`.

| case | $f(\mathcal{R}_3 > 0.90)$ | $f(\chi^2_{\mu\mu} < 4)$ | $f(N_{\rm win} > 3)$ | mean tail prob. |
|---|---|---|---|---|
| baseline fp_2d_full | 0.927 | 0.1500 | 0.0003 | 0.1267 |
| legacy Cardy reference | 0.877 | 0.1486 | 0.0003 | 0.1125 |
| $g_N$ fp_1d profile | 0.860 | 0.1486 | 0.0003 | 0.1137 |
| $g_N$ fp_2d profile | 0.860 | 0.1492 | 0.0003 | 0.1144 |

TABLE XVIII. $N_{\rm max}$ convergence check for baseline and $g_N$ fp_2d map-level summaries. Source CSV: `paper/gn_nmax_convergence.csv`.

| case | $N_{\rm max}$ | $f(\mathcal{R}_3 > 0.90)$ | $f(\chi^2_{\mu\mu} < 4)$ | $f(N_{\rm win} > 3)$ | mean tail prob. |
|---|---|---|---|---|---|
| baseline fp_2d_full | 20 | 0.927 | 0.1500 | 0.0003 | 0.126709 |
| baseline fp_2d_full | 30 | 0.927 | 0.1500 | 0.0003 | 0.126709 |
| baseline fp_2d_full | 40 | 0.927 | 0.1500 | 0.0003 | 0.126709 |
| $g_N$ fp_2d profile | 20 | 0.860 | 0.1492 | 0.0003 | 0.114352 |
| $g_N$ fp_2d profile | 30 | 0.860 | 0.1492 | 0.0003 | 0.114352 |
| $g_N$ fp_2d profile | 40 | 0.860 | 0.1492 | 0.0003 | 0.114352 |

level: the key fractions $f(\mathcal{R}_3 > 0.90)$, $f(\chi^2_{\mu\mu} < 4)$, and $f(N_{\rm win} > 3)$ remain unchanged in this scan.

### 3. Baseline Cross-Check Under *g_mode* = fp_2d_full

To implement a strict same-framework upgrade, we set `g_mode=fp_2d_full` as the baseline. Instead of direct low-$N$ substitution plus geometric continuation, this mode uses an explicit bounded microcanonical window for $N \le 3$ and an energy-gap tail prescription for $N > 3$:

$$g_N^{(\rm full)}(D) = g_3^{(\rm ps)}(D)\hat{g}_N(D), \quad \hat{g}_n = \left(\hat{g}_n^{(\rm dir)}\right)^{1-\alpha}\left(\hat{g}_n^{(\rm win)}\right)^{\alpha} \ (n = 1, 2, 3), \tag{E6}$$

with

$$\hat{g}_1^{(\rm win)} = 1 + N_{\rm ps}(\lambda_3) - N_{\rm ps}(\lambda_1),\ \hat{g}_2^{(\rm win)} = 1 + N_{\rm ps}(\lambda_3) - N_{\rm ps}(\lambda_2),\ \hat{g}_3^{(\rm win)} = 1, \tag{E7}$$

with $g_3^{(\rm ps)}(D) \equiv 1 + N_{\rm ps}(\lambda_3) - N_{\rm ps}(\lambda_1)$ from the same 2D spectrum table. For $N > 3$,

$$\hat{g}_N^{(\rm target)} = \left(\frac{\Delta N_{\rm ps}^{(N)}}{\Delta N_{\rm ps}^{(3)}}\right)^s \exp\left[-\beta\frac{\lambda_N - \lambda_3}{\lambda_3 - \lambda_2}\right], \tag{E8}$$

followed by the same per-step finite-volume clipping used in `pslt_lib.py`. In the present baseline run, we use $(\alpha, \beta, s) = (0.8, 1.1, 0.0)$. Results are summarized in Table XIX (source CSV:

TABLE XIX. Map-level baseline cross-check for $g_N$: `fp_2d_full` baseline versus legacy Cardy reference on the same $60 \times 60$ grid. Source CSV: `paper/gn_baseline_replacement.csv`.

| case | $f(\mathcal{R}_3 > 0.90)$ | $f(\chi^2_{\mu\mu} < 4)$ | $f(N_{\rm win} > 3)$ | $\Delta f(\mathcal{R}_3 > 0.90)$ vs baseline |
|---|---|---|---|---|
| baseline fp_2d_full | 0.9269 | 0.1500 | 0.0003 | 0.0000 |
| legacy Cardy reference | 0.8769 | 0.1486 | 0.0003 | -0.0500 |

`paper/gn_baseline_replacement.csv`). Relative to baseline `fp_2d_full`, the legacy Cardy reference lowers $f(\mathcal{R}_3 > 0.90)$ by 0.0500 (from 0.9269 to 0.8769), lowers $f(\chi^2_{\mu\mu} < 4)$ by 0.0014 (from 0.1500 to 0.1486), and keeps $f(N_{\rm win} > 3) \simeq 2.78 \times 10^{-4}$. Therefore, the map-level main conclusion is retained under this baseline switch.

Across $N_{\rm max} = 20, 30, 40$, both rows in Table XIX are numerically stable in this setup (source CSV: `paper/gn_baseline_replacement_nmax.csv`).

#### 4. Geometry-Informed and Microscopic Lindblad Scan-Ready Module for Open-System $\chi$

As a methodological bridge (not a replacement of Appendix D), we consider a two-level open-system model in the standard GKSL/Lindblad form [10, 11]:

$$\dot{\rho} = -i[H, \rho] + \gamma_\phi \mathcal{D}[\sigma_z]\rho + \gamma_{\rm mix}\mathcal{D}[\sigma_x]\rho, \qquad H = \begin{pmatrix} 0 & \Delta/2 \\ \Delta/2 & 0 \end{pmatrix}. \tag{E9}$$

As external motivation for this diagnostic direction, recent entanglement-based flavor analyses report that minimizing scattering-generated entanglement can qualitatively reproduce observed CKM/PMNS patterns [12]. In this manuscript we use that connection only as qualitative support for coherence-sensitive diagnostics, not as a baseline closure assumption. For this module we set $\gamma_{\rm mix}(D) \equiv M^{(H)}_{LR}(D)$ from Appendix D, and define a geometry-informed dephasing proxy

$$\gamma_\phi(D) \equiv \sqrt{\left\langle (\delta V)^2 \right\rangle_\rho}, \qquad \delta V = V_{\rm eff} - \bar{V}_{\rm eff}(r), \tag{E10}$$

with cylindrical weighting $\langle \cdots \rangle_\rho$. The minimal Lindblad evolution is generated by `lindblad_chi_minimal.py`, and geometry-profile extraction is handled by `extract_chi_open_system_geometry.py`. These scripts output reproducible diagnostics such as

$$C_{\rm max} \equiv \max_t |\rho_{LR}(t)|, \qquad P_{\rm max} \equiv \max_t \rho_{RR}(t), \tag{E11}$$

and a normalized proxy $\chi^{(\rm proxy)}_{\rm eff} = 2\gamma_{\rm mix} C_{\rm max}/\bar{\Gamma}_N$. Results for $D = \{6, 12, 18\}$ are summarized in Table XX. Extending the same profile to $D = 4, \ldots, 20$ gives a stable geometry-informed ratio band $\chi_{\rm open}/\chi_{LR} \in [0.0852, 0.2127]$ with mean 0.1269 (source CSV: `output/chi_open_system/chi_open_ratio_`

This value belongs to the geometry-only proxy closure in this subsection; the anchor-calibrated microscopic band is reported separately in Eq. (E13) and Table XXI.

To move this module toward microscopic closure, we also extract localized-basis couplings $(g_z, g_x)$ directly from the same 2D eigenmodes and set

$$\gamma_\phi = \kappa_{\rm env}\, g_z^2 S(0), \qquad \gamma_{\rm mix} = \kappa_{\rm env}\, g_x^2 S(\Delta E), \qquad S(\omega) = \frac{2\tau_{\rm env}}{1 + (\omega\tau_{\rm env})^2}, \tag{E12}$$

via `extract_chi_open_system_micro.py`. To avoid map-level fitting, we calibrate $\kappa_{\rm env}$ with a multi-anchor rule (`calibrate_kappa_env_micro_anchor.py`): least-squares match to geometry-profile ratios on anchors $D = \{6, 9, 12, 15, 18\}$, then validate on holdout $D = \{4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20\}$. This gives $\kappa_{\rm env} = 4.31 \times 10^5$ with anchor RMSE $8.61 \times 10^{-2}$ and holdout RMSE $6.95 \times 10^{-2}$ (holdout max-abs error $1.25 \times 10^{-1}$; source CSVs: `output/chi_open_system/kappa_env_anchor_calibration.csv`, `output/chi_open_system/kappa_env_ancho`
The corresponding scan-level band is

$$\chi_{\rm open}^{\rm (micro)}/\chi_{LR} \in [0.0173, 0.3271], \qquad \langle \chi_{\rm open}^{\rm (micro)}/\chi_{LR} \rangle = 0.1380, \tag{E13}$$

from `output/chi_open_system/chi_open_micro_ratio_band.csv`. To pin down the microscopic bookkeeping, we additionally export a bridge audit with `scan_chi_open_system_micro_bridge.py`, writing `paper/chi_open_system_micro_bridge_map.csv`, `paper/chi_open_system_micro_bridge_summary.cs` and `paper/chi_open_system_micro_bridge.png`. On the calibrated $D = 4, \ldots, 20$ knot set, this audit reconstructs

$$\gamma_\phi = \kappa_{\rm env} g_z^2 S(0), \qquad \gamma_{\rm mix} = \kappa_{\rm env} g_x^2 S(\Delta E), \qquad \chi_{\rm eff}^{\rm (micro)} = 2\gamma_{\rm mix} C_{\rm max}/\Gamma_{\rm ref}$$

with numerically negligible residuals:

$$\max|\Delta\gamma_\phi| = 2.03 \times 10^{-20}, \quad \max|\Delta\gamma_{\rm mix}| = 4.40 \times 10^{-14}, \quad \max|\Delta\chi_{\rm eff}^{\rm (loader)}| = 5.58 \times 10^{-17}.$$

The corresponding bridge summary identifies the effective system as the localized two-level $(L/R)$ subspace, the current bath witness as a Gaussian Markov bath with Lorentzian PSD, and the coupling operators as the diagonal/off-diagonal pair $(\sigma_z, \sigma_x)$. This means the open-system gap is no longer in the system-to-Lindblad bookkeeping, but in the microscopic EYMH origin and normalization of the bath itself.

We can therefore push the bath normalization one step further without over-claiming a full microscopic derivation. Using `scan_chi_open_system_micro_kappa_window.py`, we treat

$$\kappa_{\rm env} \to \kappa_{\rm scale}\, \kappa_{\rm env}^{\rm (calib)}$$

as a uniform bath-normalization susceptibility that rescales both $\gamma_\phi$ and $\gamma_{\rm mix}$ while leaving the localized two-level Hamiltonian $(\Delta E, \sigma_x)$ fixed. The resulting audit writes `paper/chi_open_system_micro_kapp`

`paper/chi_open_system_micro_kappa_window_summary.csv`, and `paper/chi_open_system_micro_kappa_wind`
On the current $D = 4, \ldots, 20$ knot set, the calibration-consistent candidate window is

$$\kappa_{\rm scale} \in [0.5, 1.0],$$

while a broader scan-stable window

$$\kappa_{\rm scale} \in [0.25, 1.5]$$

keeps the map-level fractions $f(\mathcal{R}_3 > 0.90)$ and $f(\chi^2_{\mu\mu} \leq 4)$ unchanged and only relaxes the holdout tolerance slightly. In this sense, $\kappa_{\rm env}$ is no longer just a single-point fit parameter: within the present witness it acts as a bounded bath-normalization susceptibility. What still remains open is the EYMH-side derivation of its absolute normalization and of the microscopic bath degrees of freedom themselves.

The newer parent-bath audits sharpen this statement. First, `scan_chi_open_system_bath_factorization.p` and `scan_chi_open_system_kappa_absolute_audit.py` show that the witness already factorizes into a system block $(g_z^2, g_x^2)$, a bath-shape block $(S_{zz}(0), S_{xx}(\Delta E))$, and an amplitude block $\kappa_{\rm env}$, with the canonical constant normalization remaining the unique exact amplitude choice. Second, `scan_chi_open_system_parent_bath_statement.py` rewrites the rates as the projected bath block

$$K_{\rm bath} = \kappa_{\rm env} \sqrt{K_{\rm sys}}\, K_{\rm spec}\, \sqrt{K_{\rm sys}},$$

and the subsequent family, log-coordinate, normal-coordinate, and generator-affinity audits show that the canonical projected bath class is uniquely selected on the current knot set: $(m, u, v) = (0, 0, 0)$, $(p_{\rm sys}, p_{\rm spec}) = (0, 0)$, $(\zeta_{\rm sys}, \zeta_{\rm spec}) = (0, 0)$, and $(q_{ss}, q_{bb}, q_{sb}) = (0, 0, 0)$. Thus the remaining open-system gap is no longer bookkeeping or nearby-family ambiguity; it is the final parent-action statement for why the projected bath generator itself naturally lives in this affine log class. Finally, `scan_chi_open_system_parent_bath_cocycle_audit.py` turns this into a positive integrability statement: after dividing out $\kappa_{\rm env}$, the normalized parent bath block defines an exact additive cocycle in the canonical log variables, with pairwise cocycle residuals at $8.95 \times 10^{-16}$ on the identifiable $\phi$ subset and $8.64 \times 10^{-13}$ on the mixing branch, while triangle flatness defects stay at $1.11 \times 10^{-15}$ in both channels. So the affine log-generator is now supported not only by nearby-family exclusion, but also by exact cocycle/flatness closure. Pushing one step further, `scan_chi_open_system_parent_bath_potential_audit.py` shows that this cocycle is globally integrable on the current knot set: the normalized bath block admits an exact anchored scalar potential $\Phi_a = \log(B_a/B_a^{\rm ref})$ whose direct decomposition, anchor-average recovery, and nearest-neighbor chain recovery agree to machine precision (max residuals $\sim 10^{-14}$

TABLE XX. Geometry-informed open-system demonstrator at $D = \{6, 12, 18\}$. Source CSV: `output/chi_open_system/chi_open_system_geometry_D6-12-18.csv`.

| $D$ | $\gamma_\phi$ | $\gamma_{\rm mix}$ | $C_{\rm max}$ | $\chi_{\rm eff}^{\rm (proxy)}$ | $\chi_N^{(LR)}$ | ratio |
|---|---|---|---|---|---|---|
| 6 | 0.656901 | 0.227474 | 0.077658 | $6.24 \times 10^{-5}$ | $4.02 \times 10^{-4}$ | 0.155 |
| 12 | 0.744389 | 0.179054 | 0.063765 | $2.82 \times 10^{-5}$ | $2.21 \times 10^{-4}$ | 0.128 |
| 18 | 0.805854 | 0.136694 | 0.051381 | $2.19 \times 10^{-5}$ | $2.13 \times 10^{-4}$ | 0.103 |

TABLE XXI. Microscopic open-system sensitivity scan with profiled $(\gamma_\phi, \gamma_{\rm mix})$ on the $60 \times 60$ map. Source CSV: `paper/chi_open_system_micro_sensitivity.csv`.

| case | ratio min | ratio max | ratio mean |
|---|---|---|---|
| baseline localized | 1.000 | 1.000 | 1.000 |
| open-micro base $(1, 1)$ | 0.017 | 0.327 | 0.138 |
| $\phi \times 0.5$, mix×1 | 0.017 | 0.327 | 0.138 |
| $\phi \times 2$, mix×1 | 0.017 | 0.327 | 0.138 |
| $\phi \times 1$, mix×0.5 | 0.009 | 0.220 | 0.082 |
| $\phi \times 1$, mix×2 | 0.034 | 0.412 | 0.208 |

on the identifiable $\phi$ subset and $\sim 5 \times 10^{-13}$ or better on the mixing branch). Thus the projected bath witness is no longer characterized only by an affine local generator; it also carries a single-valued anchored potential, so the remaining gap is specifically the parent-action origin of this potential itself.

Across all sensitivity cases in Table XXI, the generation metrics remain unchanged on this grid ($f(\mathcal{R}_3 > 0.90) = 0.9269$, $f(\mathcal{R}_3 > 0.95) = 0.2167$, $f(N_{\rm win} > 3) \simeq 2.78 \times 10^{-4}$), while the $H \to \mu\mu$ acceptance fraction remains fixed at $f(\chi^2_{\mu\mu} < 4) = 0.1500$ in this setup. These values are model-dependent because the Lindblad structure is still an effective environment ansatz and not yet a microscopic EYMH bath derivation; therefore, `chi_mode=open_system_micro` is treated as scan-ready diagnostic only and is not propagated into baseline reported figures. Under the explicit gate file `output/chi_open_system/open_system_micro_baseline_candidate.csv`, this setup is marked as `GO_baseline_candidate` under multi-anchor + holdout gating (including holdout RMSE/max-abs thresholds), but we keep it outside baseline until the bath-side microscopic derivation is completed.

### 5. Cross-Module Migration Snapshot

To compare the two main migration directions on the same map-level metrics under the updated baseline, we aggregate three scenarios: baseline (`fp_2d_full + localized_grid`), legacy

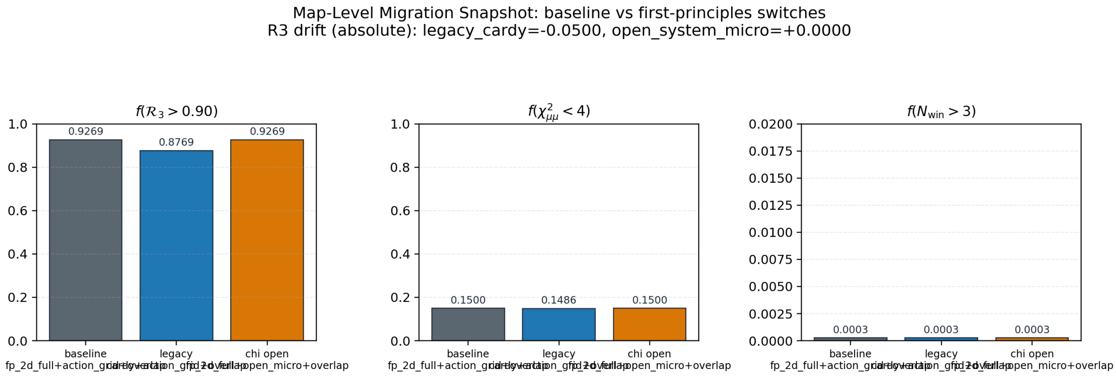


FIG. 13. Map-level migration snapshot across three scenarios. Left to right: $f(\mathcal{R}_3 > 0.90)$, $f(\chi^2_{\mu\mu} < 4)$, and $f(N_{\mathrm{win}} > 3)$.

TABLE XXII. Cross-module migration comparison from `paper/first_principles_migration_summary.csv`.

| scenario | $f(\mathcal{R}_3 > 0.90)$ | $\Delta f(\mathcal{R}_3 > 0.90)$ | $f(\chi^2_{\mu\mu} < 4)$ | $f(N_{\mathrm{win}} > 3)$ |
|---|---|---|---|---|
| baseline (`fp_2d_full + localized_grid`) | 0.927 | 0.0000 | 0.1500 | 0.0003 |
| legacy reference (`cardy + localized_grid`) | 0.877 | -0.0500 | 0.1486 | 0.0003 |
| $\chi$ migration (`fp_2d_full + open_system_micro`) | 0.927 | 0.0000 | 0.1500 | 0.0003 |

reference (`cardy + localized_grid`), and open-system $\chi$ in microscopic mode (`fp_2d_full + open_system_micro`). The reproducible script is `scan_first_principles_migration_summary.py`; outputs are `paper/first_principles_migration_summary.csv` and Fig. 13.

### 6. Reference-Anchor Sensitivity and Dynamic Selection Rule

Because Eq. (56) is normalized to a reference point $(D_0, \eta_0)$, we explicitly audit anchor dependence with two reproducible scripts: `select_reference_anchor.py` and `scan_reference_anchor_sensitivity`
The workflow is:

1. build a fixed-reference baseline map at $(D_0, \eta_0) = (10, 1)$ on the same $60 \times 60$ grid;
2. extract three candidate anchors: fixed-grid point, $\chi^2$-best [Eq. (63)], and robust-center [Eq. (64)];
3. re-anchor Eq. (56) with each candidate and recompute map summaries.

Table XXIII summarizes the selector outputs, and Fig. 14 shows the full grid-level sensitivity maps.

Across the tested reference-anchor grid $D_0 \in \{7, \ldots, 13\}$ and $\eta_0 \in \{0.6, 0.9, \ldots, 2.4\}$, we find

$$f(\chi^2_{\mu\mu} < 4) \in [0.0333,\, 0.2411], \qquad \chi^2_{\min} \in [1.52 \times 10^{-5},\, 7.47 \times 10^{-1}]. \tag{E14}$$

TABLE XXIII. Reference-anchor candidates and re-anchored map metrics. Source CSV: `paper/reference_anchor_candidates.csv`; selector JSON: `paper/reference_anchor_choice.json`.

| mode | $(D_0, \eta_0)$ | $\chi^2$ at selector point | $f(\chi^2_{\mu\mu} < 4)$ (re-anchored) | best $\chi^2$ (re-anchored) |
|---|---|---|---|---|
| fixed-grid | (9.97, 0.97) | $8.83 \times 10^{-1}$ | 0.1500 | $1.18 \times 10^{-3}$ |
| $\chi^2$-best | (4.00, 0.264) | $2.21 \times 10^{-5}$ | 0.1333 | $1.90 \times 10^{-1}$ |
| robust-center | (9.42, 3.87) | $1.39 \times 10^{-3}$ | 0.1381 | $4.74 \times 10^{-3}$ |

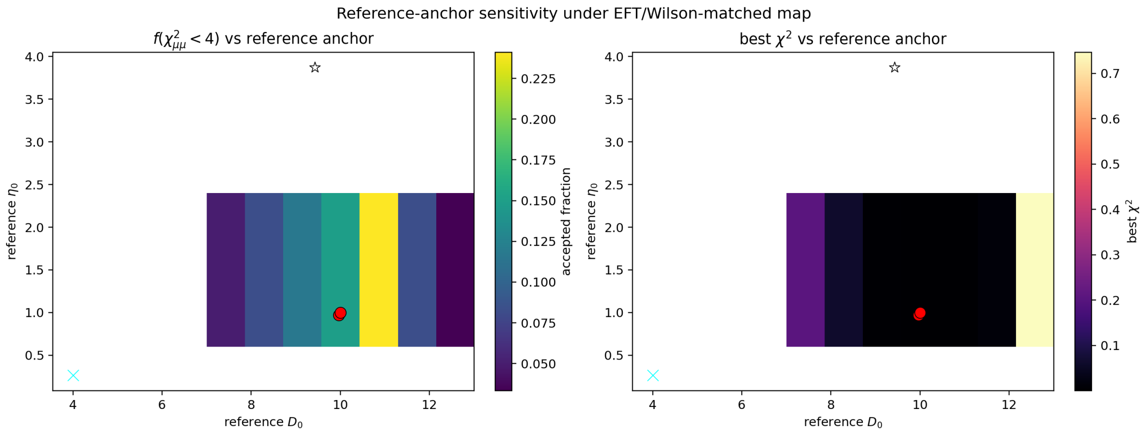


FIG. 14. Systematic sensitivity over reference anchors $(D_0, \eta_0)$ under the EFT/Wilson-matched UV+LL-RG map. Left: $f(\chi^2_{\mu\mu} < 4)$; right: best $\chi^2$. Overlaid markers indicate baseline fixed, $\chi^2$-best, and robust-center selectors. Source files: `paper/reference_anchor_sensitivity.csv`, `paper/reference_anchor_sensitivity.png`.

The baseline fixed-anchor value used in main figures is $f(\chi^2_{\mu\mu} < 4) = 0.1500$. For reporting discipline, we keep this fixed anchor in headline plots and treat dynamic-anchor outputs as sensitivity diagnostics rather than post-hoc optimization.

### 7. Dephasing-Based First-Principles Candidate for $t_{\rm coh}$

To close the finite-time module with the same action-derived spectral chain, we test the dephasing candidate

$$\Delta\omega_{12}(D) \equiv \omega_2(D) - \omega_1(D), \qquad t_{\rm coh}^{(\rm deph)}(D) \equiv \frac{\pi}{\Delta\omega_{12}(D)}, \tag{E15}$$

where $(\omega_1, \omega_2)$ are extracted from the bound-state convention $E_n = \omega_n^2 - m_0^2 < 0$ in the same 1D on-axis operator chain as Appendix B. The reproducible implementation is provided in the code directory (`extract_tcoh_dephasing_1d.py`).

For non-fine levels, the maximum relative deviation is

$$\max\left(\frac{|t_{\rm coh} - t_{\rm coh}^{(\rm fine)}|}{|t_{\rm coh}^{(\rm fine)}|}\right) = 8.86 \times 10^{-3}, \tag{E16}$$

TABLE XXIV. 1D on-axis dephasing candidate for $t_{\rm coh}$ at $D = \{6, 12, 18\}$ (coarse/mid/fine). Source CSV: `output/tcoh_fp_1d/tcoh_dephasing_D6-12-18.csv`.

| $D$ | level | $\omega_1$ | $\omega_2$ | $\Delta\omega_{12}$ | $t_{\rm coh}^{\rm (deph)}$ |
|---|---|---|---|---|---|
| 6 | coarse | 0.720883 | 0.738989 | $1.8107 \times 10^{-2}$ | $1.7350 \times 10^{2}$ |
| 6 | mid | 0.721345 | 0.739468 | $1.8123 \times 10^{-2}$ | $1.7334 \times 10^{2}$ |
| 6 | fine | 0.721500 | 0.739629 | $1.8129 \times 10^{-2}$ | $1.7329 \times 10^{2}$ |
| 12 | coarse | 0.721781 | 0.722035 | $2.5438 \times 10^{-4}$ | $1.2350 \times 10^{4}$ |
| 12 | mid | 0.722258 | 0.722513 | $2.5533 \times 10^{-4}$ | $1.2304 \times 10^{4}$ |
| 12 | fine | 0.722418 | 0.722674 | $2.5565 \times 10^{-4}$ | $1.2289 \times 10^{4}$ |
| 18 | coarse | 0.719404 | 0.719408 | $3.6700 \times 10^{-6}$ | $8.5602 \times 10^{5}$ |
| 18 | mid | 0.719885 | 0.719888 | $3.6943 \times 10^{-6}$ | $8.5039 \times 10^{5}$ |
| 18 | fine | 0.720046 | 0.720050 | $3.7025 \times 10^{-6}$ | $8.4850 \times 10^{5}$ |

TABLE XXV. Global-scan impact of the dephasing $t_{\rm coh}$ candidate (diagnostic only; not baseline).

| case | $f(\mathcal{R}_3 > 0.90)$ | $f(\mathcal{R}_3 > 0.95)$ | $f(\chi^2_{\mu\mu} < 4)$ | best $\chi^2$ |
|---|---|---|---|---|
| constant $t_{\rm coh} = 1$ | 0.983 | 0.217 | 0.1017 | $5.47 \times 10^{-5}$ |
| $\pi/\Delta\omega_{12}(D)$ | 0.983 | 0.217 | 0.1017 | $5.47 \times 10^{-5}$ |
| $\min(\pi/\Delta\omega_{12}, 10^4)$ | 0.983 | 0.217 | 0.1017 | $5.47 \times 10^{-5}$ |

which is numerically stable at the sub-$10^{-2}$ level for this benchmark set.

To assess map-level impact, we compared three cases on the same $60 \times 60$ $(D, \eta)$ grid within the legacy constant-$A_\ell$ reference chain: constant $t_{\rm coh} = 1$, raw dephasing profile from Eq. (E15), and a capped variant $\min(t_{\rm coh}^{\rm (deph)}, 10^4)$. Results are summarized in Table XXV (source CSV: `paper/tcoh_profile_impact.csv`).

The dephasing candidate leaves both $\mathcal{R}_3$ and $H \to \mu\mu$ summary fractions unchanged on this grid under the EFT/Wilson-matched diagnostic chain. Therefore, in this manuscript, Eq. (E15) is reported as a first-principles candidate benchmark only and is not propagated into the baseline reported figures.

### 8. Splitting-Prefactor First-Principles Candidate for $\eta$

To test a geometry-closed replacement direction for the overlap amplitude in Eq. (24), we use the same action-derived quantities $(\Delta E_{12}, S_1)$ and define the splitting prefactor

$$\Delta E_{12}(D) \approx A_{12}(D)\, \mathrm{e}^{-S_1(D)}, \qquad A_{12}(D) \equiv \Delta E_{12}(D)\, \mathrm{e}^{S_1(D)}. \tag{E17}$$

TABLE XXVI. 1D on-axis benchmark for $\eta$ prefactor candidates at $D = \{6, 12, 18\}$ (coarse/mid/fine). Source CSV: `output/eta_fp_1d/eta_prefactor_D6-12-18.csv`.

| $D$ | level | $\Delta E_{12}$ | $S_1$ | $A_{12}$ | $\eta_{\rm amp}$ | $\eta_{\rm prob}$ |
|---|---|---|---|---|---|---|
| 6 | coarse | $2.6433 \times 10^{-2}$ | 4.28295 | 1.91521 | 1.04764 | 1.09755 |
| 6 | mid | $2.6475 \times 10^{-2}$ | 4.29274 | 1.93709 | 1.05923 | 1.12197 |
| 6 | fine | $2.6489 \times 10^{-2}$ | 4.28158 | 1.91661 | 1.04787 | 1.09803 |
| 12 | coarse | $3.6727 \times 10^{-4}$ | 8.51270 | 1.82812 | 1.00000 | 1.00000 |
| 12 | mid | $3.6889 \times 10^{-4}$ | 8.50865 | 1.82877 | 1.00000 | 1.00000 |
| 12 | fine | $3.6944 \times 10^{-4}$ | 8.50734 | 1.82906 | 1.00000 | 1.00000 |
| 18 | coarse | $5.2804 \times 10^{-6}$ | 12.75507 | 1.82864 | 1.00028 | 1.00056 |
| 18 | mid | $5.3189 \times 10^{-6}$ | 12.75845 | 1.84820 | 1.01063 | 1.02137 |
| 18 | fine | $5.3320 \times 10^{-6}$ | 12.74576 | 1.82936 | 1.00017 | 1.00033 |

TABLE XXVII. Global-scan impact of $\eta$ prefactor candidates (diagnostic only; not baseline).

| case | $f(\mathcal{R}_3 > 0.90)$ | $f(\mathcal{R}_3 > 0.95)$ | $f(\chi^2_{\mu\mu} < 4)$ | best $\chi^2$ |
|---|---|---|---|---|
| reference $\eta_{\rm eff} = \eta$ | 0.983 | 0.217 | 0.1017 | $5.47 \times 10^{-5}$ |
| $\eta_{\rm eff} = \eta\, \eta_{\rm amp}(D)$ | 0.983 | 0.217 | 0.1022 | $6.24 \times 10^{-5}$ |
| $\eta_{\rm eff} = \eta\, \eta_{\rm prob}(D)$ | 0.983 | 0.217 | 0.1019 | $4.45 \times 10^{-6}$ |
| $\eta_{\rm eff} = \eta_{\rm amp}(D)$ | 0.983 | 0.217 | 0.0833 | $6.34 \times 10^{-2}$ |
| $\eta_{\rm eff} = \eta_{\rm prob}(D)$ | 0.983 | 0.217 | 0.0833 | $6.39 \times 10^{-2}$ |

With a reference point $D_{\rm ref} = 12$, we define two dimensionless candidates:

$$\eta_{\rm amp}(D) \equiv \frac{A_{12}(D)}{A_{12}(D_{\rm ref})}, \qquad \eta_{\rm prob}(D) \equiv \eta_{\rm amp}(D)^2. \tag{E18}$$

The reproducible script is `extract_eta_prefactor_1d.py` in the `code/` directory.

For non-fine levels in Table XXVI, we obtain

$$\max\left(\frac{|\eta_{\rm amp} - \eta_{\rm amp}^{\rm (fine)}|}{|\eta_{\rm amp}^{\rm (fine)}|}\right) = 1.08 \times 10^{-2}, \qquad \max\left(\frac{|\eta_{\rm prob} - \eta_{\rm prob}^{\rm (fine)}|}{|\eta_{\rm prob}^{\rm (fine)}|}\right) = 2.18 \times 10^{-2}, \tag{E19}$$

and the fine full-scan profile over $D = 4, \ldots, 20$ gives $\eta_{\rm amp} \in [0.9996, 1.1695]$ and $\eta_{\rm prob} \in [0.9992, 1.3677]$ (source CSV: `output/eta_fp_1d/eta_prefactor_D4-5-...-20.csv`).

To quantify map-level sensitivity, we compare a reference $\eta_{\rm eff} = \eta$ run (legacy constant-$A_\ell$ chain) against profile-scaled and fully-closed variants on the same $60 \times 60$ grid. Results are summarized in Table XXVII (source CSV: `paper/eta_profile_impact.csv`).

At current scan resolution, profile-scaled implementations remain close to the reference map-level summaries, while fully-closed implementations reduce $f(\chi^2_{\mu\mu} < 4)$ to 0.0833 in this diagnostic chain. Therefore, Eq. (E18) is reported as a first-principles candidate benchmark and is

TABLE XXVIII. 1D on-axis benchmark for barrier-leakage channel-profile candidates at $D = \{6, 12, 18\}$ (coarse/mid/fine). Source CSV: `output/superrad_fp_1d/superrad_prefactor_D6-12-18.csv` (legacy naming).

| $D$ | level | $\omega_{N_{\rm ref}}$ | $S_{N_{\rm ref},1}$ | $S_{N_{\rm ref},2}$ | $\tilde{A}_1$ | $\tilde{A}_2$ | $A_2^{(\rm fp)}/A_1^{(\rm fp)}$ |
|---|---|---|---|---|---|---|---|
| 6 | coarse | 0.738989 | 12.7478 | 20.7033 | $1.2936 \times 10^4$ | $5.5305 \times 10^4$ | $4.1247 \times 10^{-7}$ |
| 6 | mid | 0.739468 | 12.7644 | 20.7073 | $1.2452 \times 10^4$ | $5.4667 \times 10^4$ | $4.2192 \times 10^{-7}$ |
| 6 | fine | 0.739629 | 12.7711 | 20.7088 | $1.2652 \times 10^4$ | $5.6814 \times 10^4$ | $4.2589 \times 10^{-7}$ |
| 12 | coarse | 0.722035 | 17.5746 | 26.3029 | 1.0000 | 1.0000 | $9.6480 \times 10^{-8}$ |
| 12 | mid | 0.722513 | 17.5720 | 26.3009 | 1.0000 | 1.0000 | $9.6103 \times 10^{-8}$ |
| 12 | fine | 0.722674 | 17.5866 | 26.3217 | 1.0000 | 1.0000 | $9.4844 \times 10^{-8}$ |
| 18 | coarse | 0.719408 | 22.0000 | 31.0465 | $1.47 \times 10^{-4}$ | $7.9 \times 10^{-5}$ | $5.1810 \times 10^{-8}$ |
| 18 | mid | 0.719888 | 22.0060 | 31.0550 | $1.45 \times 10^{-4}$ | $7.8 \times 10^{-5}$ | $5.1411 \times 10^{-8}$ |
| 18 | fine | 0.720050 | 22.0043 | 31.0573 | $1.50 \times 10^{-4}$ | $8.0 \times 10^{-5}$ | $5.0954 \times 10^{-8}$ |

not propagated into baseline reported figures.

### 9. Action-Derived First-Principles Candidate for Barrier-Leakage Channel Scaling

To test a geometry-closed replacement direction for the channel normalization in Eq. (32), we keep the same 1D action-derived chain and define channel-resolved barrier actions

$$U_\ell(z; D) \equiv U(z; D) + \frac{\ell(\ell+1)}{z^2 + \varepsilon^2}, \qquad S_{N,\ell}(D) \equiv \int_{z_-}^{z_+} \mathrm{d}z \, \sqrt{U_\ell(z; D) - E_N(D)}, \tag{E20}$$

with $E_N = \omega_N^2 - m_0^2$ in bound-state convention. We then define geometry-driven channel rates

$$\Gamma^{(\rm geo)}_{N,\ell}(D) \equiv \omega_N(D) \, e^{-2S_{N,\ell}(D)}, \tag{E21}$$

and infer effective channel normalizations by matching to Eq. (32):

$$A_\ell^{(\rm fp)}(D; N) \equiv \frac{\Gamma^{(\rm geo)}_{N,\ell}(D)}{\omega_N(D) \, (\omega_N(D) M_*)^{4\ell+4}}. \tag{E22}$$

Using $(D_{\rm ref}, N_{\rm ref}) = (12, 2)$, we define profile factors

$$\tilde{A}_\ell(D) \equiv \frac{A_\ell^{(\rm fp)}(D; N_{\rm ref})}{A_\ell^{(\rm fp)}(D_{\rm ref}; N_{\rm ref})}. \tag{E23}$$

The reproducible script is `extract_superrad_prefactor_1d.py` in the `code/` directory (legacy naming).

For non-fine rows in Table XXVIII, the maximum relative deviations are

$$\max\left(\frac{|\tilde{A}_1 - \tilde{A}_1^{(\rm fine)}|}{|\tilde{A}_1^{(\rm fine)}|}\right) = 3.22 \times 10^{-2}, \quad \max\left(\frac{|\tilde{A}_2 - \tilde{A}_2^{(\rm fine)}|}{|\tilde{A}_2^{(\rm fine)}|}\right) = 3.78 \times 10^{-2}. \tag{E24}$$

TABLE XXIX. Global-scan impact of barrier-leakage channel-profile candidates (diagnostic only; not baseline).

| case | $f(\mathcal{R}_3 > 0.90)$ | $f(\mathcal{R}_3 > 0.95)$ | $f(\chi^2_{\mu\mu} < 4)$ | best $\chi^2$ |
|---|---|---|---|---|
| legacy const $(A_1, A_2) = (1, 1)$ | 0.9833 | 0.217 | 0.0833 | $5.10 \times 10^{-2}$ |
| action-profile $(A_1, A_2) = (\tilde{A}_1, \tilde{A}_2)$ | 0.9269 | 0.217 | 0.0833 | $5.10 \times 10^{-2}$ |
| $(A_1, A_2) = 0.5(\tilde{A}_1, \tilde{A}_2)$ | 0.9061 | 0.217 | 0.0833 | $5.10 \times 10^{-2}$ |
| $(A_1, A_2) = 2(\tilde{A}_1, \tilde{A}_2)$ | 0.9469 | 0.217 | 0.0833 | $5.10 \times 10^{-2}$ |

The fine full-scan profile over $D = 4, \ldots, 20$ yields $\tilde{A}_1 \in [8.54 \times 10^{-6}, 4.10 \times 10^{5}]$ and $\tilde{A}_2 \in [3.95 \times 10^{-6}, 4.92 \times 10^{6}]$ (source files in `output/superrad_fp_1d/`), indicating a very stiff $D$-dependence in this preliminary closure.

To quantify map-level sensitivity, we compare a legacy constant-$A_\ell$ reference $(A_1, A_2) = (1, 1)$ against profile-scaled implementations on the same $60 \times 60$ grid. Results are summarized in Table XXIX (source CSV: `paper/superrad_profile_impact.csv`).

At current scan resolution, replacing the legacy constant-$A_\ell$ reference by the action-derived profile induces non-negligible shifts in $\mathcal{R}_3$ coverage, while the $H \to \mu\mu$ acceptance area in this diagnostic chain remains unchanged. In the updated global scans of this manuscript, Eq. (E23) is used in baseline mode, and $(A_1, A_2) = (1, 1)$ is retained only as a comparator.

## Appendix F: Ansatz Ablation Study

To demonstrate that the three-generation structure is robust and not an artifact of a particular ansatz choice, we compare the baseline EFT-operator-normalized $B_N(D)$ profile (this work) against the alternative "inverse Yukawa" ansatz explored in earlier versions.

### 1. Inverse Yukawa Ansatz (Ablation A)

The inverse Yukawa ansatz sets $B_N^{(\mathrm{inv})} \propto 1/\sum_{f \in \mathrm{Gen}\ N} y_f$, yielding:

$$B_1^{(\mathrm{inv})} \approx 10^5, \quad B_2^{(\mathrm{inv})} \approx 125, \quad B_3^{(\mathrm{inv})} = 1. \tag{F1}$$

This massive hierarchy ($B_1 \gg B_2 \gg B_3$) directly encodes the answer by over-enhancing layer 1. While it can produce $\mathcal{R}_3 > 95\%$, the mechanism is less transparent: the visibility factor, rather than the entropic/kinetic competition, drives the result.

### 2. Comparison

- **Baseline (EFT-operator-normalized profile)**: $B_N(D)$ remains smooth and bounded across the scanned range, with shape inherited from tracked microcanonical overlap extraction and operator normalization. The three-generation structure emerges from the $g_N$–$\Gamma_N$ competition modulated by these profile differences.

- **Ablation A (Inverse Yukawa)**: $B_1 \gg B_2 \gg B_3$. Layer 1 is artificially boosted; the selection mechanism is less falsifiable.

We recommend the baseline as the primary closure due to its greater transparency and falsifiability.

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